1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION
A	0000891242	PRODUCTION RELEASED

K74 MLB

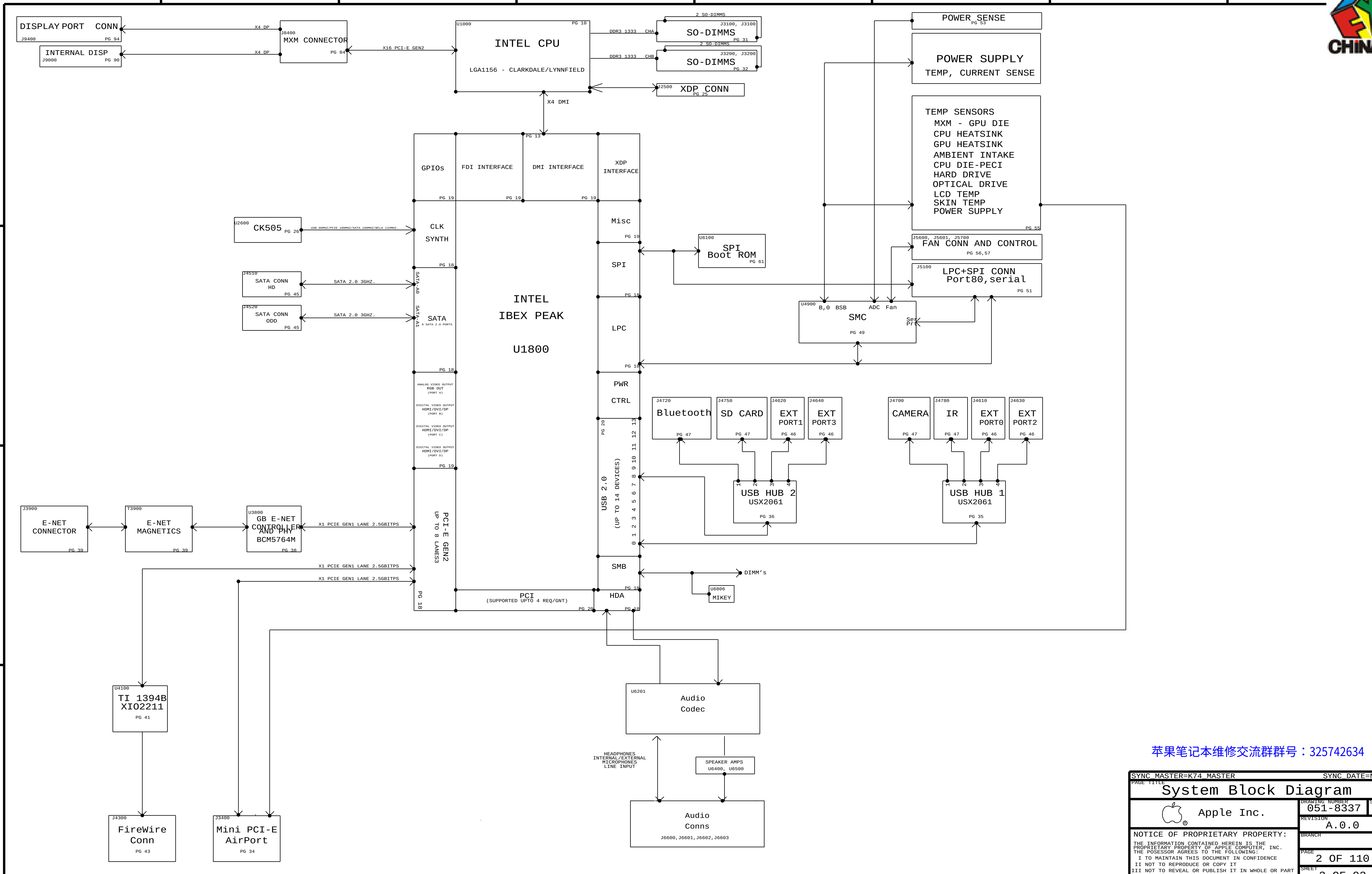
LAST_MODIFIED=Tue Apr 13 17:17:57 2010

DRAWING
TITLE=K22
ABBREV=DRAWING
LAST_MODIFIED=Tue Apr 13 17:17:57 2010

DRAWING TITLE: SCH,K74,MLB

Apple Inc.

DRAWING NUMBER: 051-8337
SIZE: D
REVISION: A.0.0
BRANCH:
PAGE: 1 OF 110
SHEET: 1 OF 92



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

System Block Diagram

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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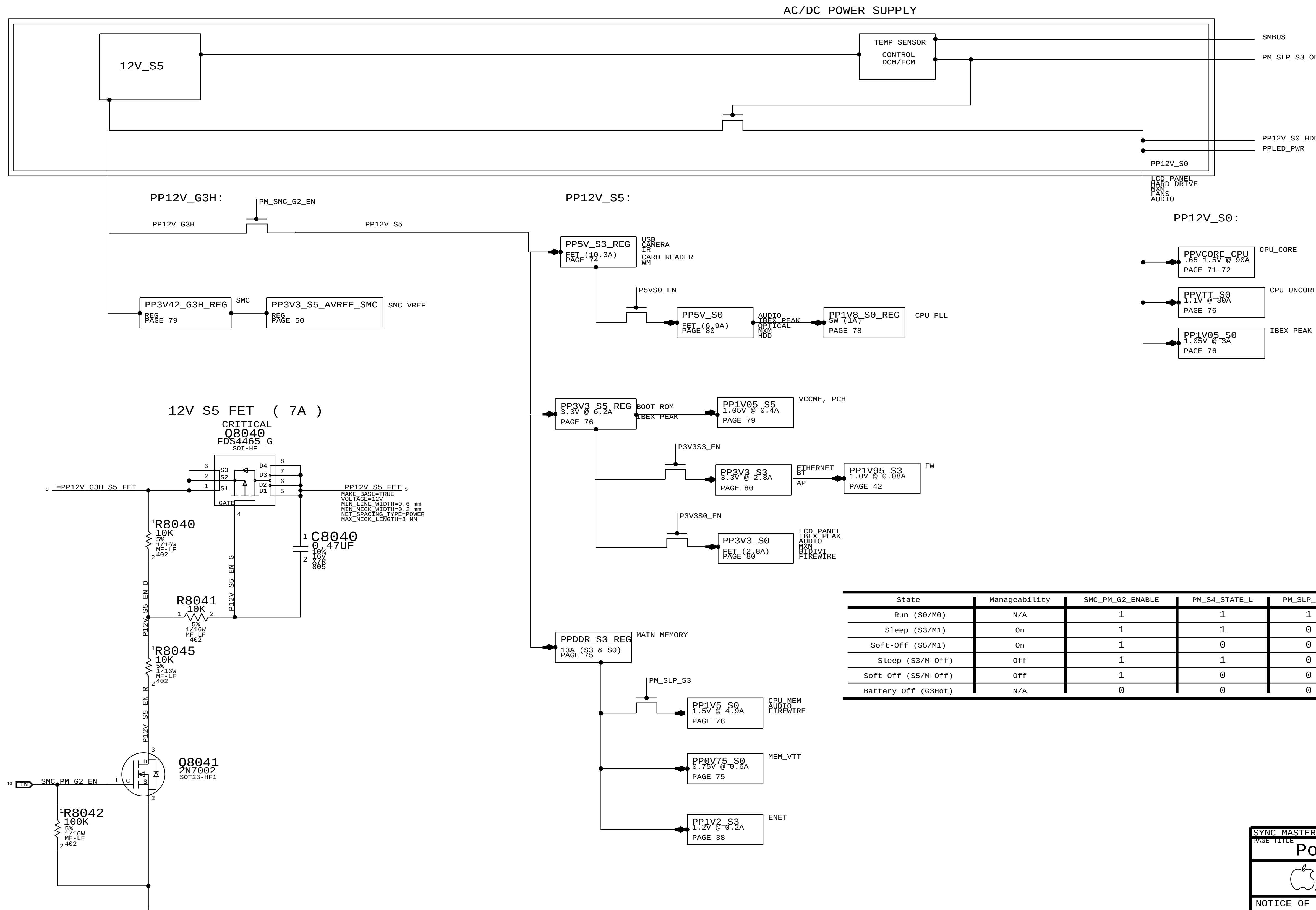

State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_M_L
Run (S0/M0)	N/A	1	1	1	1	1
Sleep (S3/M1)	On	1	1	0	1	1
Soft-Off (S5/M1)	On	1	0	0	1	1
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

Power Block Diagram

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-1107	PCBA,MLB,DEV,K74	DEVELOPMENT,DEV_GROUP
639-0698	PCBA,MLB,K74,2.93GHZ,CKD	K74,2P93GHZ_CKD_CPU,BASIC,CLARKDALE_73W
639-0707	PCBA,MLB,K74,3.06GHZ,CKD	K74,3P06GHZ_CKD_CPU,BASIC,CLARKDALE_73W
639-0808	PCBA,MLB,K74,3.20GHZ,CKD	K74,3P20GHZ_CKD_CPU,BASIC,CLARKDALE_73W
639-0695	PCBA,MLB,K74,3.46GHZ,CKD	K74,3P46GHZ_CKD_CPU,BASIC,CLARKDALE_73W
639-0991	PCBA,MLB,K74,3.60GHZ,CKD	K74,3P60GHZ_CKD_CPU,BASIC,CLARKDALE_73W
639-0694	PCBA,MLB,K74,2.53GHZ,LFD	K74,2P53GHZ_LFD_CPU,BASIC,LYNNFIELD_82W

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC	COMMON,ALTERNATE,XDP,MXM,XDP_CPU_BPM,PCH_VRM,BUF_CLK,HUB_USX2061,FW_TI_INT_VREG,BCM5764M,SD_USB,METAL_IO,PRODUCTION
DEV_GROUP	XDP_CONN,LPCPLUS,MOJOMUX,CPU_1V5_SENSE,VREFMRGN

CPU SOCKET & ILM SUB-BOMS

ALTERNATE SOCKET VENDORS MUST USE MATCHING ILM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0063	1	SOCKET,LGA1156,CPU-LF	U1000	CRITICAL	MOLEX_SOCKET
604-1161	1	ASSY,PURCHASED,ILM,MOLEX,K74	ILM	CRITICAL	MOLEX_SOCKET
511S0069	1	SOCKET,LGA1156,CPU-LF	U1000	CRITICAL	FOXCONN_SOCKET
604-1246	1	ASSY,PURCHASED,ILM,MOLEX,K74	ILM	CRITICAL	FOXCONN_SOCKET

BOM NUMBER	BOM NAME	BOM OPTIONS
607-6694	SUB ASSY,CPU SOCKET,K74,MOLEX	MOLEX_SOCKET
607-6693	SUB ASSY,CPU SOCKET,K74,FOXCONN	FOXCONN_SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
607-6694	1	MOLEX CPU SOCKET AND ILM	SKT_ILM	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
607-6693	607-6694		SKT_ILM	FOXCONN ALTERNATE

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

RAW: 335S0663

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S3828	1	IC,IBEX PEAK PRQ,DESKTOP,FCBGA,PCH,P425	U1800	CRITICAL	
359S0157	1	IC,SLG2AP108,CLK GEN,CK505,QFN3	U2600	CRITICAL	BUF_CLK
341T0230	1	IC,EFI BOOTROM,K74/K75	U6100	CRITICAL	
338S0765	1	IC,XIO2211ZAY,1394B_PCIE,PHY/LINK	U4100	CRITICAL	
825-7122	1	MLB LABEL,48.0X4.8	X14	CRITICAL	
343S0493	1	IC,BCM5764M,ENET,8X8	U3900	CRITICAL	BCM5764M
343S0494	1	IC,BCM57765A,ENET&SD,8X8	U3900	CRITICAL	BCM57765
341T0269	1	ENET 1MBIT FLASH,CII,K74/K75	U3990	CRITICAL	BCM5764M
341T0246	1	ENET 1MBIT FLASH,CIV,K74/K75	U3990	CRITICAL	BCM57765

CPUS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S3837	1	CKD,Q3GR,QS,2.93,73W,1333,C2,4M,LGA	CPU	CRITICAL	2P93GHZ_CKD_CPU
337S3912	1	CKD,SLBTD,PRQ,3.06,73W,1333,K0,4M,LGA	CPU	CRITICAL	3P06GHZ_CKD_CPU
337S3911	1	CKD,SLBUD,PRQ,3.20,73W,1333,K0,4M,LGA	CPU	CRITICAL	3P20GHZ_CKD_CPU
337S3900	1	CKD,SLBLT,PRQ,3.46,73W,1333,C2,4M,LGA	CPU	CRITICAL	3P46GHZ_CKD_CPU
337S3910	1	CKD,SLBTM,PRQ,3.60,73W,1333,K0,4M,LGA	CPU	CRITICAL	3P60GHZ_CKD_CPU
337S3862	1	LFD,Q3C6,QS,2.53,82W,1333,B1,8M,LGA	CPU	CRITICAL	2P53GHZ_LFD_CPU

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
337S3898	337S3912	3P06GHZ_CKD_CPU	CPU	C2,PRQ,3.06 GHZ CKD

K74 PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-8337	1	SCH,MLB,K74	SCH1		
820-2784	1	PCBF,MLB,K74	MLB1		
341T0231	1	IC,SMC,K74	U4900	CRITICAL	K74

ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0339	197S0179		Y4190	FIREWIRE OSCILLATOR
128S0298	128S0293		C1670,C7260,C7444	

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
BOM Configuration

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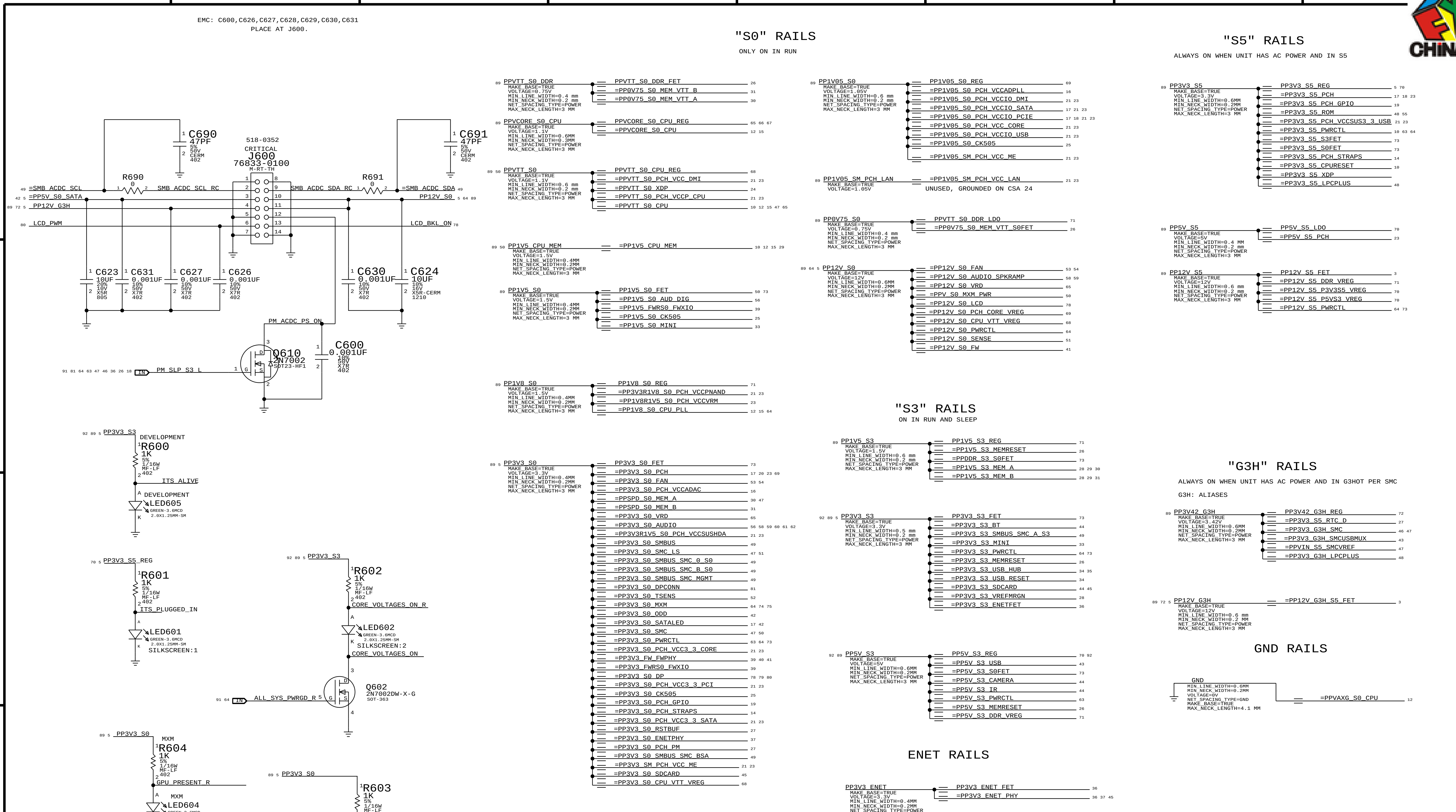

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PAGE TITLE: Power Conn / Alias

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DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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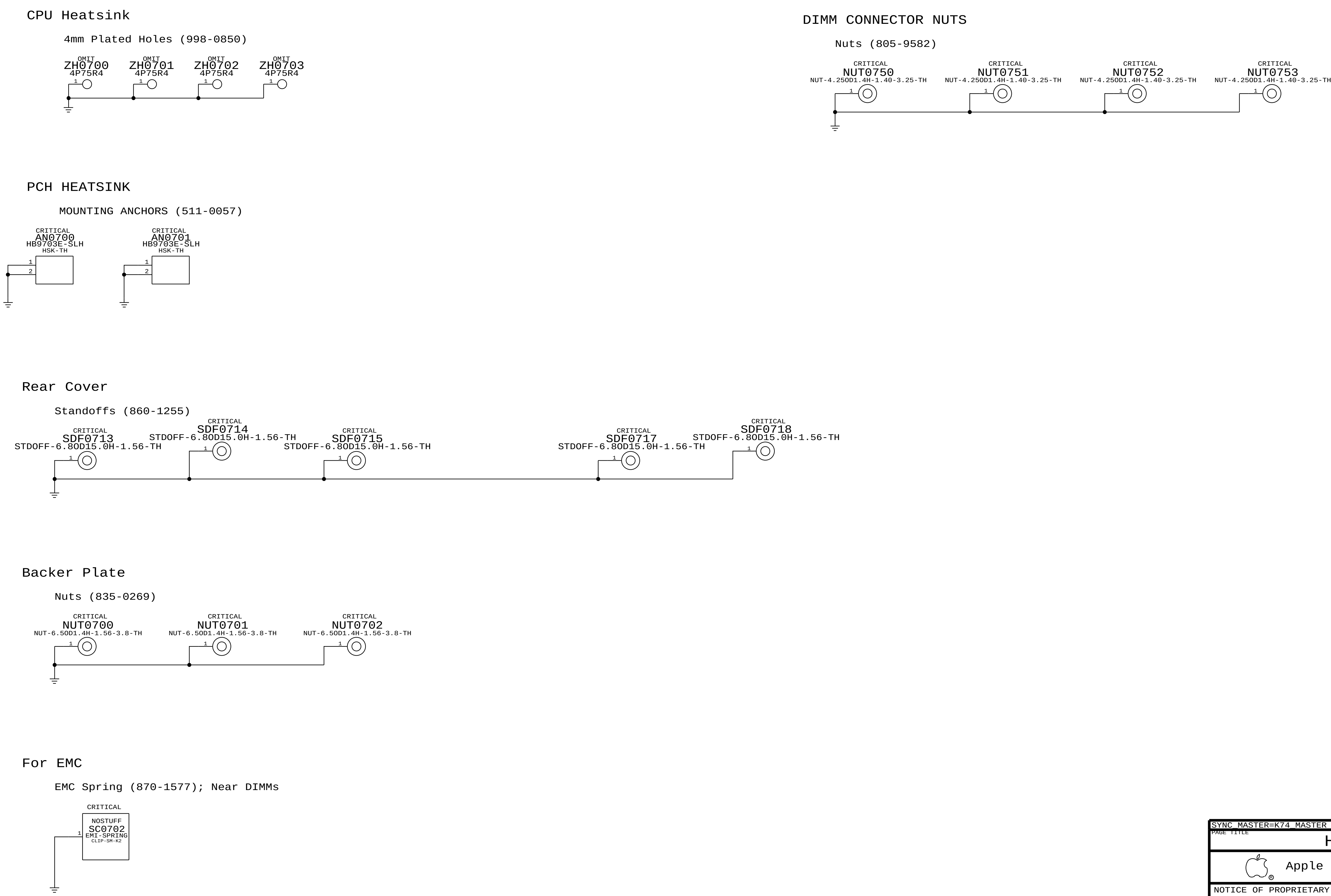

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

Holes

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8	7	6	5	4	3	2	1

UNUSED CPU SIGNALS

Page	Test point		Net alias	Properties
9	TP_CPU_RSVD<41..29>	=	NC_CPU_RSVD<41..29>	MAKE_BASE=TRUE NO_TEST=TRUE
9	TP_CPU_RSVD<26..1>	=	NC_CPU_RSVD<26..1>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_CPU_FC_AE38	=	NC_CPU_FC_AE38	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_CPU_FC_AG40	=	NC_CPU_FC_AG40	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED PCI ALIASES

Page	Test point		Net alias	Properties
19	TP_PCI_AD<31..0>	=	NC_PCI_AD<31..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCI_C_BE_L<3..0>	=	NC_PCI_C_BE_L<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCI_PAR	=	NC_PCI_PAR	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_PCI_RESET_L	=	NC_PCI_RESET_L	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCIE_CLK100M_XDPP	=	NC_PCIE_CLK100M_XDPP	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCIE_CLK100M_XDPN	=	NC_PCIE_CLK100M_XDPN	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_DMI_CLK100M_LAP	=	NC_DMI_CLK100M_LAP	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_DMI_CLK100M_LAN	=	NC_DMI_CLK100M_LAN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_LPC_DREQ1_L	=	NC_LPC_DREQ1_L	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_LPC_DREQ0_L	=	NC_LPC_DREQ0_L	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED NAND ALIASES

Page	Test point		Net alias	Properties
19	TP_NV_CE_L<3..0>	=	NC_NV_CE_L<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_DQS<1..0>	=	NC_NV_DQS<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_DQ<15..0>	=	NC_NV_DQ<15..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_RCOMP	=	NC_NV_RCOMP	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_RB_L	=	NC_NV_RB_L	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_WR_RE_L<1..0>	=	NC_NV_WR_RE_L<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_WE_CK_L<1..0>	=	NC_NV_WE_CK_L<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_ALE	=	NC_NV_ALE	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_NV_CLE	=	NC_NV_CLE	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MEM ALIASES

Page	Test point		Net alias	Properties
11	TP_MEM_A_CS_L<7..4>	=	NC_MEM_A_CS_L<7..4>	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_MEM_A_DQ_CB<7..0>	=	NC_MEM_A_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_MEM_A_DQS_N<8>	=	NC_MEM_A_DQSN<8>	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_MEM_A_DQS_P<8>	=	NC_MEM_A_DQSP<8>	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_MEM_B_CS_L<7..4>	=	NC_MEM_B_CS_L<7..4>	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_MEM_B_DQ_CB<7..0>	=	NC_MEM_B_DQ_CB<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_MEM_B_DQS_N<8>	=	NC_MEM_B_DQSN<8>	MAKE_BASE=TRUE NO_TEST=TRUE
11	TP_MEM_B_DQS_P<8>	=	NC_MEM_B_DQSP<8>	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MISC ALIASES

Page	Test point		Net alias	Properties
17	TP_HDA_SDIN1	=	NC_HDA_SDIN1	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_HDA_SDIN2	=	NC_HDA_SDIN2	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_HDA_SDIN3	=	NC_HDA_SDIN3	MAKE_BASE=TRUE NO_TEST=TRUE
	TP_JTAG_XDP_TRST_L	=	NC_JTAG_XDP_TRST_L	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCH_PWM0	=	NC_PCH_PWM0	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCH_PWM1	=	NC_PCH_PWM1	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCH_PWM2	=	NC_PCH_PWM2	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCH_PWM3	=	NC_PCH_PWM3	MAKE_BASE=TRUE NO_TEST=TRUE
20	TP_PCH_SST	=	NC_PCH_SST	MAKE_BASE=TRUE NO_TEST=TRUE
9	SNS_CPU_THERMD_N	=	NC_SNS_CPU_THERMDN	MAKE_BASE=TRUE NO_TEST=TRUE
9	SNS_CPU_THERMD_P	=	NC_SNS_CPU_THERMDP	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED PCIE ALIASES

Page	Test point		Net alias	Properties
17	TP_PCIE_T28_D2R_N<3..0>	=	NC_PCIE_T28_D2RN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_PCIE_T28_D2R_P<3..0>	=	NC_PCIE_T28_D2RP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_PCIE_T28_R2D_C_N<3..0>	=	NC_PCIE_T28_R2D_CN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_PCIE_T28_R2D_C_P<3..0>	=	NC_PCIE_T28_R2D_CP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_PCIE_CLK100M_T28_N	=	NC_PCIE_CLK100M_T28N	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_PCIE_CLK100M_T28_P	=	NC_PCIE_CLK100M_T28P	MAKE_BASE=TRUE NO_TEST=TRUE
17	PCIE_EXCARD_D2R_P	=	NC_PCIE_EXCARD_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
17	PCIE_EXCARD_D2R_N	=	NC_PCIE_EXCARD_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
17	PCIE_EXCARD_R2D_C_P	=	NC_PCIE_EXCARD_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
17	PCIE_EXCARD_R2D_C_N	=	NC_PCIE_EXCARD_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
17	PCIE_CLK100M_EXCARD_P	=	NC_PCIE_CLK100M_EXCARDP	MAKE_BASE=TRUE NO_TEST=TRUE
17	PCIE_CLK100M_EXCARD_N	=	NC_PCIE_CLK100M_EXCARDN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_PCIE_CLK100M_PE5P	=	NC_PCIE_CLK100M_PE5P	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_PCIE_CLK100M_PE5N	=	NC_PCIE_CLK100M_PE5N	MAKE_BASE=TRUE NO_TEST=TRUE
17	DMI_MIDBUS_CLK100M_P	=	NC_DMI_MIDBUS_CLK100MP	MAKE_BASE=TRUE NO_TEST=TRUE
17	DMI_MIDBUS_CLK100M_N	=	NC_DMI_MIDBUS_CLK100MN	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED USB ALIASES

Page	Test point		Net alias	Properties
19	TP_USB_1N	=	NC_USB_1N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_1P	=	NC_USB_1P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_2N	=	NC_USB_2N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_2P	=	NC_USB_2P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_3N	=	NC_USB_3N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_3P	=	NC_USB_3P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_4N	=	NC_USB_4N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_4P	=	NC_USB_4P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_5N	=	NC_USB_5N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_5P	=	NC_USB_5P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_6N	=	NC_USB_6N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_6P	=	NC_USB_6P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_7N	=	NC_USB_7N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_7P	=	NC_USB_7P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_9N	=	NC_USB_9N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_9P	=	NC_USB_9P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_10N	=	NC_USB_10N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_10P	=	NC_USB_10P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_11N	=	NC_USB_11N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_11P	=	NC_USB_11P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_12N	=	NC_USB_12N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_12P	=	NC_USB_12P	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_13N	=	NC_USB_13N	MAKE_BASE=TRUE NO_TEST=TRUE
19	TP_USB_13P	=	NC_USB_13P	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED DISPLAY ALIASES

Page	Test point		Net alias	Properties
18	TP_CRT_IG_DDC_CLK	=	NC_CRT_IG_DDC_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CRT_IG_DDC_DATA	=	NC_CRT_IG_DDC_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CRT_IG_RED	=	NC_CRT_IG_RED	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CRT_IG_GREEN	=	NC_CRT_IG_GREEN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CRT_IG_BLUE	=	NC_CRT_IG_BLUE	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CRT_IG_HSYNC	=	NC_CRT_IG_HSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_CRT_IG_VSYNC	=	NC_CRT_IG_VSYNC	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_B_MLN<3..0>	=	NC_DP_IG_B_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_B_MLP<3..0>	=	NC_DP_IG_B_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_B_AUX_N	=	NC_DP_IG_B_AUXN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_B_AUX_P	=	NC_DP_IG_B_AUXP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_B_HPD	=	NC_DP_IG_B_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_B_DDC_CLK	=	NC_DP_IG_B_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_B_DDC_DATA	=	NC_DP_IG_B_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_C_MLN<3..0>	=	NC_DP_IG_C_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_C_MLP<3..0>	=	NC_DP_IG_C_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_C_AUX_N	=	NC_DP_IG_C_AUXN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_C_AUX_P	=	NC_DP_IG_C_AUXP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_C_HPD	=	NC_DP_IG_C_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_C_CTRL_CLK	=	NC_DP_IG_C_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_C_CTRL_DATA	=	NC_DP_IG_C_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_D_MLN<3..0>	=	NC_DP_IG_D_MLN<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_D_MLP<3..0>	=	NC_DP_IG_D_MLP<3..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_D_AUXN	=	NC_DP_IG_D_AUXN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_D_AUXP	=	NC_DP_IG_D_AUXP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_D_HPD	=	NC_DP_IG_D_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_D_CTRL_CLK	=	NC_DP_IG_D_CTRL_CLK	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_DP_IG_D_CTRL_DATA	=	NC_DP_IG_D_CTRL_DATA	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_GFX_VID<0..6>	=	NC_GFX_VID<0..6>	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_GFX_VSENSE_N	=	NC_GFX_VSENSEN	MAKE_BASE=TRUE NO_TEST=TRUE
12	TP_GFX_VSENSE_P	=	NC_GFX_VSENSEP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SDVO_TVCLKINN	=	NC_SDVO_TVCLKINN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SDVO_TVCLKINP	=	NC_SDVO_TVCLKINP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SDVO_STALLN	=	NC_SDVO_STALLN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SDVO_STALLP	=	NC_SDVO_STALLP	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SDVO_INTN	=	NC_SDVO_INTN	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_SDVO_INTP	=	NC_SDVO_INTP	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED FDI ALIASES

Page	Test point		Net alias	Properties
9	TP_CPU_FDI_TX_N<7..0>	=	NC_CPU_FDI_TXN<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
9	TP_CPU_FDI_TX_P<7..0>	=	NC_CPU_FDI_TXP<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_FDI_RX_N<7..0>	=	NC_PCH_FDI_RXN<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_FDI_RX_P<7..0>	=	NC_PCH_FDI_RXP<7..0>	MAKE_BASE=TRUE NO_TEST=TRUE
9	TP_CPU_FDI_FSYNC<1..0>	=	NC_CPU_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_FDI_FSYNC<1..0>	=	NC_PCH_FDI_FSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
9	TP_CPU_FDI_LSYNC<1..0>	=	NC_CPU_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_FDI_LSYNC<1..0>	=	NC_PCH_FDI_LSYNC<1..0>	MAKE_BASE=TRUE NO_TEST=TRUE
9	TP_CPU_FDI_INT	=	NC_CPU_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE
18	TP_PCH_FDI_INT	=	NC_PCH_FDI_INT	MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED SATA ALIASES

Page	Test point		Net alias	Properties
17	TP_SATA_D_D2RN	=	NC_SATA_D_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_D_D2RP	=	NC_SATA_D_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_D_R2D_CN	=	NC_SATA_D_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_D_R2D_CP	=	NC_SATA_D_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_E_D2RN	=	NC_SATA_E_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_E_D2RP	=	NC_SATA_E_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_E_R2D_CN	=	NC_SATA_E_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_E_R2D_CP	=	NC_SATA_E_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_F_D2RN	=	NC_SATA_F_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_F_D2RP	=	NC_SATA_F_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_F_R2D_CN	=	NC_SATA_F_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_F_R2D_CP	=	NC_SATA_F_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_SSD_D2R_N	=	NC_SATA_SSD_D2RN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_SSD_D2R_P	=	NC_SATA_SSD_D2RP	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_SSD_R2D_C_N	=	NC_SATA_SSD_R2D_CN	MAKE_BASE=TRUE NO_TEST=TRUE
17	TP_SATA_SSD_R2D_C_P	=	NC_SATA_SSD_R2D_CP	MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: UNUSED SIGNAL ALIAS

Apple Inc.

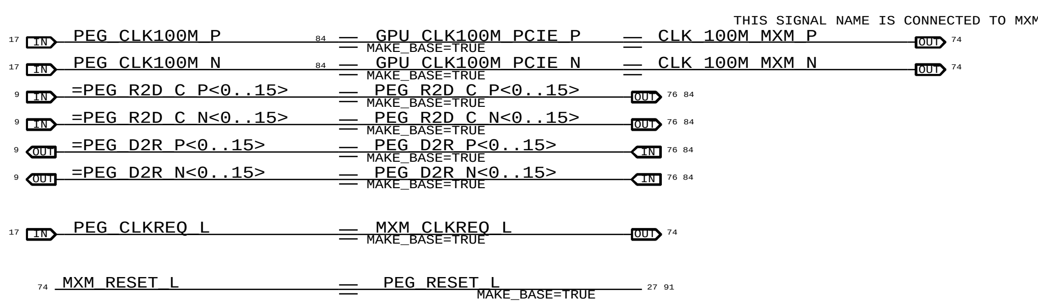
DRAWING NUMBER: 051-8337 SIZE: D

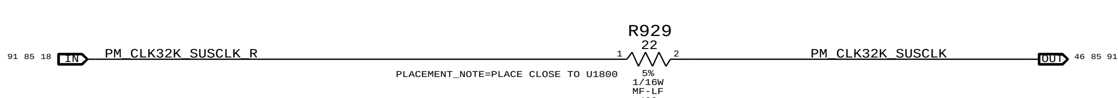
REVISION: A.0.0

BRANCH

8	7	6	5	4	3	2	1

PEG Slot Support





SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

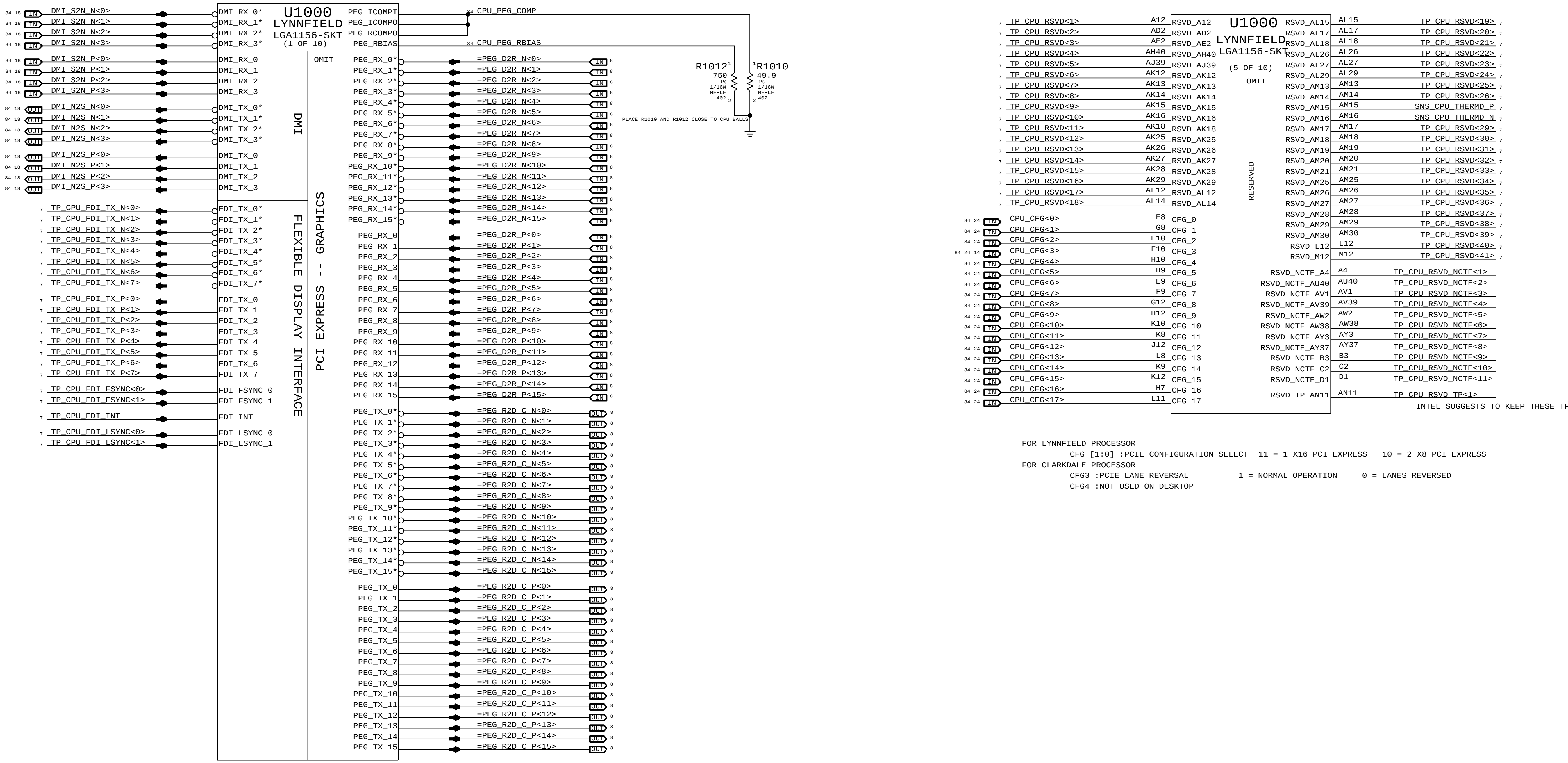
Signal Aliases

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

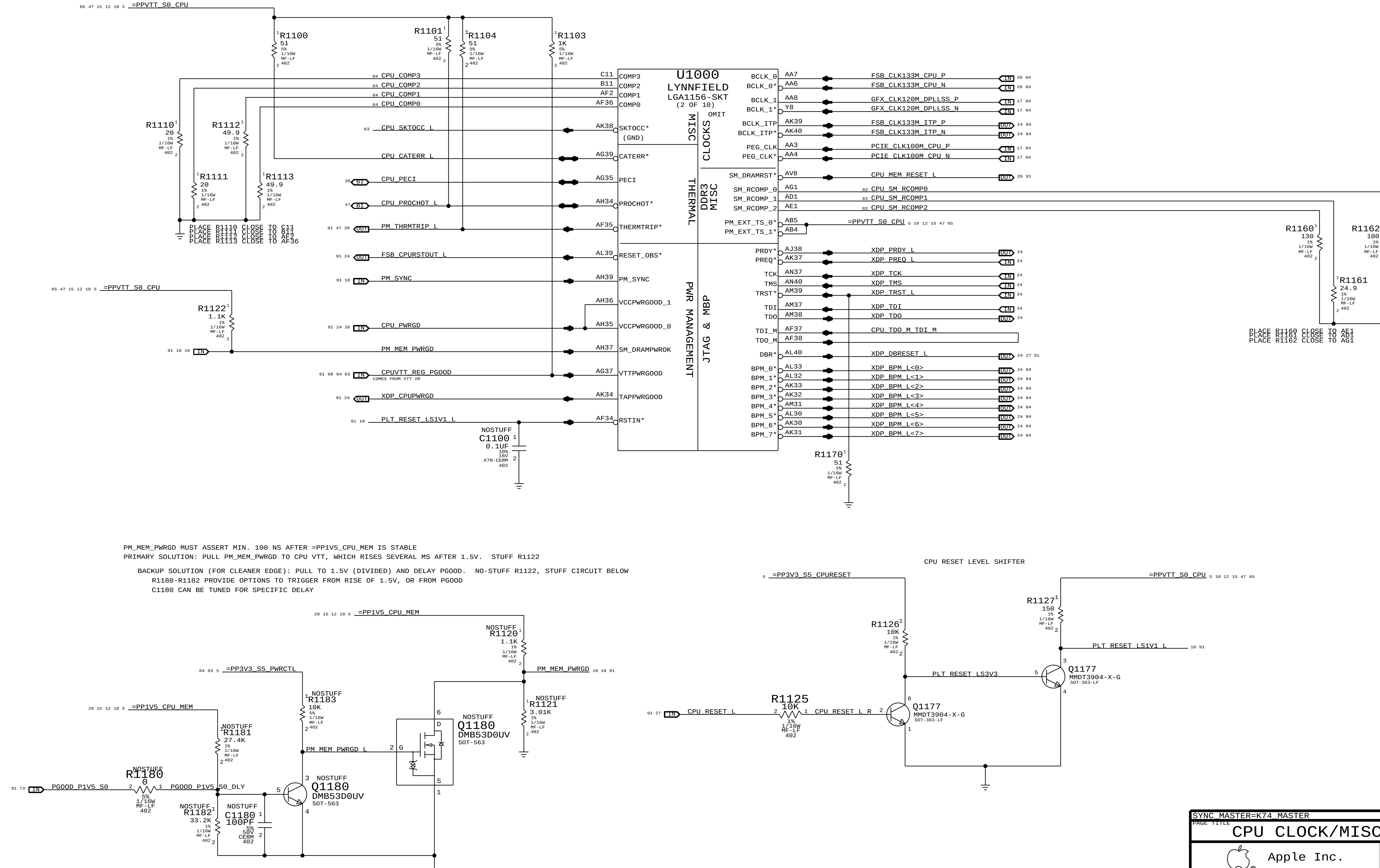
PAGE TITLE
CPU DMI/PEG/FDI/RSVD

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

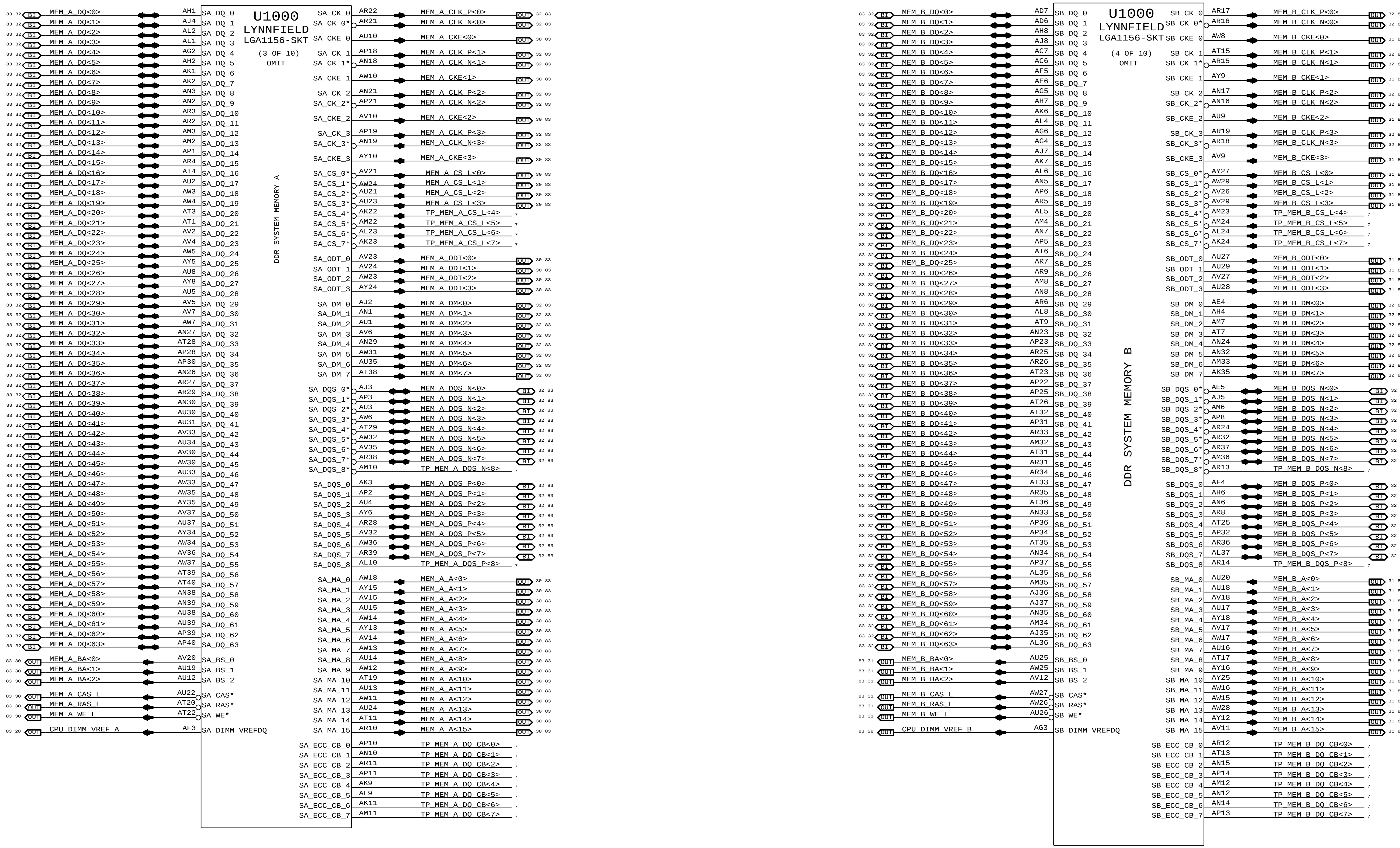
PAGE TITLE
CPU CLOCK/MISC/JTAG

DRAWING NUMBER 051-8337 SIZE D

Apple Inc.

REVISION A.0.0

BRANCH



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

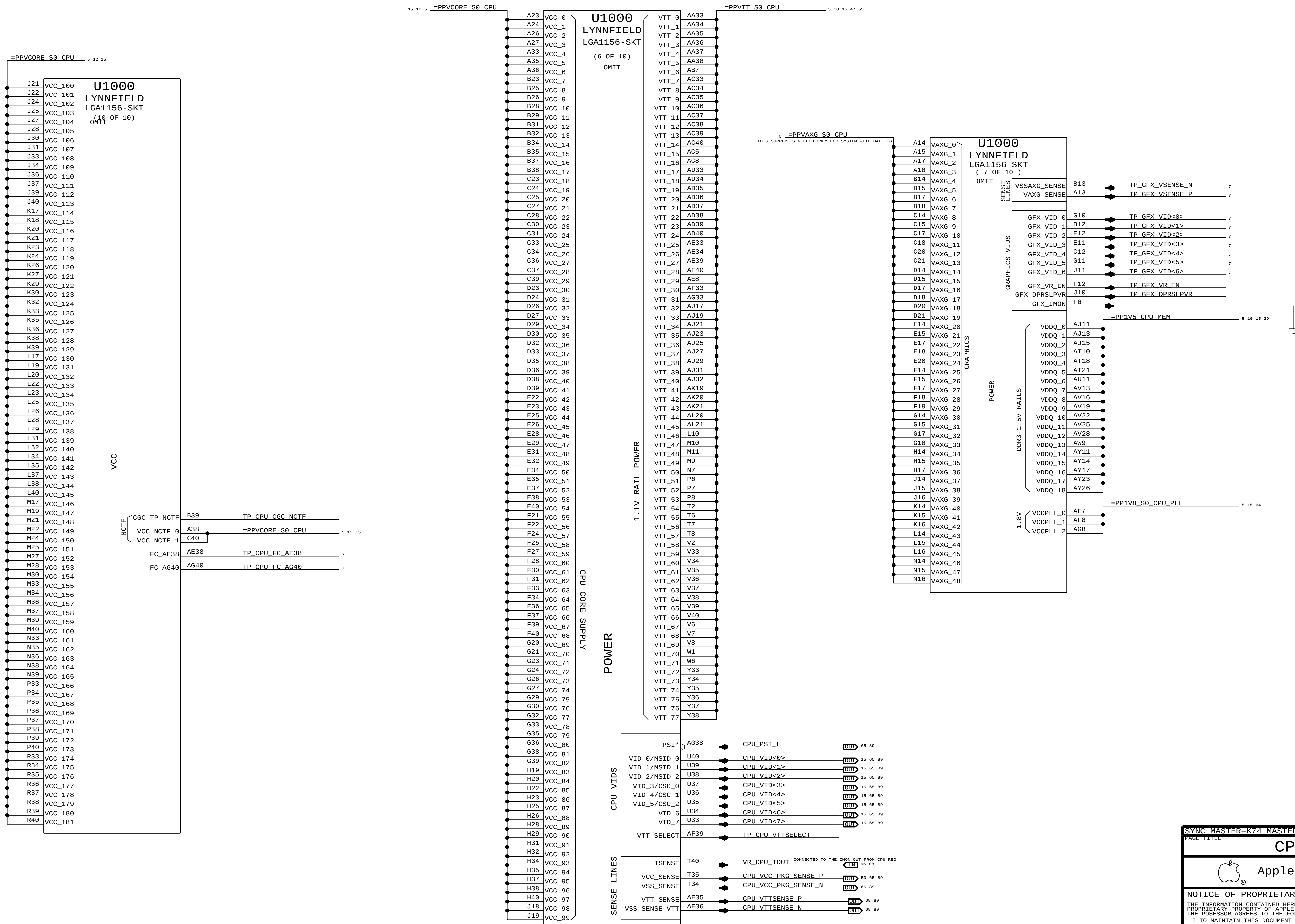
CPU DDR3 INTERFACES

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=K74_MASTER

SYNC_DATE=N/A

PAGE TITLE

CPU POWER

Apple Inc.

DRAWING NUMBER 051-8337

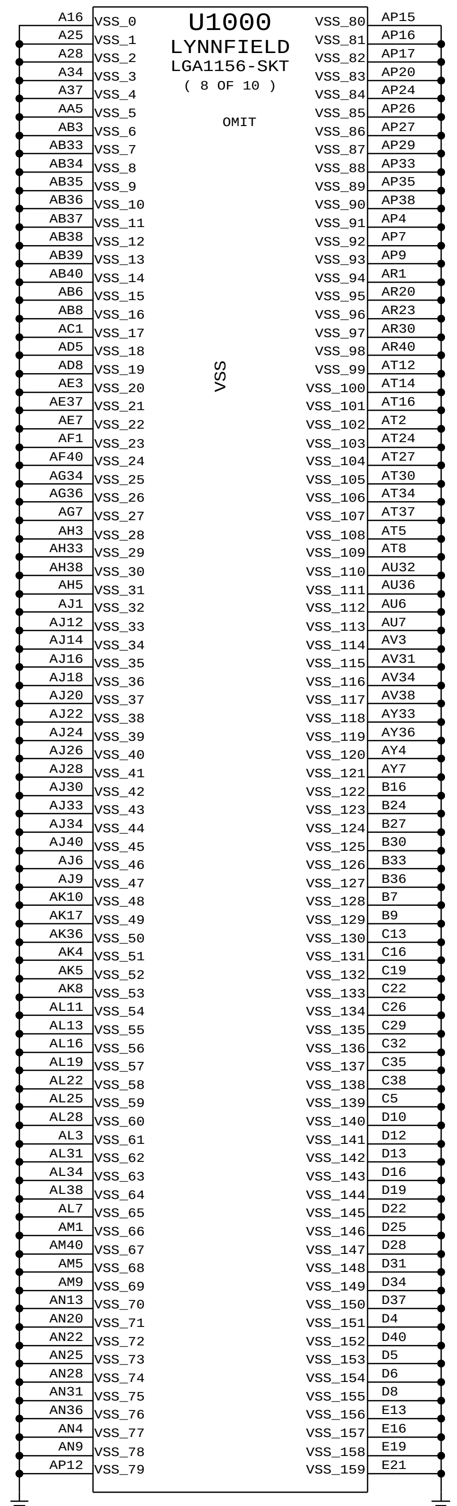
SIZE D

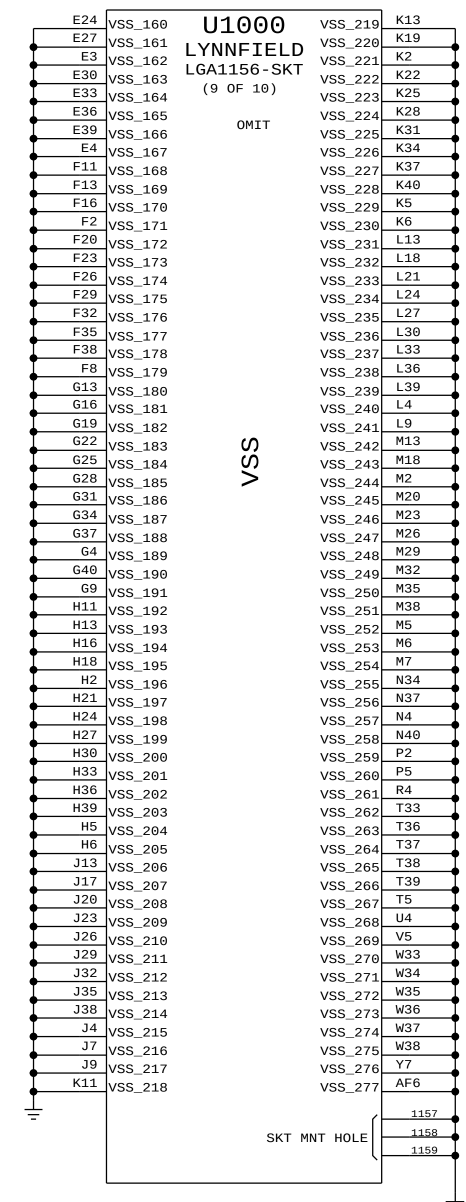
REVISION A.0.0

BRANCH

PAGE 13 OF 110

SHEET 12 OF 92





SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

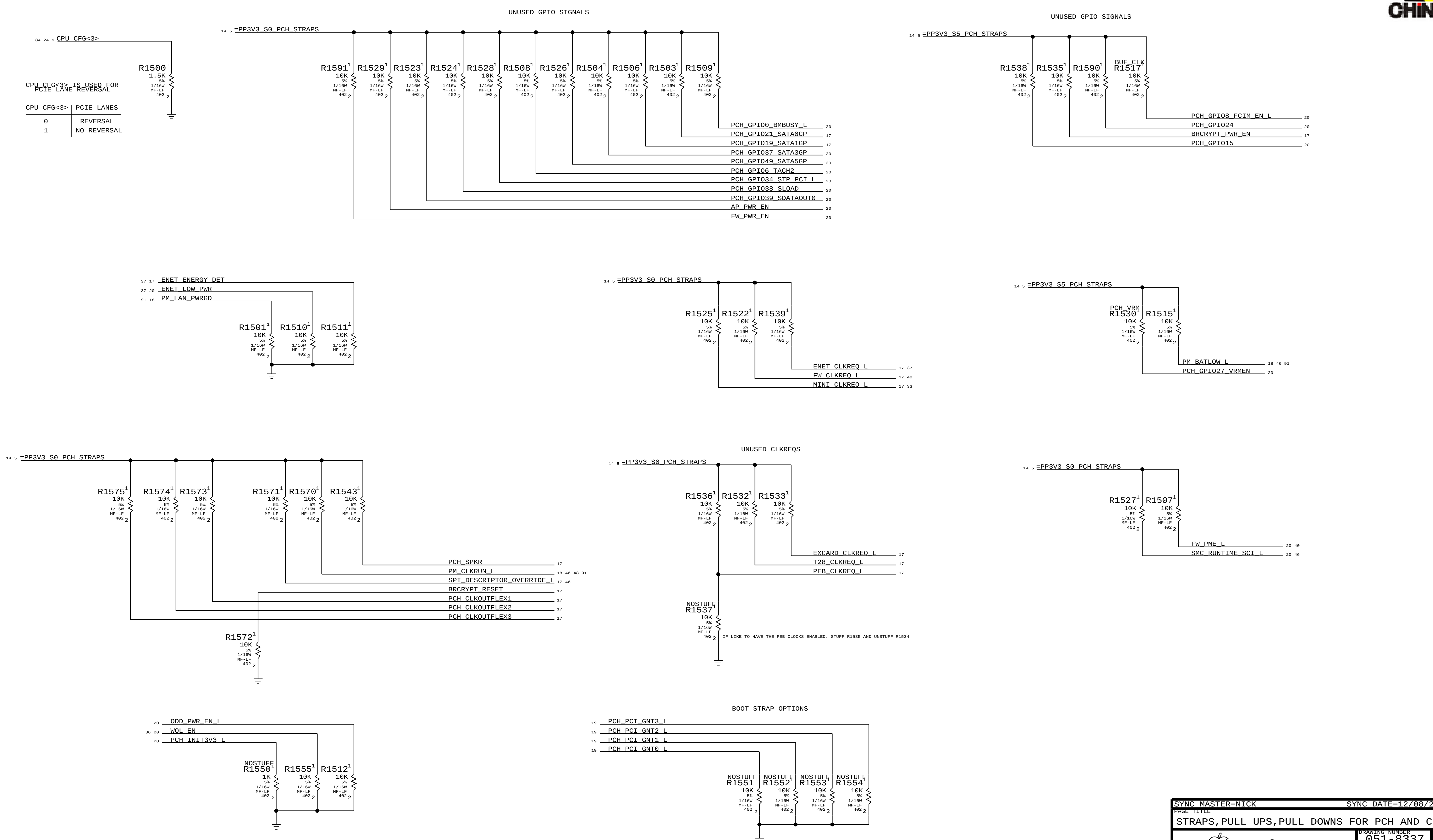
CPU GROUNDS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=NICK SYNC_DATE=12/08/2009

PAGE TITLE
STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU

Apple Inc.

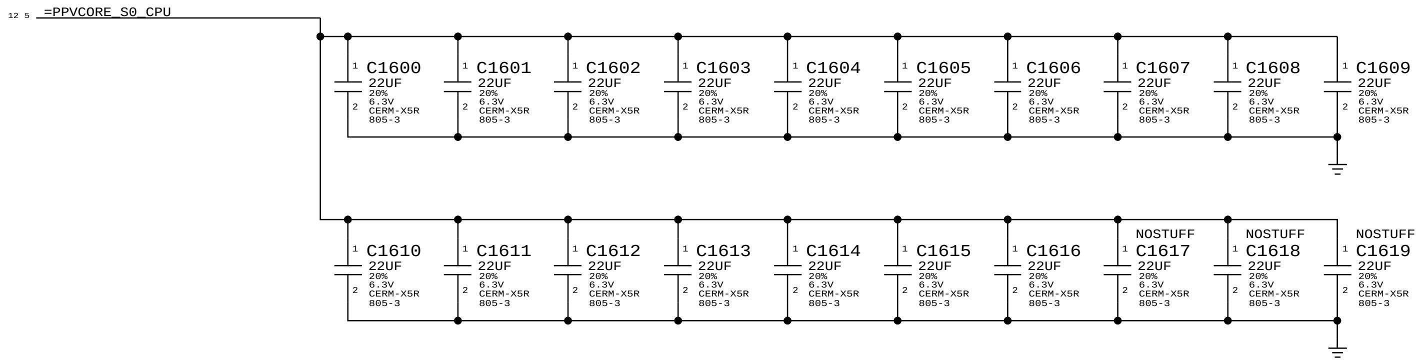
DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

CPU VCORE DECOUPLING

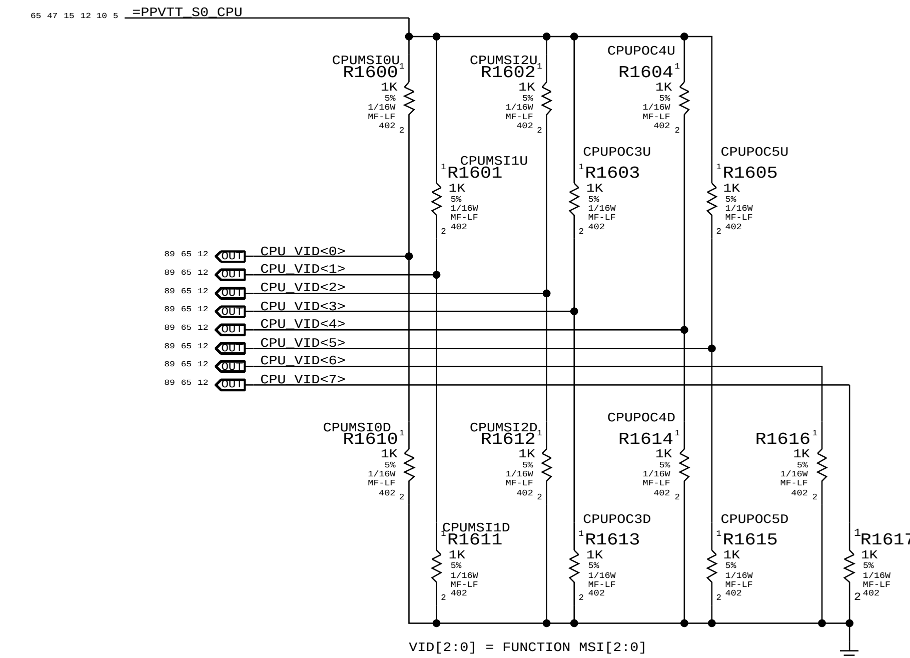
INTEL RECOMMENDATION 17X 22UF 0805



BULK CAPS ON CPU VREG PAGE 74

CPU Power On Configuration (POC) Straps

Intel recommends all option straps should be provided in layout



VID[2:0] = FUNCTION MSI[2:0]
VID[5:3] = IMON CONFIG DEFAULT
VID[6] = Reserved (0)
VID[7] = VRD SELECT (0)

VTT (CPU Uncore) DECOUPLING

8X 22UF 0805, 7X 10UF 0805 INTEL RECOMMENDATION 9X22UF 0805

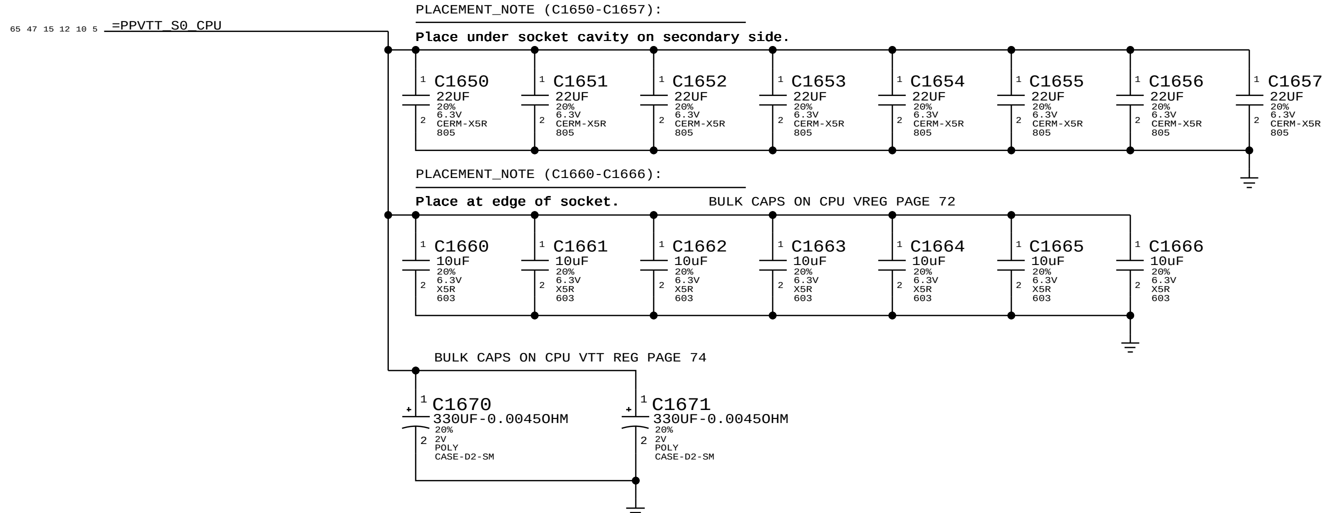
PLACEMENT_NOTE (C1650-C1657):

Place under socket cavity on secondary side.

PLACEMENT_NOTE (C1660-C1666):

Place at edge of socket. BULK CAPS ON CPU VREG PAGE 72

BULK CAPS ON CPU VTT REG PAGE 74



MSI - MARKET SEGMENT IDENTIFICATION
PREVENTS THE PLATFORM BOOTING USING A HIGHER POWERED CPU

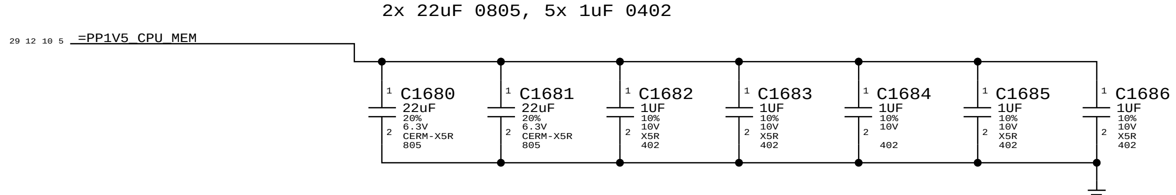
BOM GROUP	IMAX @ 900mV	CPU Gain Setting	BOM OPTIONS	Equivalent Gain
CPUPOC_IMAX_DIS		000	CPUPOC5D,CPUPOC4D,CPUPOC3D	
CPUPOC_IMAX_0_40	40A	001	CPUPOC5D,CPUPOC4D,CPUPOC3U	45
CPUPOC_IMAX_40_60	60A	010	CPUPOC5D,CPUPOC4U,CPUPOC3D	30
CPUPOC_IMAX_60_80	80A	011	CPUPOC5D,CPUPOC4U,CPUPOC3U	22.5
CPUPOC_IMAX_80_100	100A	100	CPUPOC5U,CPUPOC4D,CPUPOC3D	18
CPUPOC_IMAX_100_120	120A	101	CPUPOC5U,CPUPOC4D,CPUPOC3U	15
CPUPOC_IMAX_120_140	140A	110	CPUPOC5U,CPUPOC4U,CPUPOC3D	12.857
CPUPOC_IMAX_140_180	180A	111	CPUPOC5U,CPUPOC4U,CPUPOC3U	10

BOM GROUP	BOM OPTIONS
CLARKDALE_73W	CKD,CPUPOC_IMAX_60_80,CPUMSI2U,CPUMSI1D,CPUMSI0U
LYNNFIELD_82W	LFD,CPUPOC_IMAX_60_80,CPUMSI2U,CPUMSI1D,CPUMSI0U
LYNNFIELD_95W	LFD,CPUPOC_IMAX_100_120,CPUMSI2U,CPUMSI1U,CPUMSI0D

NOTE: BOM Configurations should not call out CPUPOCnU/D BOMOPTIONs directly.
Instead call out appropriate BOM GROUP defined in tables above.

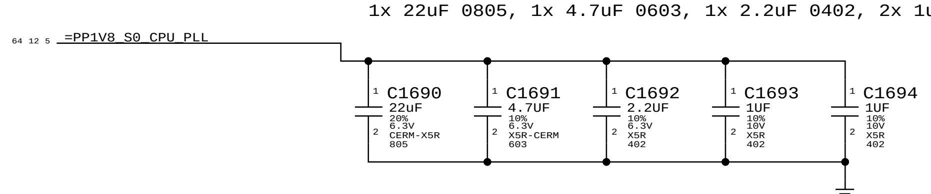
Memory (CPU VCCDDR) DECOUPLING

2x 22uF 0805, 5x 1uF 0402



PLL (CPU VCCSFR) DECOUPLING

1x 22uF 0805, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402



SYNC_MASTER=NICK SYNC_DATE=12/08/2009

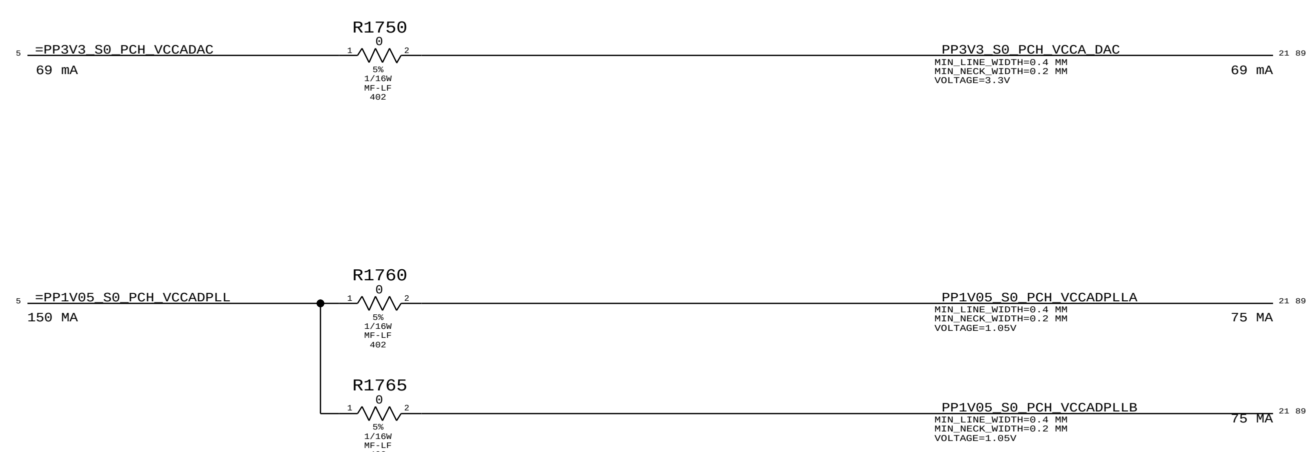
PAGE TITLE: CPU NON-GFX DECOUPLING

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

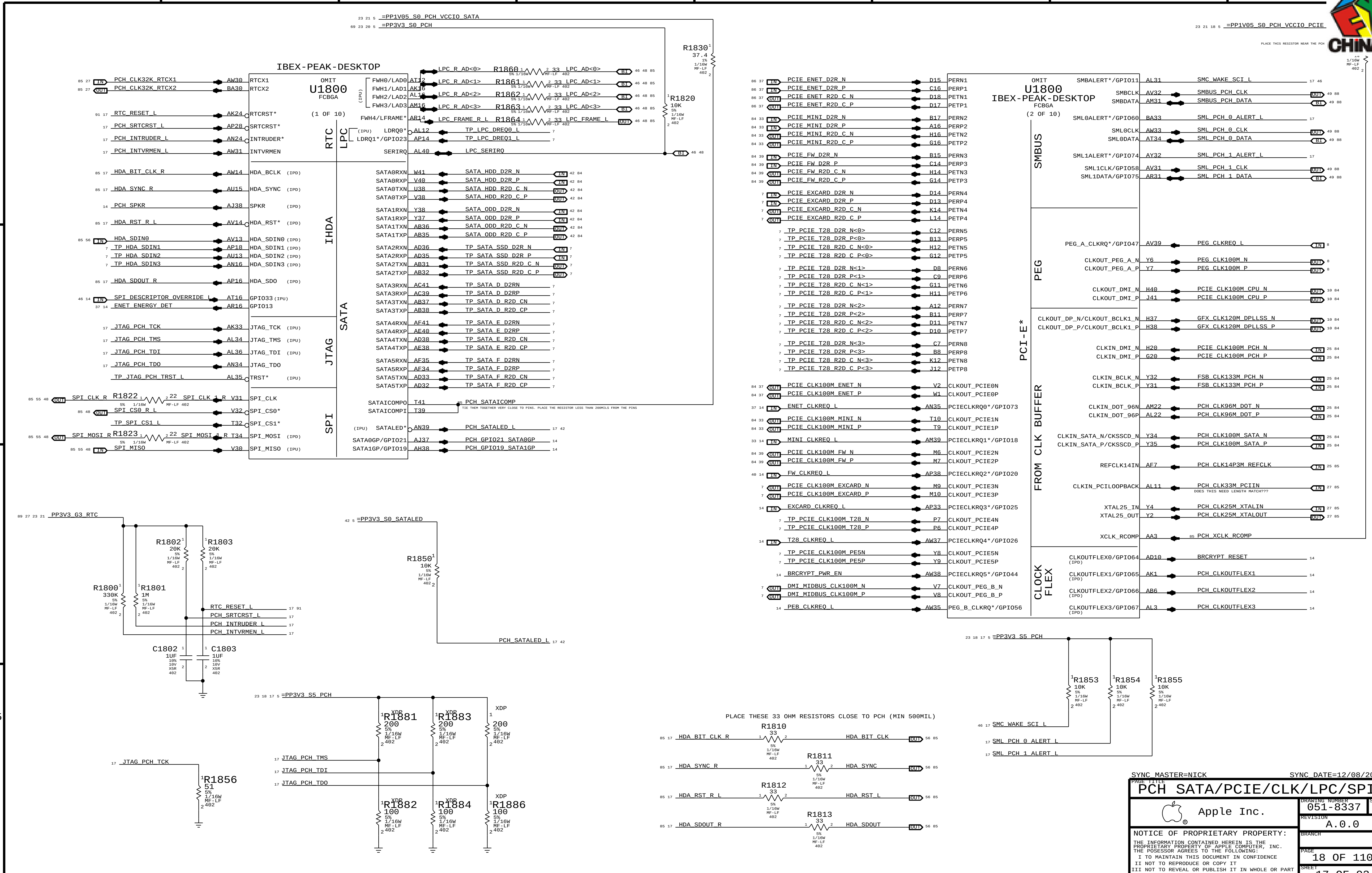
CPU/PCH GFX DECOUPLING

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=NICK

SYNC_DATE=12/08/2009

PAGE TITLE

PCH SATA/PCIE/CLK/LPC/SPI

Apple Inc.

DRAWING NUMBER 051-8337

SIZE D

REVISION A.0.0

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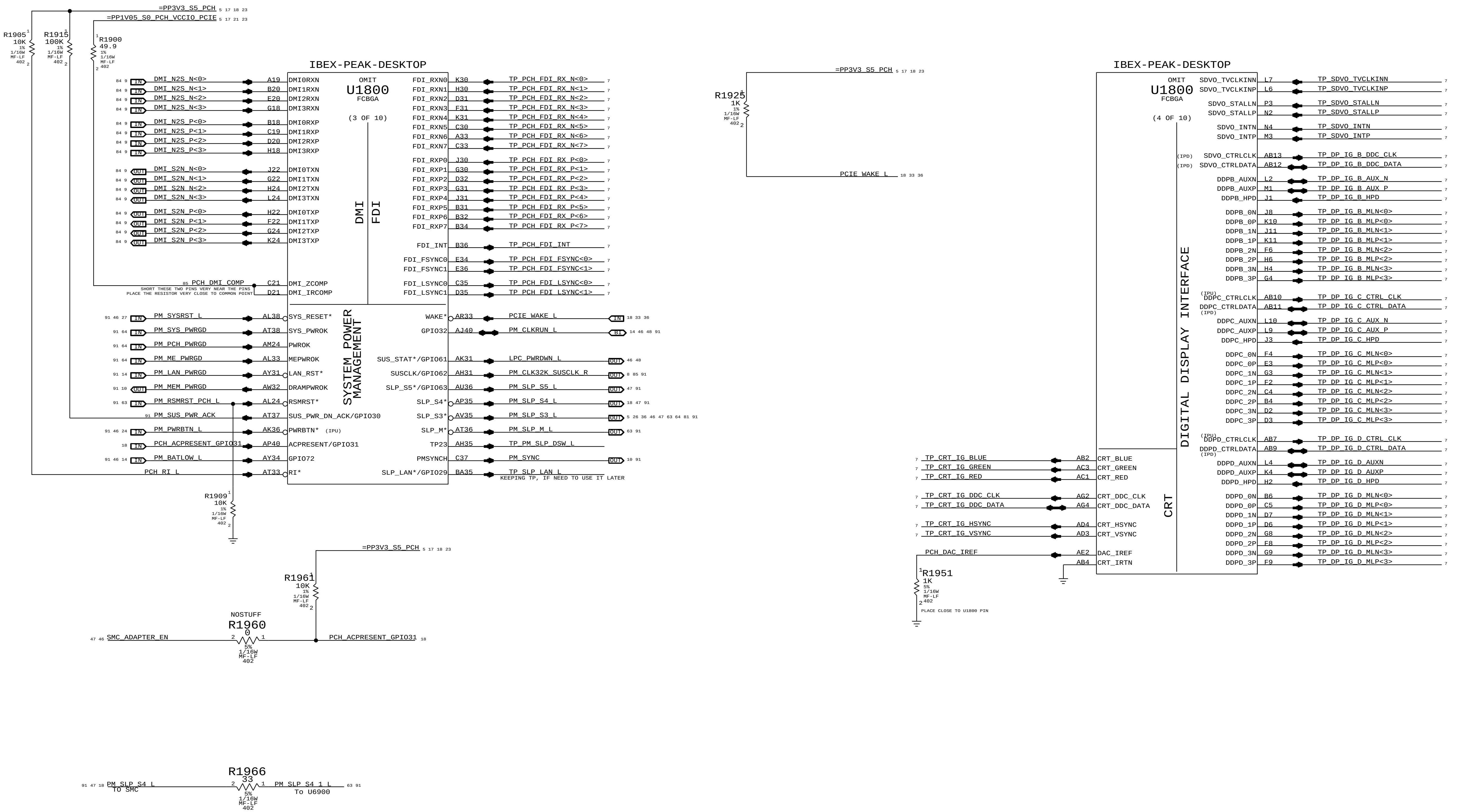


SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

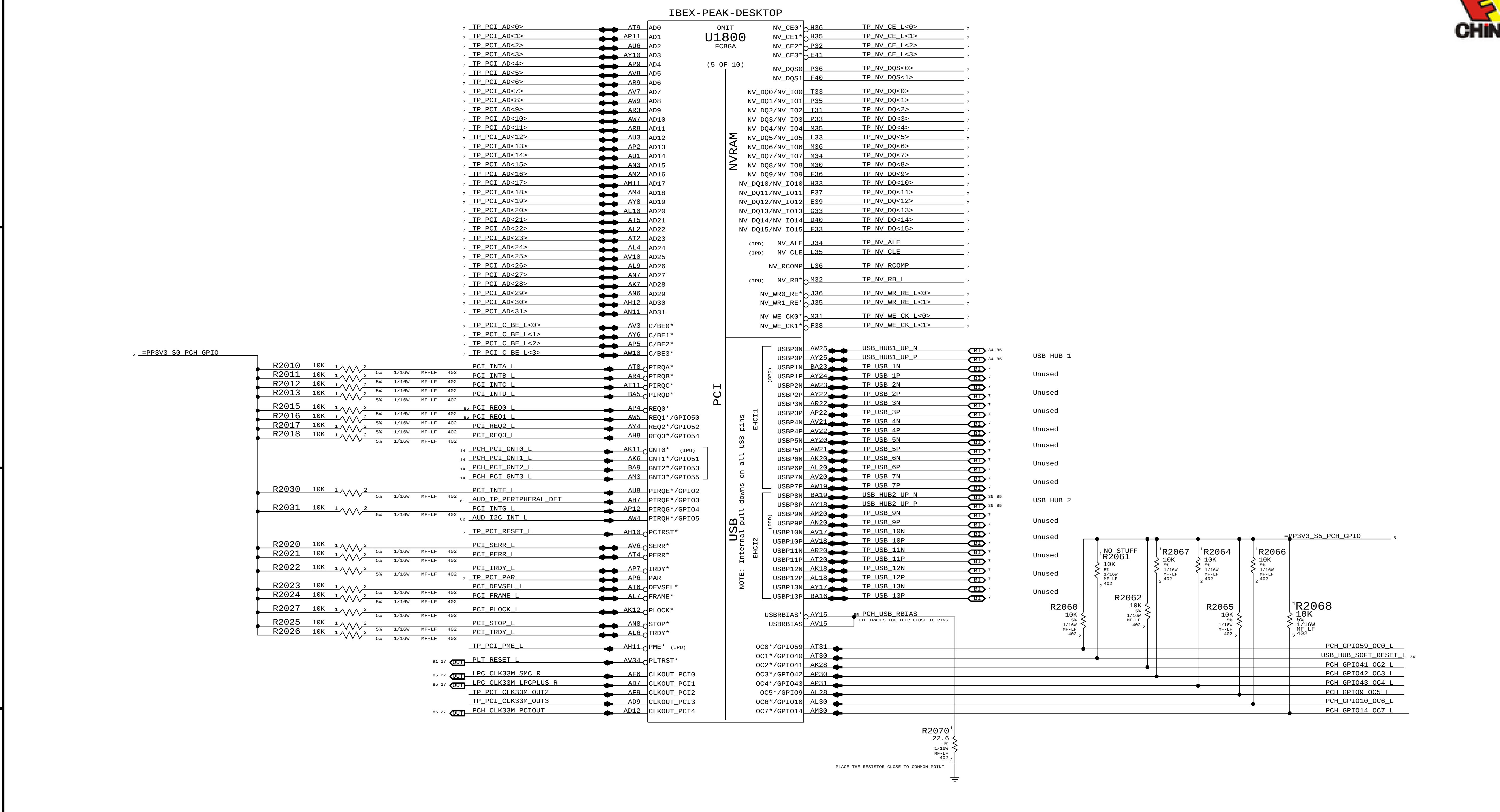
PCH DMI/FDI/GRAPHICS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

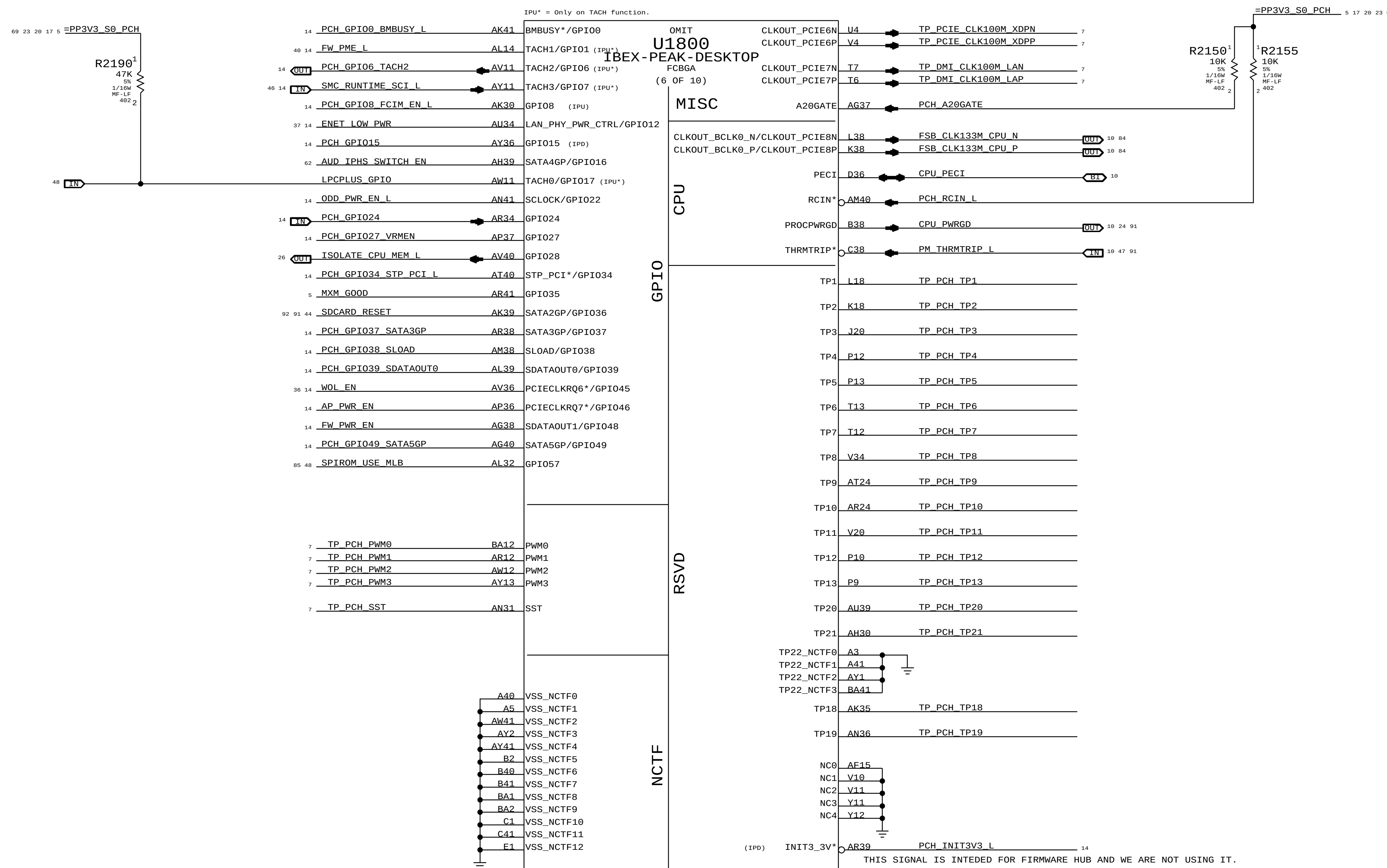


SYNC_MASTER=NICK SYNC_DATE=12/08/2009

PAGE TITLE
PCH PCI/FLASHCACHE/USB

Apple Inc.

DRAWING NUMBER	051-8337
SIZE	D
REVISION	A.0.0
BRANCH	
PAGE	20 OF 110
SHEET	19 OF 92



SYNC_MASTER=K23F SYNC_DATE=11/30/2009

PAGE TITLE

PCH MISC

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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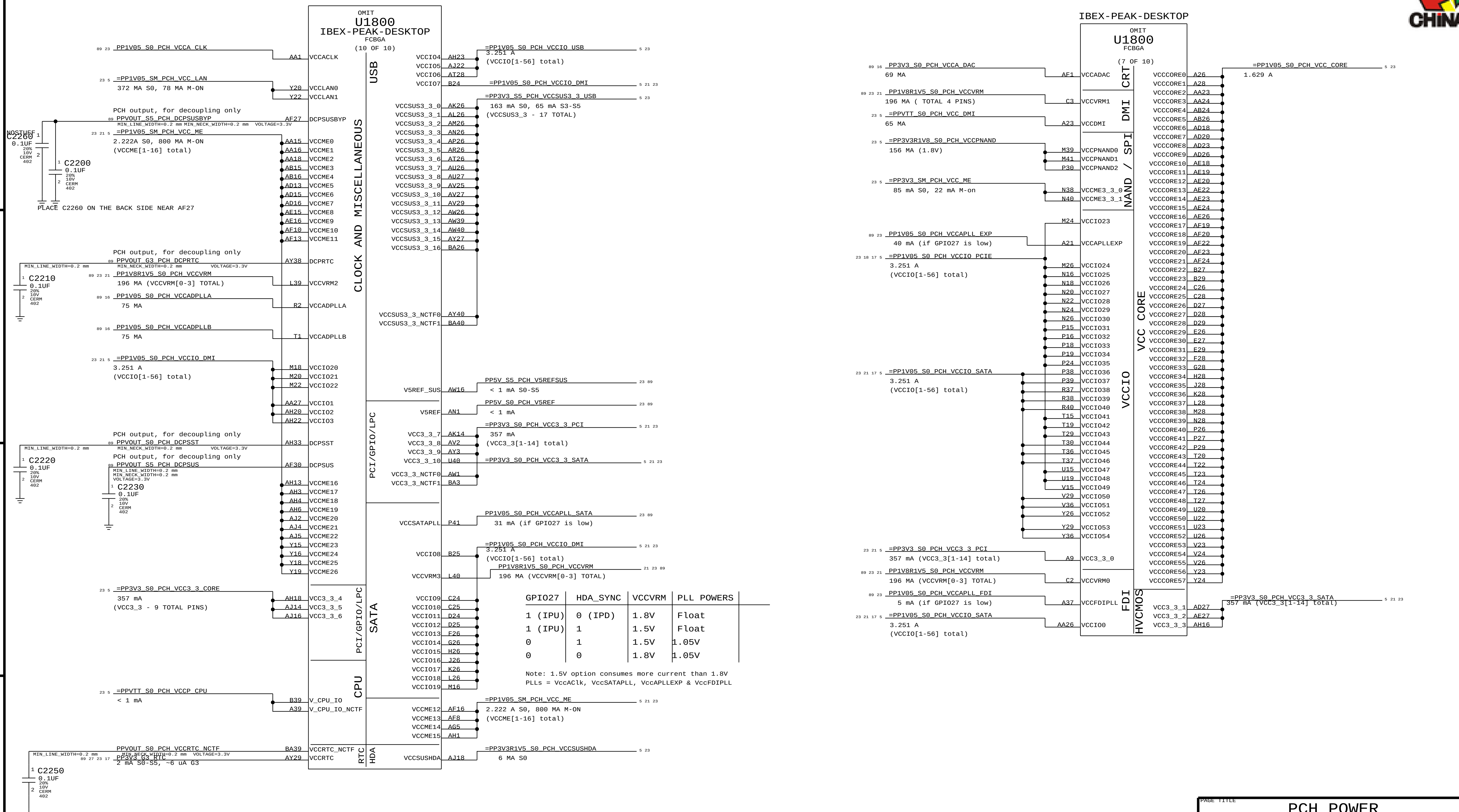


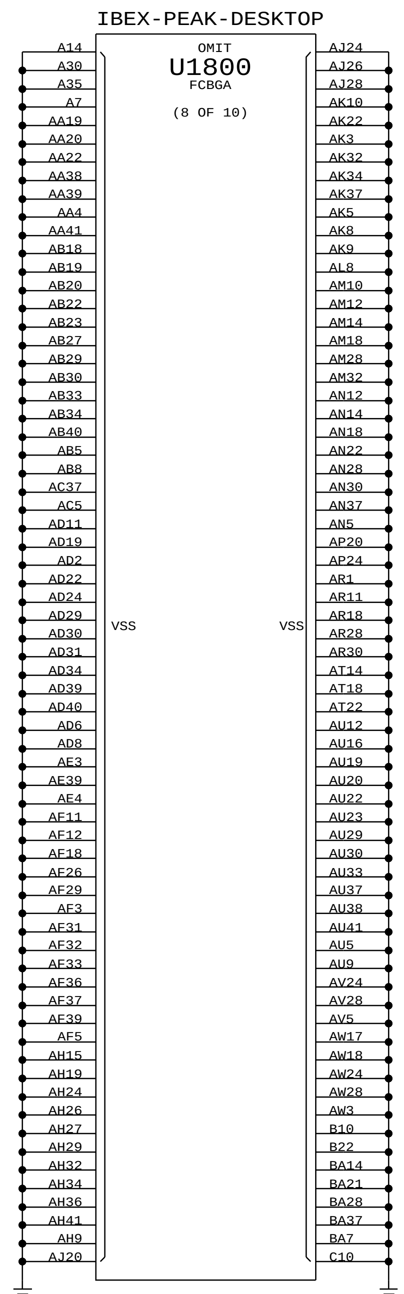
GPIO27	HDA_SYNC	VCCVRM	PLL POWERS
1 (IPU)	0 (IPD)	1.8V	Float
1 (IPU)	1	1.5V	Float
0	1	1.5V	1.05V
0	0	1.8V	1.05V

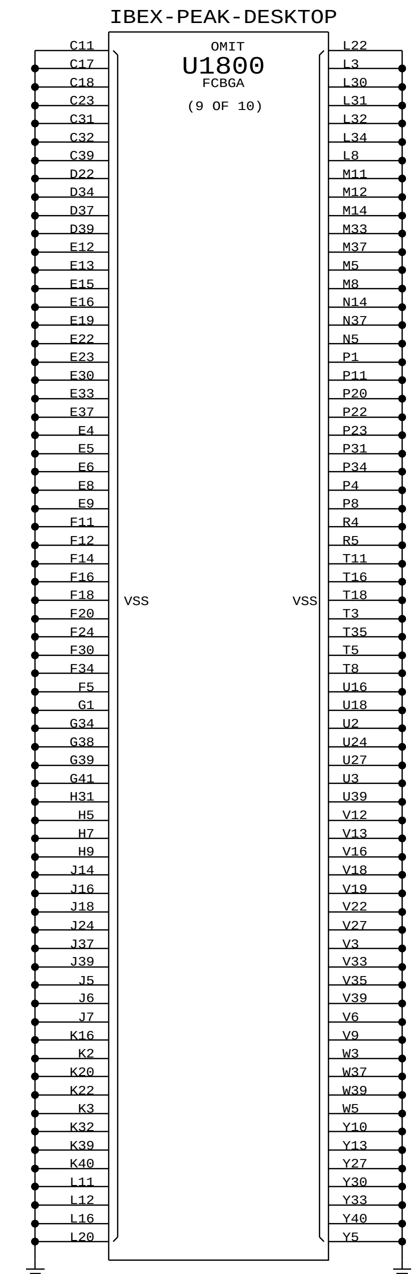
Note: 1.5V option consumes more current than 1.8V
PLLs = VccAClk, VccSATAPLL, VccAPLLEXP & VccFDIPLL

PAGE TITLE
PCH POWER

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D
REVISION A.0.0
BRANCH
PAGE 22 OF 110
SHEET 21 OF 92





SYNC_MASTER=K23F SYNC_DATE=11/30/2009

PAGE TITLE
PCH GROUNDS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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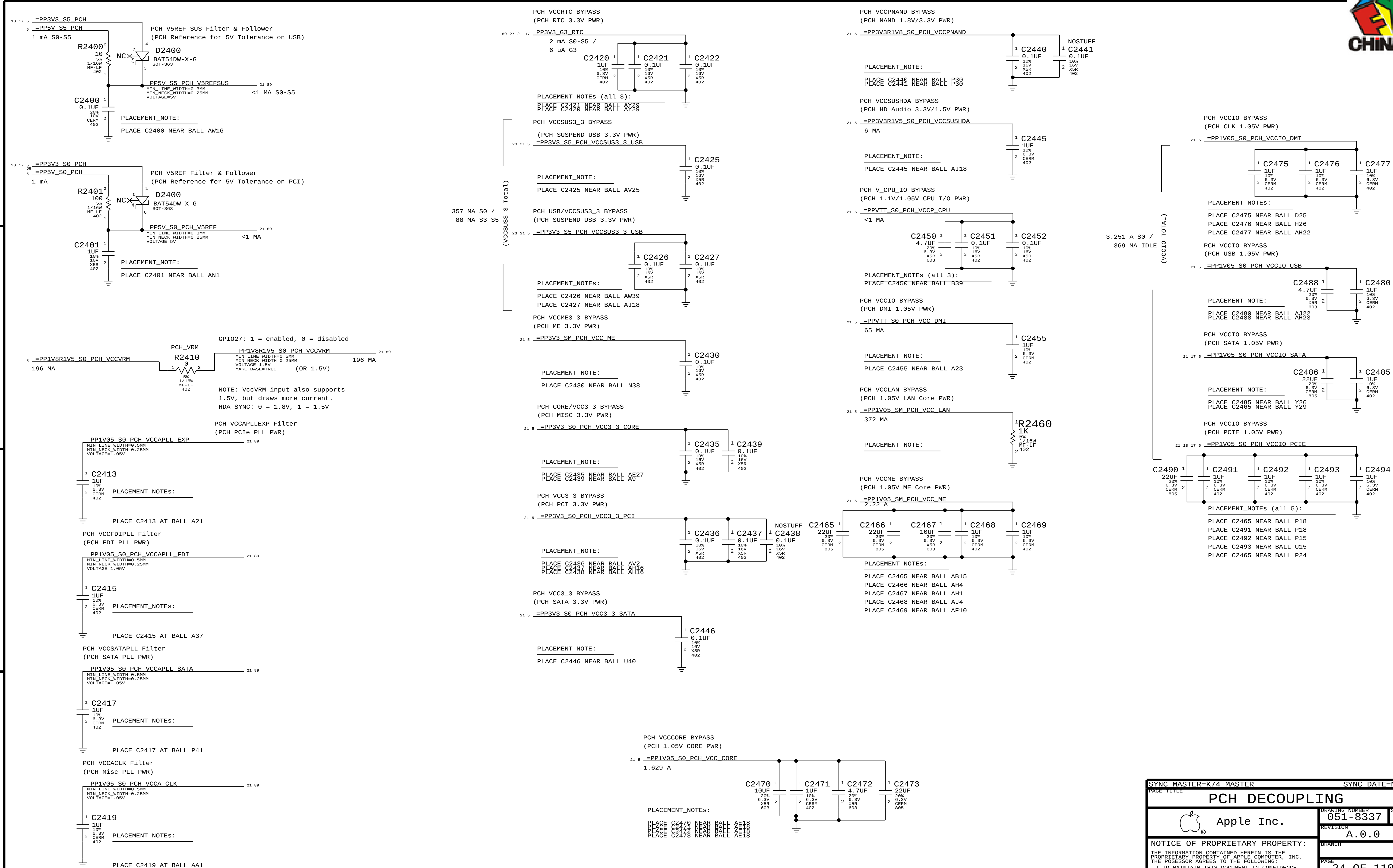


Current numbers from Ibex Peak EDS Spec Update rev 0.71, doc #386904 (Table 8-3). Pre-Silicon Mobile Estimates.

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

PCH DECOUPLING

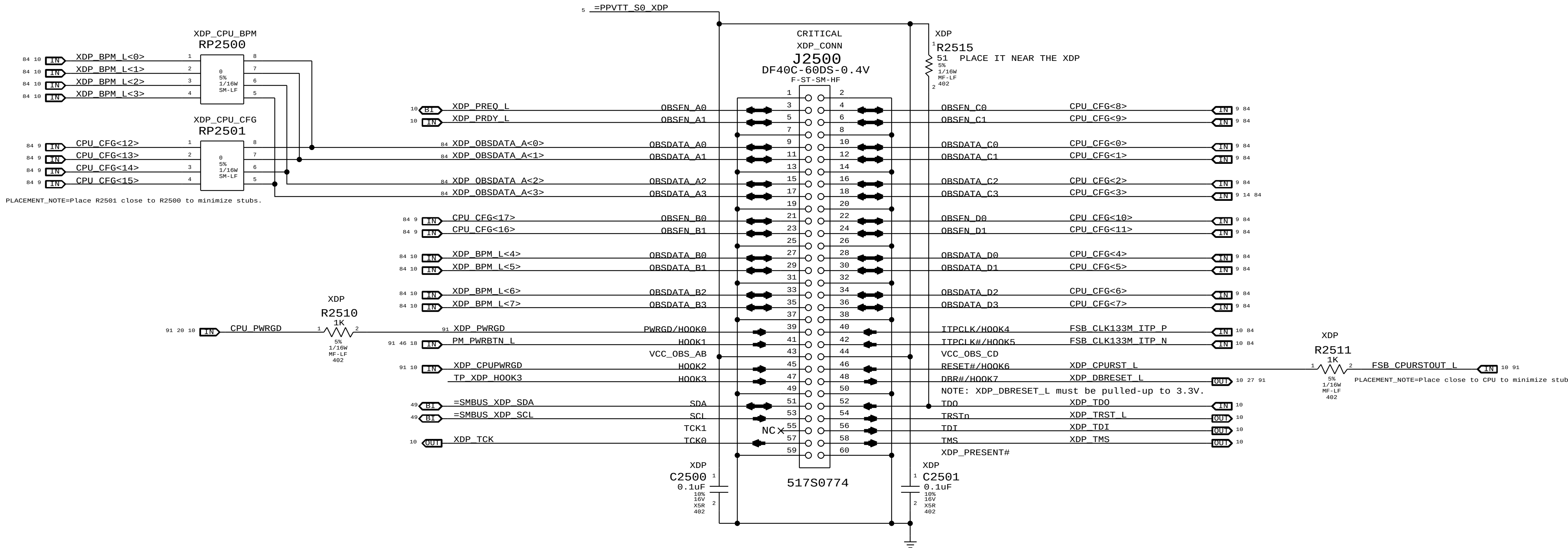
Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

PROCESSOR MINI XDP



SYNC_MASTER=NICK SYNC_DATE=12/08/2009

PAGE TITLE
EXTENDED DEBUG PORT(XDP)

Apple Inc.

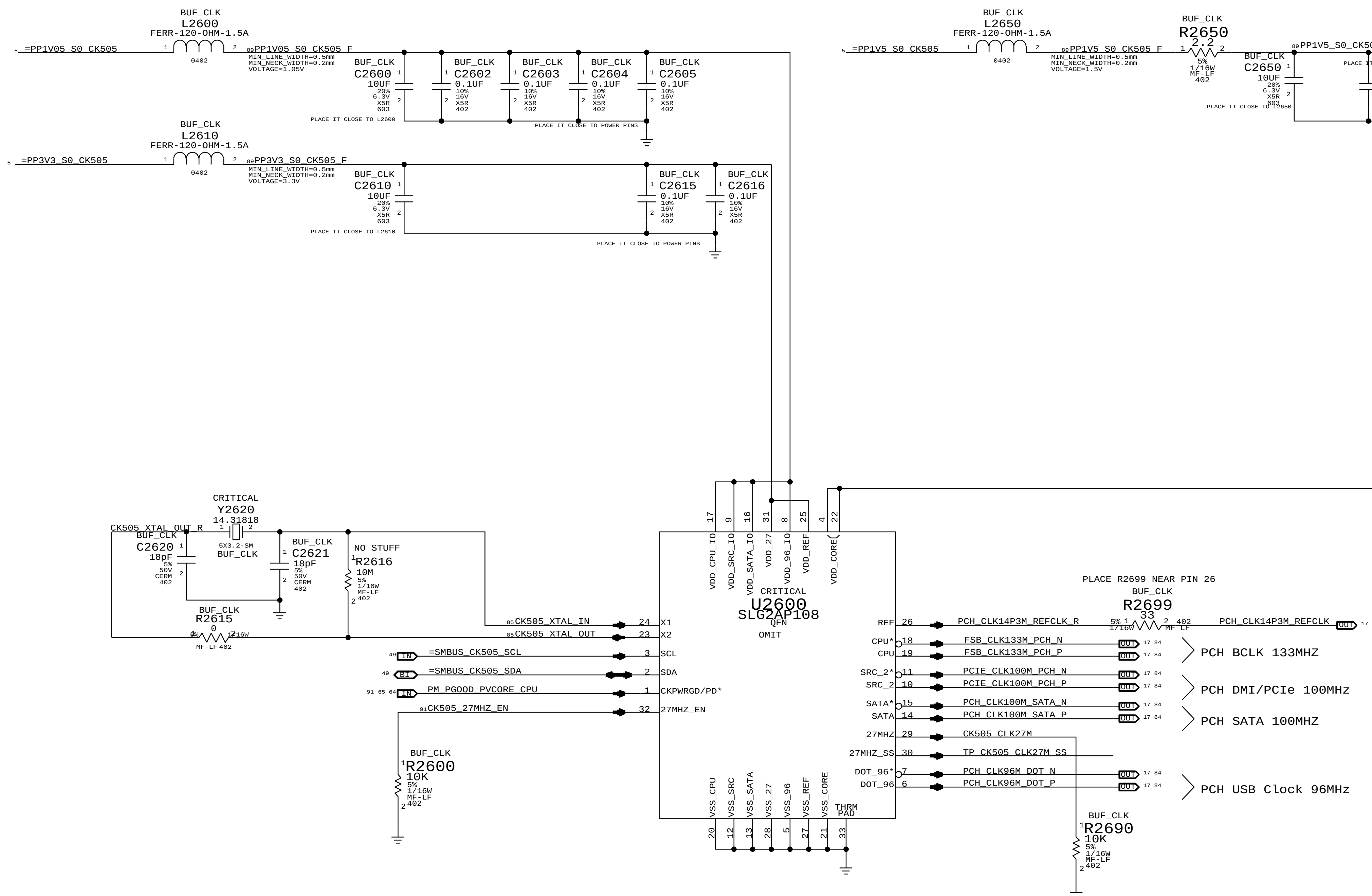
DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

PAGE 25 OF 110

SHEET 24 OF 92



SYNC_MASTER=K23F SYNC_DATE=11/30/2009

PAGE TITLE: CLOCK (CK505)

Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

BRANCH

PAGE: 26 OF 110

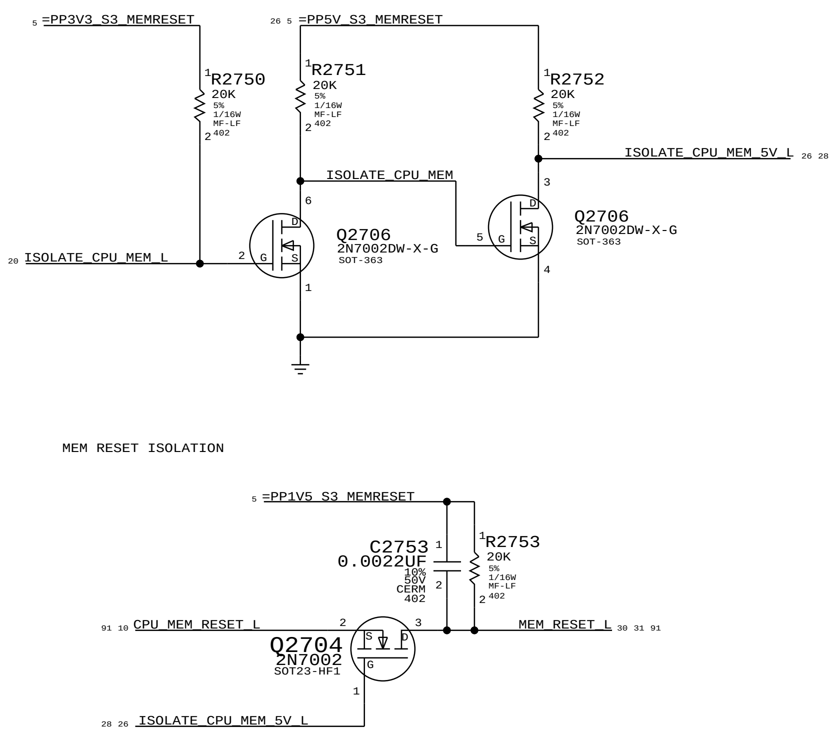
SHEET: 25 OF 92

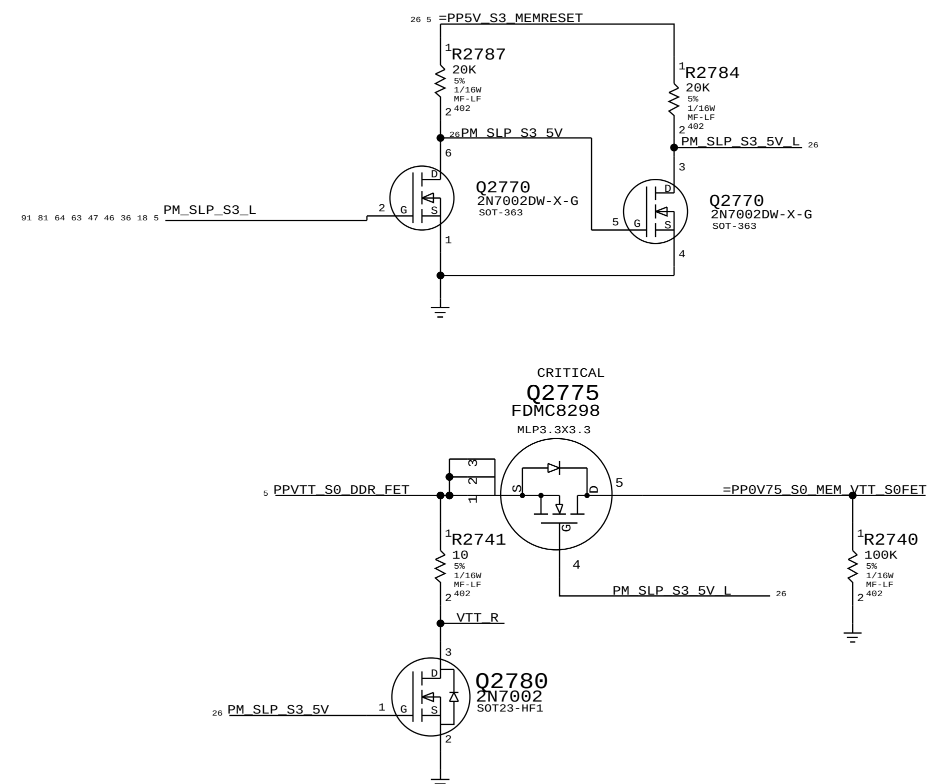
DDR3 RESET Support

LFD CANNOT CONTROL THIS SIGNAL DIRECTLY SINCE IT MUST BE HIGH IN SLEEP AND CPU MEM RAILS ARE NOT POWERED IN SLEEP.

	CPU_RESET_L	ISOLATE_L	ME
S5	0	3.3V	[illegible]
S0	0	3.3V	[illegible]
S0	1.5V	3.3V	1.5V
S3	0	0	1.5V
S0	1.5V	3.3V	1.5V
S5	0	3.3V	0

BUFFER ISOLATE_CPU_MEM_L TO 5V





SYNC_MASTER=MATT SYNC_DATE=01/06/2010

PAGE TITLE

DDR3 RESET

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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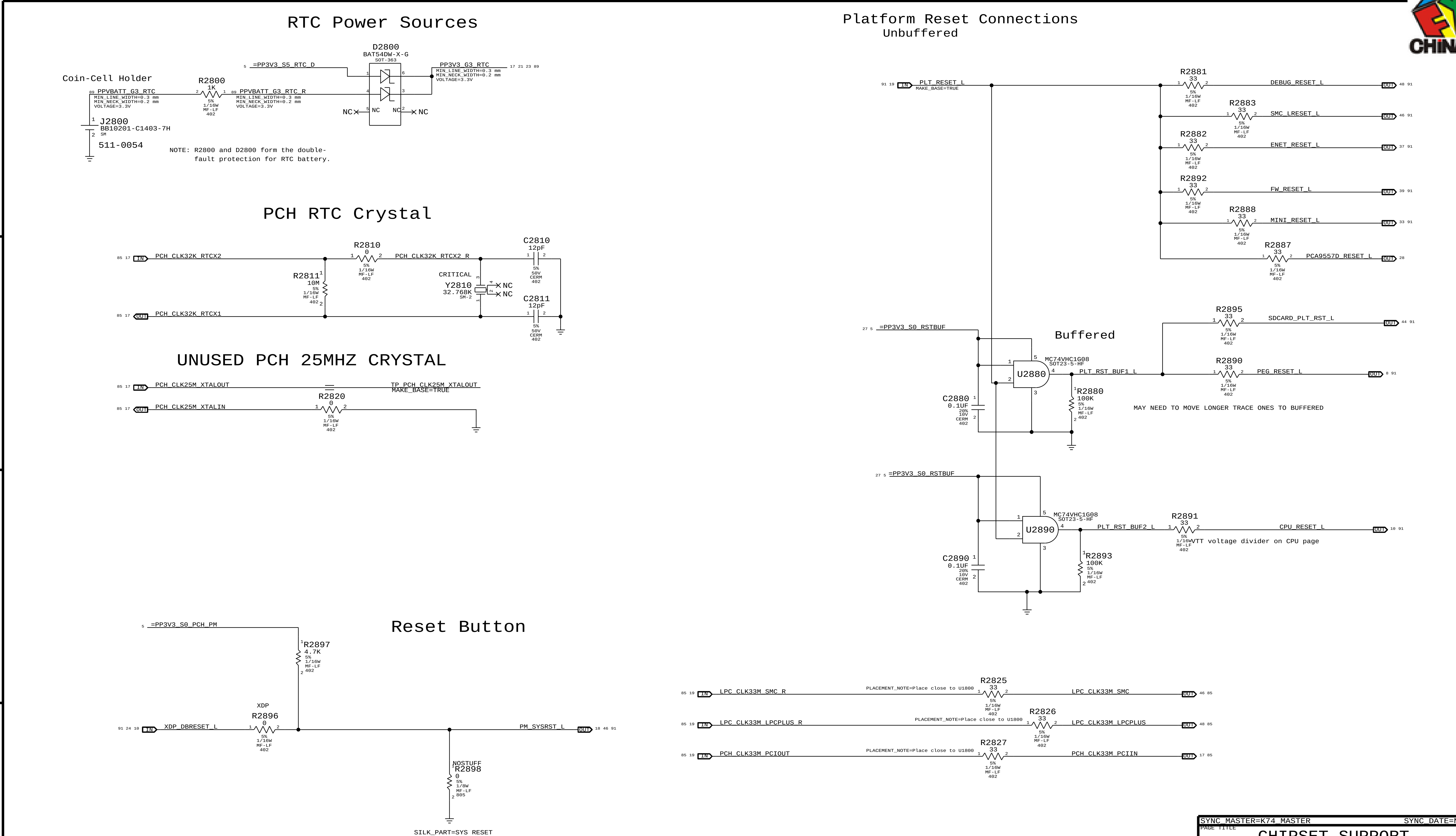
NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

CHIPSET SUPPORT

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

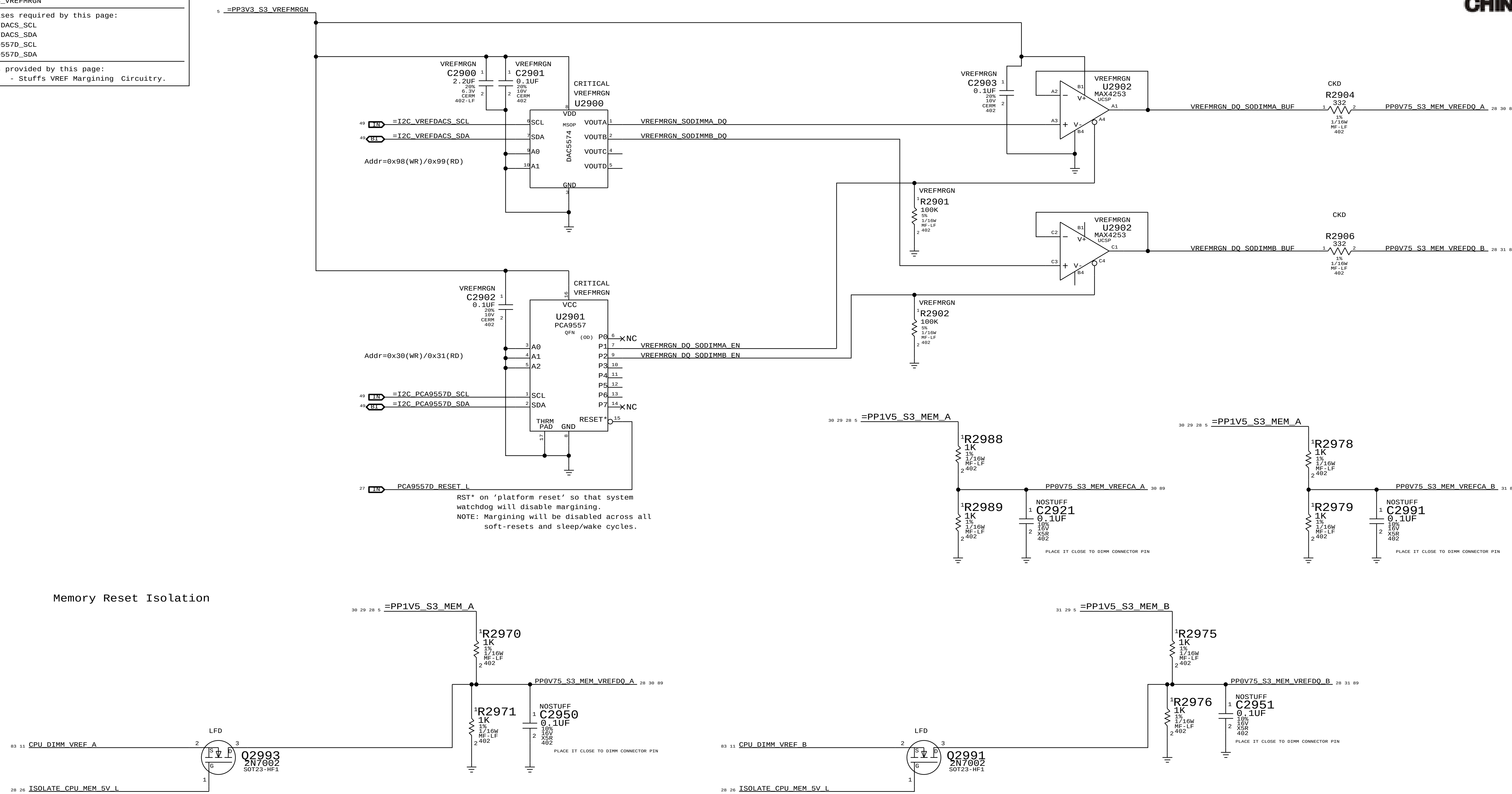
BRANCH

Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN - Stuffs VREF Margining Circuitry.



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)				1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)				1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)				0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)				+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output				8.59mV / step @ output	1.51mV / step @ output

SYNC_MASTER=MATT SYNC_DATE=01/06/2010

PAGE TITLE
DDR3 Vref Margining

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

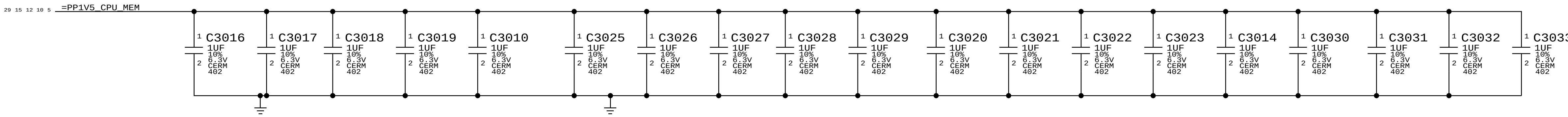
REVISION A.0.0

BRANCH

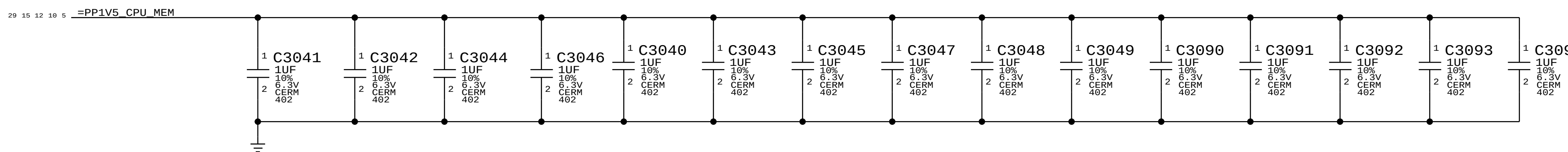
DIMM A (FURTHER FROM CPU)

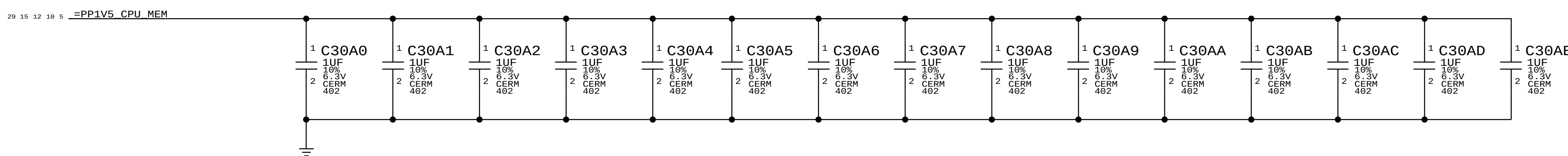
CAPS TO COUPLE CPU 1V5_MEM

DIMM B (CLOSER TO CPU)

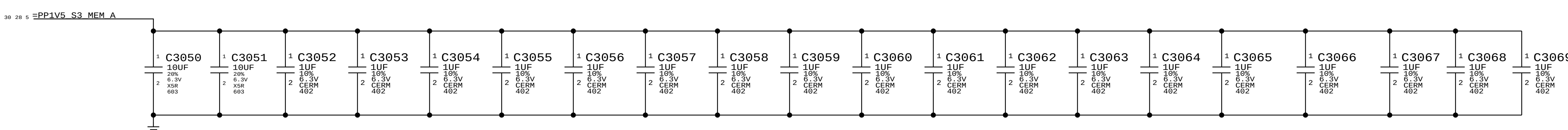


EXTRA DECOUPLING CAPS FOR CPU MEM RAIL



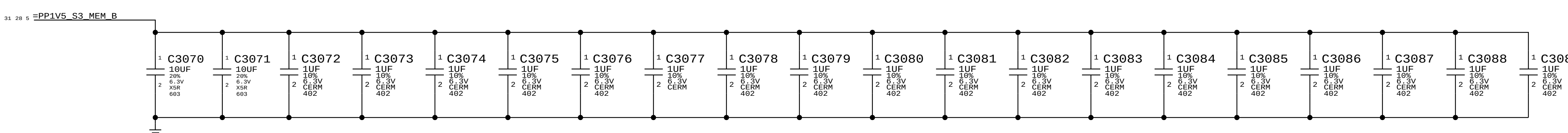


DECOUPLING CAPS FOR DIMM ON CHANNEL A - AT CONNECTOR



DECOUPLING CAPS FOR DIMM ON CHANNEL B - AT CONNECTOR

402



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

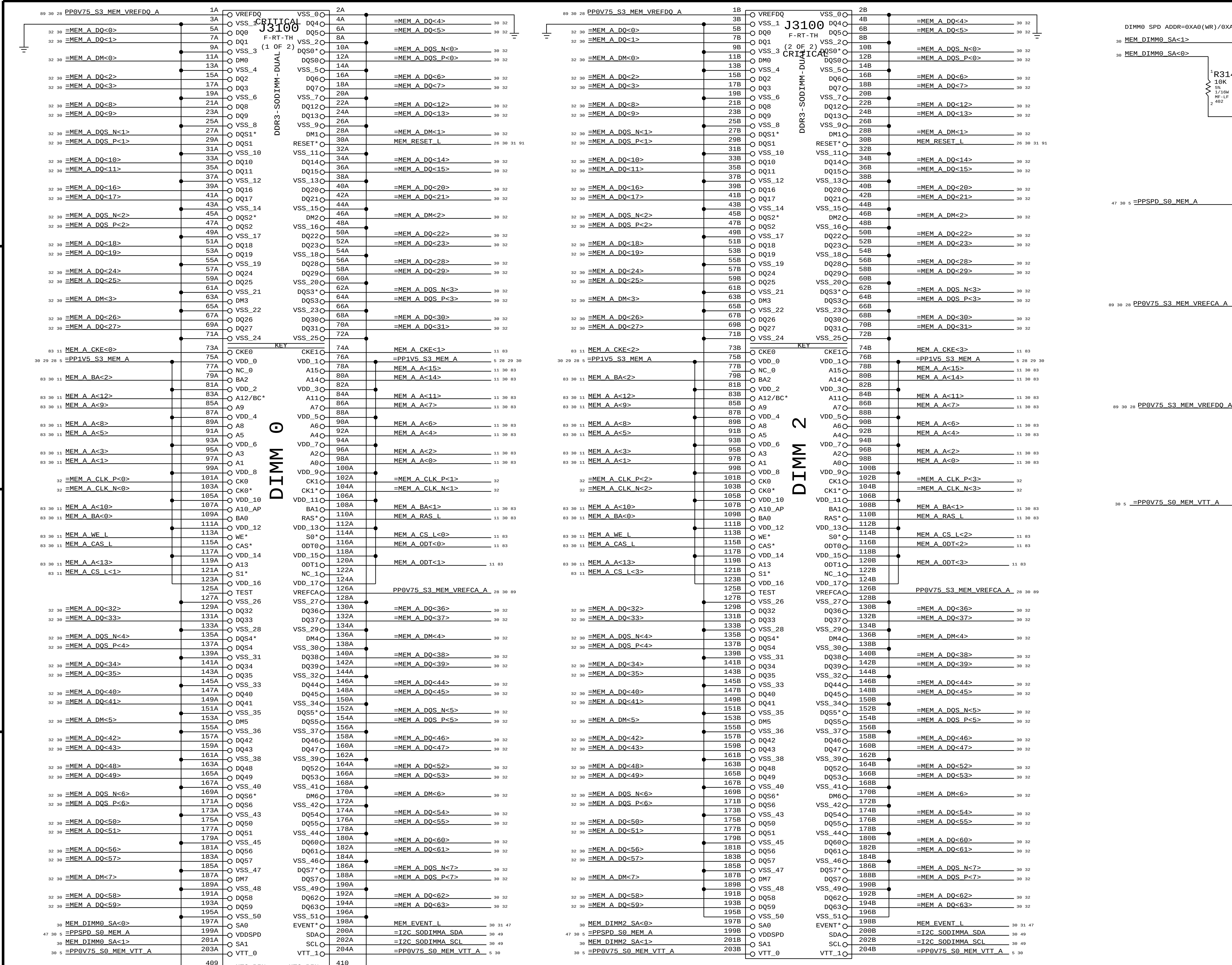
PAGE TITLE: MEMORY CAPS

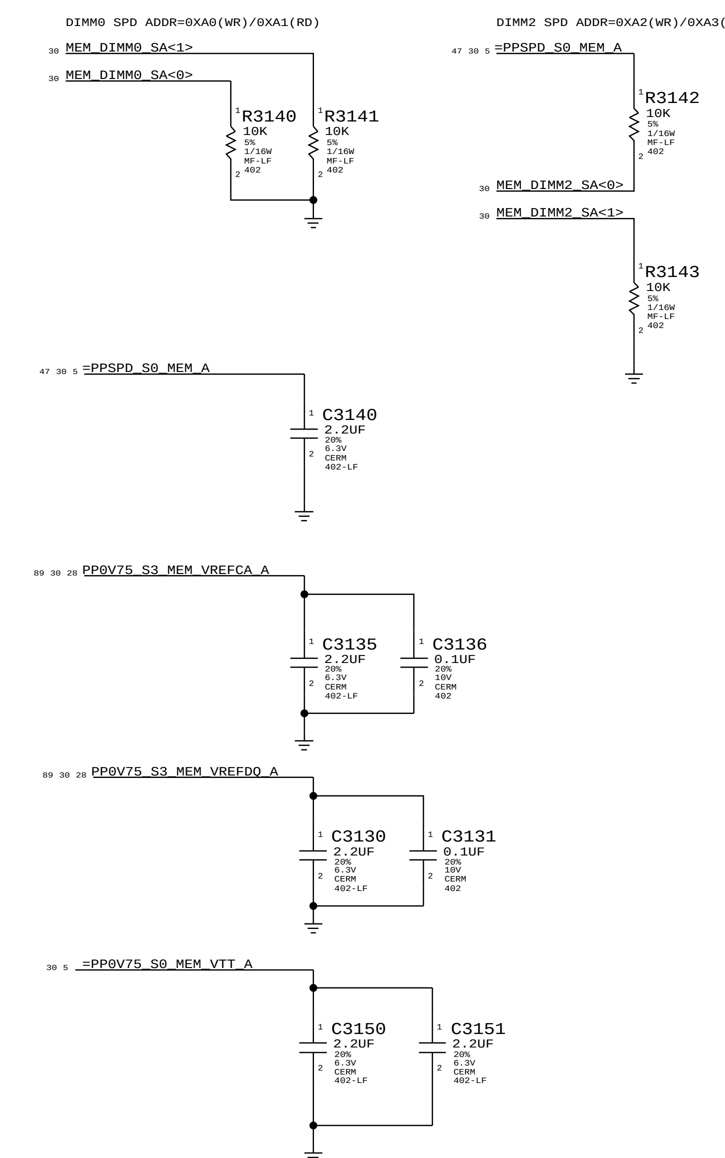
Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

BRANCH





Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL — - ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

DDR3 SO-DIMMs 0 & 2

Apple Inc.

DRAWING NUMBER	051-8337
SIZE	D
REVISION	A.0.0
BRANCH	
PAGE	31 OF 110
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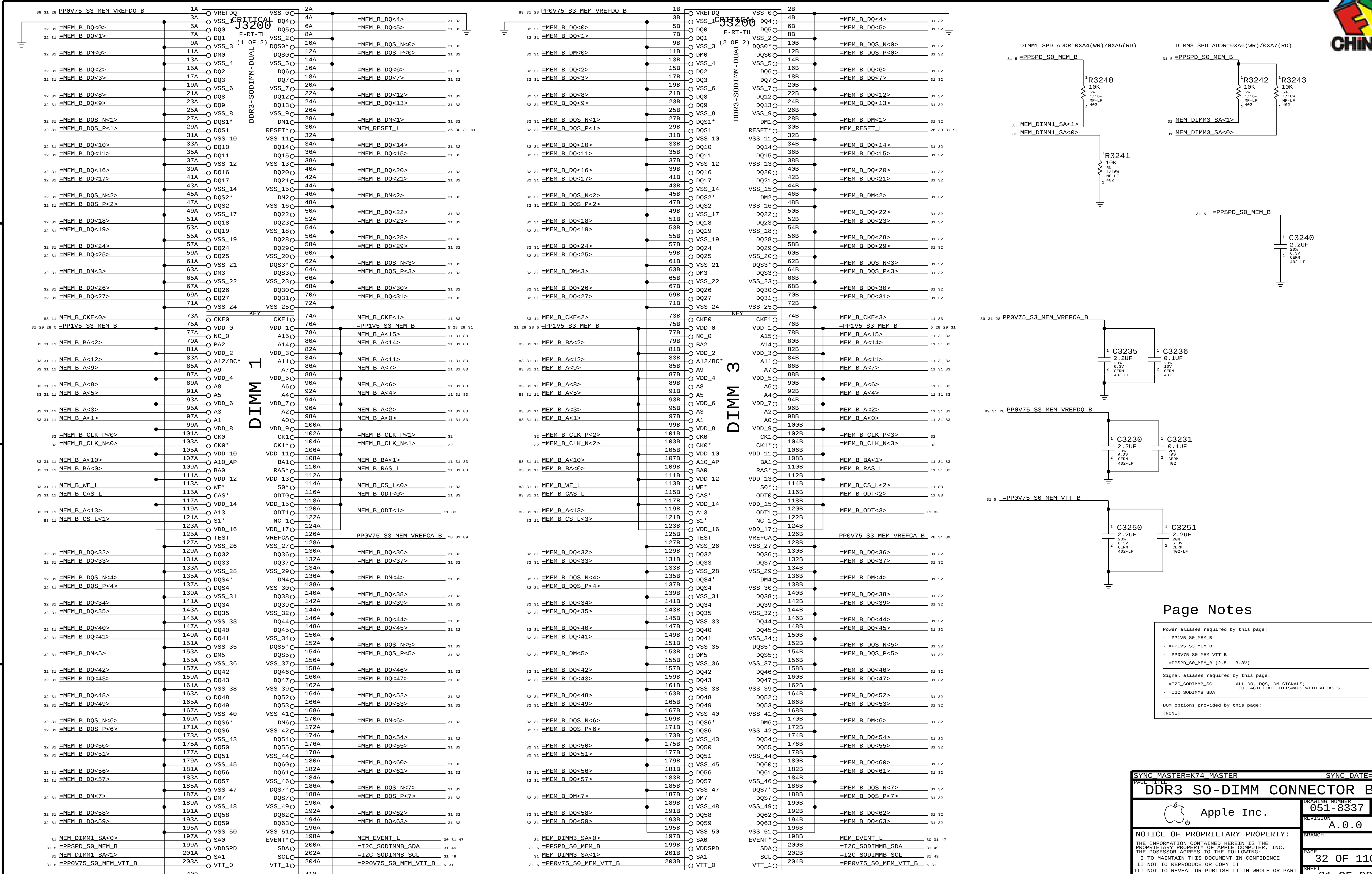


Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMB_SCL
- =I2C_SODIMMB_SDA

- ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES

BOM options provided by this page:

(NONE)

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

DDR3 SO-DIMM CONNECTOR B

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

CPU CHANNEL A DQS 0 -> DIMM A DQS 7

83 11	MEM_A_DQS_N<0>	MAKE_BASE=TRUE	=MEM_A_DQS_N<7>	30
83 11	MEM_A_DQS_P<0>	MAKE_BASE=TRUE	=MEM_A_DQS_P<7>	30
83 11	MEM_A_DM<0>	MAKE_BASE=TRUE	=MEM_A_DM<7>	30
83 11	MEM_A_DQ<7>	MAKE_BASE=TRUE	=MEM_A_DQ<60>	30
83 11	MEM_A_DQ<6>	MAKE_BASE=TRUE	=MEM_A_DQ<56>	30
83 11	MEM_A_DQ<5>	MAKE_BASE=TRUE	=MEM_A_DQ<58>	30
83 11	MEM_A_DQ<4>	MAKE_BASE=TRUE	=MEM_A_DQ<59>	30
83 11	MEM_A_DQ<3>	MAKE_BASE=TRUE	=MEM_A_DQ<57>	30
83 11	MEM_A_DQ<2>	MAKE_BASE=TRUE	=MEM_A_DQ<61>	30
83 11	MEM_A_DQ<1>	MAKE_BASE=TRUE	=MEM_A_DQ<62>	30
83 11	MEM_A_DQ<0>	MAKE_BASE=TRUE	=MEM_A_DQ<63>	30

CPU CHANNEL A DQS 1 -> DIMM A DQS 6

83 11	MEM_A_DQS_N<1>	MAKE_BASE=TRUE	=MEM_A_DQS_N<6>	30
83 11	MEM_A_DQS_P<1>	MAKE_BASE=TRUE	=MEM_A_DQS_P<6>	30
83 11	MEM_A_DM<1>	MAKE_BASE=TRUE	=MEM_A_DM<6>	30
83 11	MEM_A_DQ<15>	MAKE_BASE=TRUE	=MEM_A_DQ<49>	30
83 11	MEM_A_DQ<14>	MAKE_BASE=TRUE	=MEM_A_DQ<52>	30
83 11	MEM_A_DQ<13>	MAKE_BASE=TRUE	=MEM_A_DQ<51>	30
83 11	MEM_A_DQ<12>	MAKE_BASE=TRUE	=MEM_A_DQ<50>	30
83 11	MEM_A_DQ<11>	MAKE_BASE=TRUE	=MEM_A_DQ<48>	30
83 11	MEM_A_DQ<10>	MAKE_BASE=TRUE	=MEM_A_DQ<53>	30
83 11	MEM_A_DQ<9>	MAKE_BASE=TRUE	=MEM_A_DQ<55>	30
83 11	MEM_A_DQ<8>	MAKE_BASE=TRUE	=MEM_A_DQ<54>	30

CPU CHANNEL A DQS 2 -> DIMM A DQS 5

83 11	MEM_A_DQS_N<2>	MAKE_BASE=TRUE	=MEM_A_DQS_N<5>	30
83 11	MEM_A_DQS_P<2>	MAKE_BASE=TRUE	=MEM_A_DQS_P<5>	30
83 11	MEM_A_DM<2>	MAKE_BASE=TRUE	=MEM_A_DM<5>	30
83 11	MEM_A_DQ<23>	MAKE_BASE=TRUE	=MEM_A_DQ<40>	30
83 11	MEM_A_DQ<22>	MAKE_BASE=TRUE	=MEM_A_DQ<45>	30
83 11	MEM_A_DQ<21>	MAKE_BASE=TRUE	=MEM_A_DQ<43>	30
83 11	MEM_A_DQ<20>	MAKE_BASE=TRUE	=MEM_A_DQ<42>	30
83 11	MEM_A_DQ<19>	MAKE_BASE=TRUE	=MEM_A_DQ<41>	30
83 11	MEM_A_DQ<18>	MAKE_BASE=TRUE	=MEM_A_DQ<44>	30
83 11	MEM_A_DQ<17>	MAKE_BASE=TRUE	=MEM_A_DQ<46>	30
83 11	MEM_A_DQ<16>	MAKE_BASE=TRUE	=MEM_A_DQ<47>	30

CPU CHANNEL A DQS 3 -> DIMM A DQS 4

83 11	MEM_A_DQS_N<3>	MAKE_BASE=TRUE	=MEM_A_DQS_N<4>	30
83 11	MEM_A_DQS_P<3>	MAKE_BASE=TRUE	=MEM_A_DQS_P<4>	30
83 11	MEM_A_DM<3>	MAKE_BASE=TRUE	=MEM_A_DM<4>	30
83 11	MEM_A_DQ<31>	MAKE_BASE=TRUE	=MEM_A_DQ<37>	30
83 11	MEM_A_DQ<30>	MAKE_BASE=TRUE	=MEM_A_DQ<33>	30
83 11	MEM_A_DQ<29>	MAKE_BASE=TRUE	=MEM_A_DQ<35>	30
83 11	MEM_A_DQ<28>	MAKE_BASE=TRUE	=MEM_A_DQ<34>	30
83 11	MEM_A_DQ<27>	MAKE_BASE=TRUE	=MEM_A_DQ<36>	30
83 11	MEM_A_DQ<26>	MAKE_BASE=TRUE	=MEM_A_DQ<32>	30
83 11	MEM_A_DQ<25>	MAKE_BASE=TRUE	=MEM_A_DQ<38>	30
83 11	MEM_A_DQ<24>	MAKE_BASE=TRUE	=MEM_A_DQ<39>	30

CPU CHANNEL A DQS 4 -> DIMM A DQS 3

83 11	MEM_A_DQS_N<4>	MAKE_BASE=TRUE	=MEM_A_DQS_N<3>	30
83 11	MEM_A_DQS_P<4>	MAKE_BASE=TRUE	=MEM_A_DQS_P<3>	30
83 11	MEM_A_DM<4>	MAKE_BASE=TRUE	=MEM_A_DM<3>	30
83 11	MEM_A_DQ<39>	MAKE_BASE=TRUE	=MEM_A_DQ<25>	30
83 11	MEM_A_DQ<38>	MAKE_BASE=TRUE	=MEM_A_DQ<29>	30
83 11	MEM_A_DQ<37>	MAKE_BASE=TRUE	=MEM_A_DQ<27>	30
83 11	MEM_A_DQ<36>	MAKE_BASE=TRUE	=MEM_A_DQ<31>	30
83 11	MEM_A_DQ<35>	MAKE_BASE=TRUE	=MEM_A_DQ<28>	30
83 11	MEM_A_DQ<34>	MAKE_BASE=TRUE	=MEM_A_DQ<24>	30
83 11	MEM_A_DQ<33>	MAKE_BASE=TRUE	=MEM_A_DQ<26>	30
83 11	MEM_A_DQ<32>	MAKE_BASE=TRUE	=MEM_A_DQ<30>	30

CPU CHANNEL A DQS 5 -> DIMM A DQS 2

83 11	MEM_A_DQS_N<5>	MAKE_BASE=TRUE	=MEM_A_DQS_N<2>	30
83 11	MEM_A_DQS_P<5>	MAKE_BASE=TRUE	=MEM_A_DQS_P<2>	30
83 11	MEM_A_DM<5>	MAKE_BASE=TRUE	=MEM_A_DM<2>	30
83 11	MEM_A_DQ<47>	MAKE_BASE=TRUE	=MEM_A_DQ<17>	30
83 11	MEM_A_DQ<46>	MAKE_BASE=TRUE	=MEM_A_DQ<16>	30
83 11	MEM_A_DQ<45>	MAKE_BASE=TRUE	=MEM_A_DQ<22>	30
83 11	MEM_A_DQ<44>	MAKE_BASE=TRUE	=MEM_A_DQ<23>	30
83 11	MEM_A_DQ<43>	MAKE_BASE=TRUE	=MEM_A_DQ<21>	30
83 11	MEM_A_DQ<42>	MAKE_BASE=TRUE	=MEM_A_DQ<20>	30
83 11	MEM_A_DQ<41>	MAKE_BASE=TRUE	=MEM_A_DQ<18>	30
83 11	MEM_A_DQ<40>	MAKE_BASE=TRUE	=MEM_A_DQ<19>	30

CPU CHANNEL A DQS 6 -> DIMM A DQS 1

83 11	MEM_A_DQS_N<6>	MAKE_BASE=TRUE	=MEM_A_DQS_N<1>	30
83 11	MEM_A_DQS_P<6>	MAKE_BASE=TRUE	=MEM_A_DQS_P<1>	30
83 11	MEM_A_DM<6>	MAKE_BASE=TRUE	=MEM_A_DM<1>	30
83 11	MEM_A_DQ<55>	MAKE_BASE=TRUE	=MEM_A_DQ<12>	30
83 11	MEM_A_DQ<54>	MAKE_BASE=TRUE	=MEM_A_DQ<13>	30
83 11	MEM_A_DQ<53>	MAKE_BASE=TRUE	=MEM_A_DQ<10>	30
83 11	MEM_A_DQ<52>	MAKE_BASE=TRUE	=MEM_A_DQ<15>	30
83 11	MEM_A_DQ<51>	MAKE_BASE=TRUE	=MEM_A_DQ<8>	30
83 11	MEM_A_DQ<50>	MAKE_BASE=TRUE	=MEM_A_DQ<9>	30
83 11	MEM_A_DQ<49>	MAKE_BASE=TRUE	=MEM_A_DQ<11>	30
83 11	MEM_A_DQ<48>	MAKE_BASE=TRUE	=MEM_A_DQ<14>	30

CPU CHANNEL A DQS 7 -> DIMM A DQS 0

83 11	MEM_A_DQS_N<7>	MAKE_BASE=TRUE	=MEM_A_DQS_N<0>	30
83 11	MEM_A_DQS_P<7>	MAKE_BASE=TRUE	=MEM_A_DQS_P<0>	30
83 11	MEM_A_DM<7>	MAKE_BASE=TRUE	=MEM_A_DM<0>	30
83 11	MEM_A_DQ<63>	MAKE_BASE=TRUE	=MEM_A_DQ<5>	30
83 11	MEM_A_DQ<62>	MAKE_BASE=TRUE	=MEM_A_DQ<4>	30
83 11	MEM_A_DQ<61>	MAKE_BASE=TRUE	=MEM_A_DQ<3>	30
83 11	MEM_A_DQ<60>	MAKE_BASE=TRUE	=MEM_A_DQ<2>	30
83 11	MEM_A_DQ<59>	MAKE_BASE=TRUE	=MEM_A_DQ<1>	30
83 11	MEM_A_DQ<58>	MAKE_BASE=TRUE	=MEM_A_DQ<0>	30
83 11	MEM_A_DQ<57>	MAKE_BASE=TRUE	=MEM_A_DQ<7>	30
83 11	MEM_A_DQ<56>	MAKE_BASE=TRUE	=MEM_A_DQ<6>	30

CPU CHANNEL B DQS 0 -> DIMM B DQS 7

83 11	MEM_B_DQS_N<0>	MAKE_BASE=TRUE	=MEM_B_DQS_N<7>	31
83 11	MEM_B_DQS_P<0>	MAKE_BASE=TRUE	=MEM_B_DQS_P<7>	31
83 11	MEM_B_DM<0>	MAKE_BASE=TRUE	=MEM_B_DM<7>	31
83 11	MEM_B_DQ<7>	MAKE_BASE=TRUE	=MEM_B_DQ<61>	31
83 11	MEM_B_DQ<6>	MAKE_BASE=TRUE	=MEM_B_DQ<60>	31
83 11	MEM_B_DQ<5>	MAKE_BASE=TRUE	=MEM_B_DQ<62>	31
83 11	MEM_B_DQ<4>	MAKE_BASE=TRUE	=MEM_B_DQ<63>	31
83 11	MEM_B_DQ<3>	MAKE_BASE=TRUE	=MEM_B_DQ<56>	31
83 11	MEM_B_DQ<2>	MAKE_BASE=TRUE	=MEM_B_DQ<57>	31
83 11	MEM_B_DQ<1>	MAKE_BASE=TRUE	=MEM_B_DQ<58>	31
83 11	MEM_B_DQ<0>	MAKE_BASE=TRUE	=MEM_B_DQ<59>	31

CPU CHANNEL B DQS 1 -> DIMM B DQS 6

83 11	MEM_B_DQS_N<1>	MAKE_BASE=TRUE	=MEM_B_DQS_N<6>	31
83 11	MEM_B_DQS_P<1>	MAKE_BASE=TRUE	=MEM_B_DQS_P<6>	31
83 11	MEM_B_DM<1>	MAKE_BASE=TRUE	=MEM_B_DM<6>	31
83 11	MEM_B_DQ<15>	MAKE_BASE=TRUE	=MEM_B_DQ<51>	31
83 11	MEM_B_DQ<14>	MAKE_BASE=TRUE	=MEM_B_DQ<54>	31
83 11	MEM_B_DQ<13>	MAKE_BASE=TRUE	=MEM_B_DQ<53>	31
83 11	MEM_B_DQ<12>	MAKE_BASE=TRUE	=MEM_B_DQ<52>	31
83 11	MEM_B_DQ<11>	MAKE_BASE=TRUE	=MEM_B_DQ<48>	31
83 11	MEM_B_DQ<10>	MAKE_BASE=TRUE	=MEM_B_DQ<49>	31
83 11	MEM_B_DQ<9>	MAKE_BASE=TRUE	=MEM_B_DQ<50>	31
83 11	MEM_B_DQ<8>	MAKE_BASE=TRUE	=MEM_B_DQ<55>	31

CPU CHANNEL B DQS 2 -> DIMM B DQS 5

83 11	MEM_B_DQS_N<2>	MAKE_BASE=TRUE	=MEM_B_DQS_N<5>	31
83 11	MEM_B_DQS_P<2>	MAKE_BASE=TRUE	=MEM_B_DQS_P<5>	31
83 11	MEM_B_DM<2>	MAKE_BASE=TRUE	=MEM_B_DM<5>	31
83 11	MEM_B_DQ<23>	MAKE_BASE=TRUE	=MEM_B_DQ<45>	31
83 11	MEM_B_DQ<22>	MAKE_BASE=TRUE	=MEM_B_DQ<40>	31
83 11	MEM_B_DQ<21>	MAKE_BASE=TRUE	=MEM_B_DQ<42>	31
83 11	MEM_B_DQ<20>	MAKE_BASE=TRUE	=MEM_B_DQ<47>	31
83 11	MEM_B_DQ<19>	MAKE_BASE=TRUE	=MEM_B_DQ<44>	31
83 11	MEM_B_DQ<18>	MAKE_BASE=TRUE	=MEM_B_DQ<41>	31
83 11	MEM_B_DQ<17>	MAKE_BASE=TRUE	=MEM_B_DQ<46>	31
83 11	MEM_B_DQ<16>	MAKE_BASE=TRUE	=MEM_B_DQ<43>	31

CPU CHANNEL B DQS 3 -> DIMM B DQS 4

83 11	MEM_B_DQS_N<3>	MAKE_BASE=TRUE	=MEM_B_DQS_N<4>	31
83 11	MEM_B_DQS_P<3>	MAKE_BASE=TRUE	=MEM_B_DQS_P<4>	31
83 11	MEM_B_DM<3>	MAKE_BASE=TRUE	=MEM_B_DM<4>	31
83 11	MEM_B_DQ<31>	MAKE_BASE=TRUE	=MEM_B_DQ<32>	31
83 11	MEM_B_DQ<30>	MAKE_BASE=TRUE	=MEM_B_DQ<39>	31
83 11	MEM_B_DQ<29>	MAKE_BASE=TRUE	=MEM_B_DQ<37>	31
83 11	MEM_B_DQ<28>	MAKE_BASE=TRUE	=MEM_B_DQ<38>	31
83 11	MEM_B_DQ<27>	MAKE_BASE=TRUE	=MEM_B_DQ<35>	31
83 11	MEM_B_DQ<26>	MAKE_BASE=TRUE	=MEM_B_DQ<34>	31
83 11	MEM_B_DQ<25>	MAKE_BASE=TRUE	=MEM_B_DQ<36>	31
83 11	MEM_B_DQ<24>	MAKE_BASE=TRUE	=MEM_B_DQ<33>	31

CPU CHANNEL B DQS 4 -> DIMM B DQS 3

83 11	MEM_B_DQS_N<4>	MAKE_BASE=TRUE	=MEM_B_DQS_N<3>	31
83 11	MEM_B_DQS_P<4>	MAKE_BASE=TRUE	=MEM_B_DQS_P<3>	31
83 11	MEM_B_DM<4>	MAKE_BASE=TRUE	=MEM_B_DM<3>	31
83 11	MEM_B_DQ<39>	MAKE_BASE=TRUE	=MEM_B_DQ<28>	31
83 11	MEM_B_DQ<38>	MAKE_BASE=TRUE	=MEM_B_DQ<24>	31
83 11	MEM_B_DQ<37>	MAKE_BASE=TRUE	=MEM_B_DQ<31>	31
83 11	MEM_B_DQ<36>	MAKE_BASE=TRUE	=MEM_B_DQ<30>	31
83 11	MEM_B_DQ<35>	MAKE_BASE=TRUE	=MEM_B_DQ<29>	31
83 11	MEM_B_DQ<34>	MAKE_BASE=TRUE	=MEM_B_DQ<25>	31
83 11	MEM_B_DQ<33>	MAKE_BASE=TRUE	=MEM_B_DQ<27>	31
83 11	MEM_B_DQ<32>	MAKE_BASE=TRUE	=MEM_B_DQ<26>	31

CPU CHANNEL B DQS 5 -> DIMM B DQS 2

83 11	MEM_B_DQS_N<5>	MAKE_BASE=TRUE	=MEM_B_DQS_N<2>	31
83 11	MEM_B_DQS_P<5>	MAKE_BASE=TRUE	=MEM_B_DQS_P<2>	31
83 11	MEM_B_DM<5>	MAKE_BASE=TRUE	=MEM_B_DM<2>	31
83 11	MEM_B_DQ<47>	MAKE_BASE=TRUE	=MEM_B_DQ<18>	31
83 11	MEM_B_DQ<46>	MAKE_BASE=TRUE	=MEM_B_DQ<20>	31
83 11	MEM_B_DQ<45>	MAKE_BASE=TRUE	=MEM_B_DQ<23>	31
83 11	MEM_B_DQ<44>	MAKE_BASE=TRUE	=MEM_B_DQ<22>	31
83 11	MEM_B_DQ<43>	MAKE_BASE=TRUE	=MEM_B_DQ<19>	31
83 11	MEM_B_DQ<42>	MAKE_BASE=TRUE	=MEM_B_DQ<21>	31
83 11	MEM_B_DQ<41>	MAKE_BASE=TRUE	=MEM_B_DQ<17>	31
83 11	MEM_B_DQ<40>	MAKE_BASE=TRUE	=MEM_B_DQ<16>	31

CPU CHANNEL B DQS 6 -> DIMM B DQS 1

83 11	MEM_B_DQS_N<6>	MAKE_BASE=TRUE	=MEM_B_DQS_N<1>	31
83 11	MEM_B_DQS_P<6>	MAKE_BASE=TRUE	=MEM_B_DQS_P<1>	31
83 11	MEM_B_DM<6>	MAKE_BASE=TRUE	=MEM_B_DM<1>	31
83 11	MEM_B_DQ<55>	MAKE_BASE=TRUE	=MEM_B_DQ<8>	31
83 11	MEM_B_DQ<54>	MAKE_BASE=TRUE	=MEM_B_DQ<14>	31
83 11	MEM_B_DQ<53>	MAKE_BASE=TRUE	=MEM_B_DQ<10>	31
83 11	MEM_B_DQ<52>	MAKE_BASE=TRUE	=MEM_B_DQ<11>	31
83 11	MEM_B_DQ<51>	MAKE_BASE=TRUE	=MEM_B_DQ<9>	31
83 11	MEM_B_DQ<50>	MAKE_BASE=TRUE	=MEM_B_DQ<15>	31
83 11	MEM_B_DQ<49>	MAKE_BASE=TRUE	=MEM_B_DQ<12>	31
83 11	MEM_B_DQ<48>	MAKE_BASE=TRUE	=MEM_B_DQ<13>	31

CPU CHANNEL B DQS 7 -> DIMM B DQS 0

83 11	MEM_B_DQS_N<7>	MAKE_BASE=TRUE	=MEM_B_DQS_N<0>	31
83 11	MEM_B_DQS_P<7>	MAKE_BASE=TRUE	=MEM_B_DQS_P<0>	31
83 11	MEM_B_DM<7>	MAKE_BASE=TRUE	=MEM_B_DM<0>	31
83 11	MEM_B_DQ<63>	MAKE_BASE=TRUE	=MEM_B_DQ<2>	31
83 11	MEM_B_DQ<62>	MAKE_BASE=TRUE	=MEM_B_DQ<1>	31
83 11	MEM_B_DQ<61>	MAKE_BASE=TRUE	=MEM_B_DQ<3>	31
83 11	MEM_B_DQ<60>	MAKE_BASE=TRUE	=MEM_B_DQ<7>	31
83 11	MEM_B_DQ<59>	MAKE_BASE=TRUE	=MEM_B_DQ<4>	31
83 11	MEM_B_DQ<58>	MAKE_BASE=TRUE	=MEM_B_DQ<5>	31
83 11	MEM_B_DQ<57>	MAKE_BASE=TRUE	=MEM_B_DQ<6>	31
83 11	MEM_B_DQ<56>	MAKE_BASE=TRUE	=MEM_B_DQ<0>	31

MEMORY CLOCK ALIASING

83 11	MEM_A_CLK_P<0>	MAKE_BASE=TRUE	=MEM_A_CLK_P<0>	30
83 11	MEM_A_CLK_N<0>	MAKE_BASE=TRUE	=MEM_A_CLK_N<0>	30
83 11	MEM_A_CLK_P<1>	MAKE_BASE=TRUE	=MEM_A_CLK_P<1>	30
83 11	MEM_A_CLK_N<1>	MAKE_BASE=TRUE	=MEM_A_CLK_N<1>	30
83 11	MEM_A_CLK_P<2>	MAKE_BASE=TRUE	=MEM_A_CLK_P<2>	30
83 11	MEM_A_CLK_N<2>	MAKE_BASE=TRUE	=MEM_A_CLK_N<2>	30
83 11	MEM_A_CLK_P<3>	MAKE_BASE=TRUE	=MEM_A_CLK_P<3>	30
83 11	MEM_A_CLK_N<3>	MAKE_BASE=TRUE	=MEM_A_CLK_N<3>	30

83 11	MEM_B_CLK_P<0>	MAKE_BASE=TRUE	=MEM_B_CLK_P<0>	31
83 11	MEM_B_CLK_N<0>	MAKE_BASE=TRUE	=MEM_B_CLK_N<0>	31
83 11	MEM_B_CLK_P<1>	MAKE_BASE=TRUE	=MEM_B_CLK_P<1>	31
83 11	MEM_B_CLK_N<1>	MAKE_BASE=TRUE	=MEM_B_CLK_N<1>	31
83 11	MEM_B_CLK_P<2>	MAKE_BASE=TRUE	=MEM_B_CLK_P<2>	31
83 11	MEM_B_CLK_N<2>	MAKE_BASE=TRUE	=MEM_B_CLK_N<2>	31
83 11	MEM_B_CLK_P<3>	MAKE_BASE=TRUE	=MEM_B_CLK_P<3>	31
83 11	MEM_B_CLK_N<3>	MAKE_BASE=TRUE	=MEM_B_CLK_N<3>	31

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

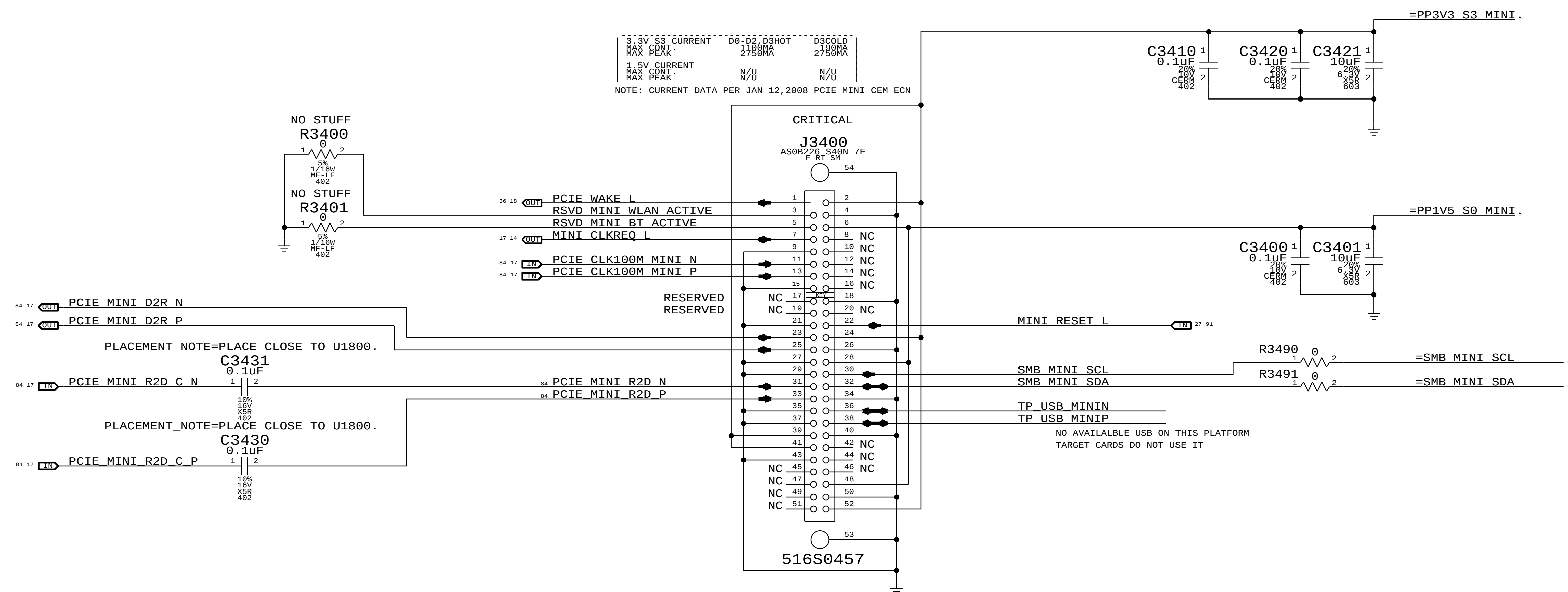
DDR3 ALIAS AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

PCI-E MiniCard Connector

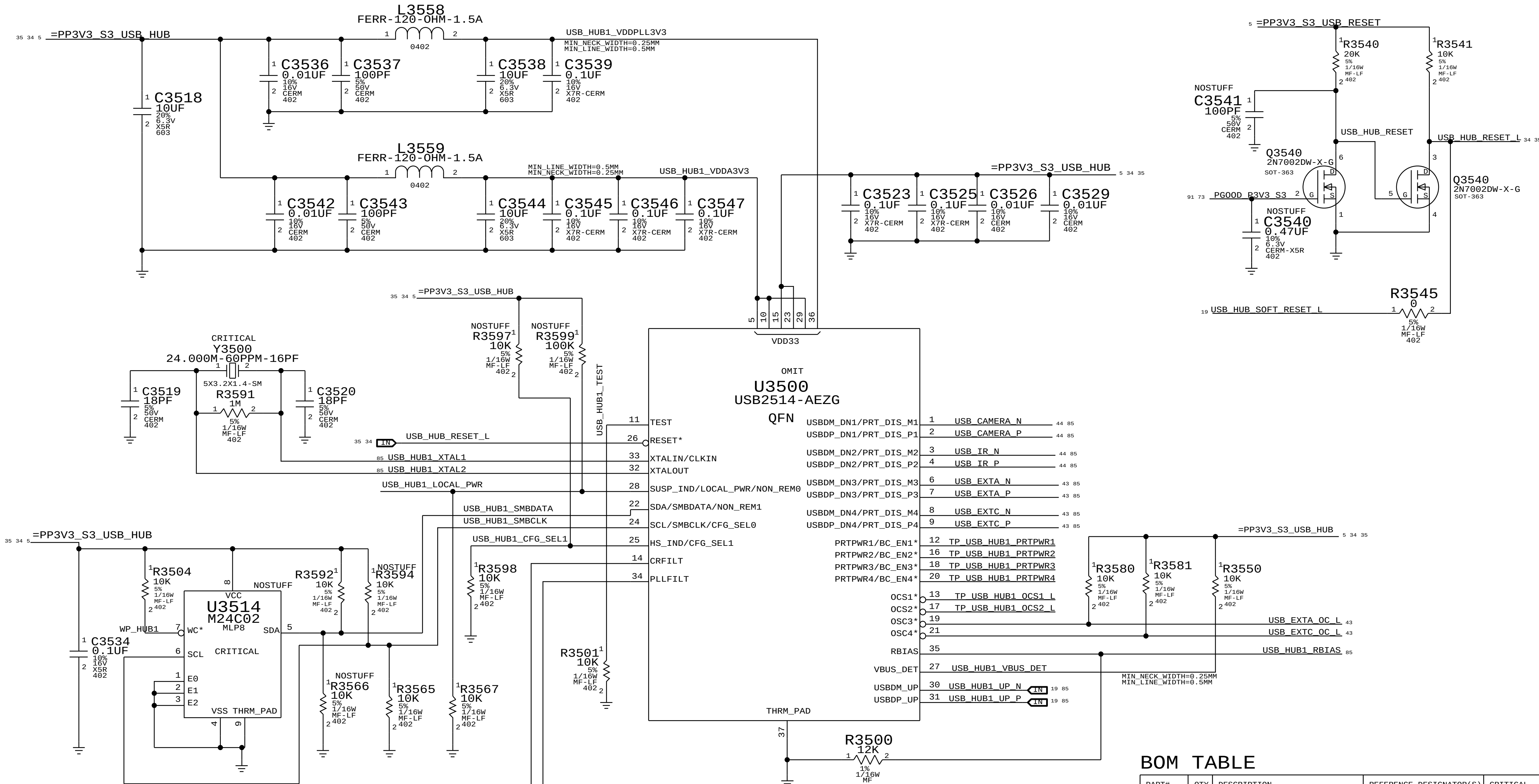
Apple Inc.

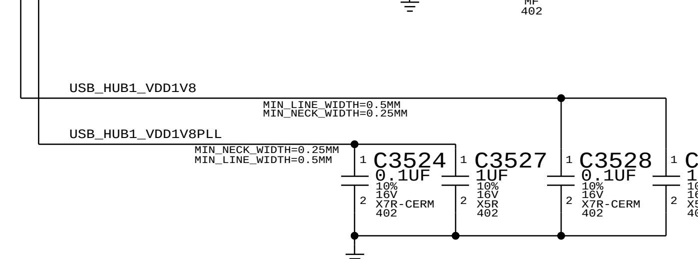
DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

USB HUB-1





SEL1	SEL0	DESCRIPTION
DEFAULT K23F ==> 0	0	Internal Default with Self powered Operation
0	1	SMBUS Slave Config
1	0	Internal Default with Bus powered Operation
1	1	EEPROM Supported

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are Non removable
0	1	Port1 is non removable
DEFAULT K23F ==> 1	0	Port 1 and 2 are non removable
1	1	Port1,2 and 3 are non Removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0721	2	SMSC USX2061-AEZG	U3500,U3600	CRITICAL	HUB_USX2061
338S0824	2	SMSC USB2514B	U3500,U3600	CRITICAL	HUB_USB2514B

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: USB HUB 1

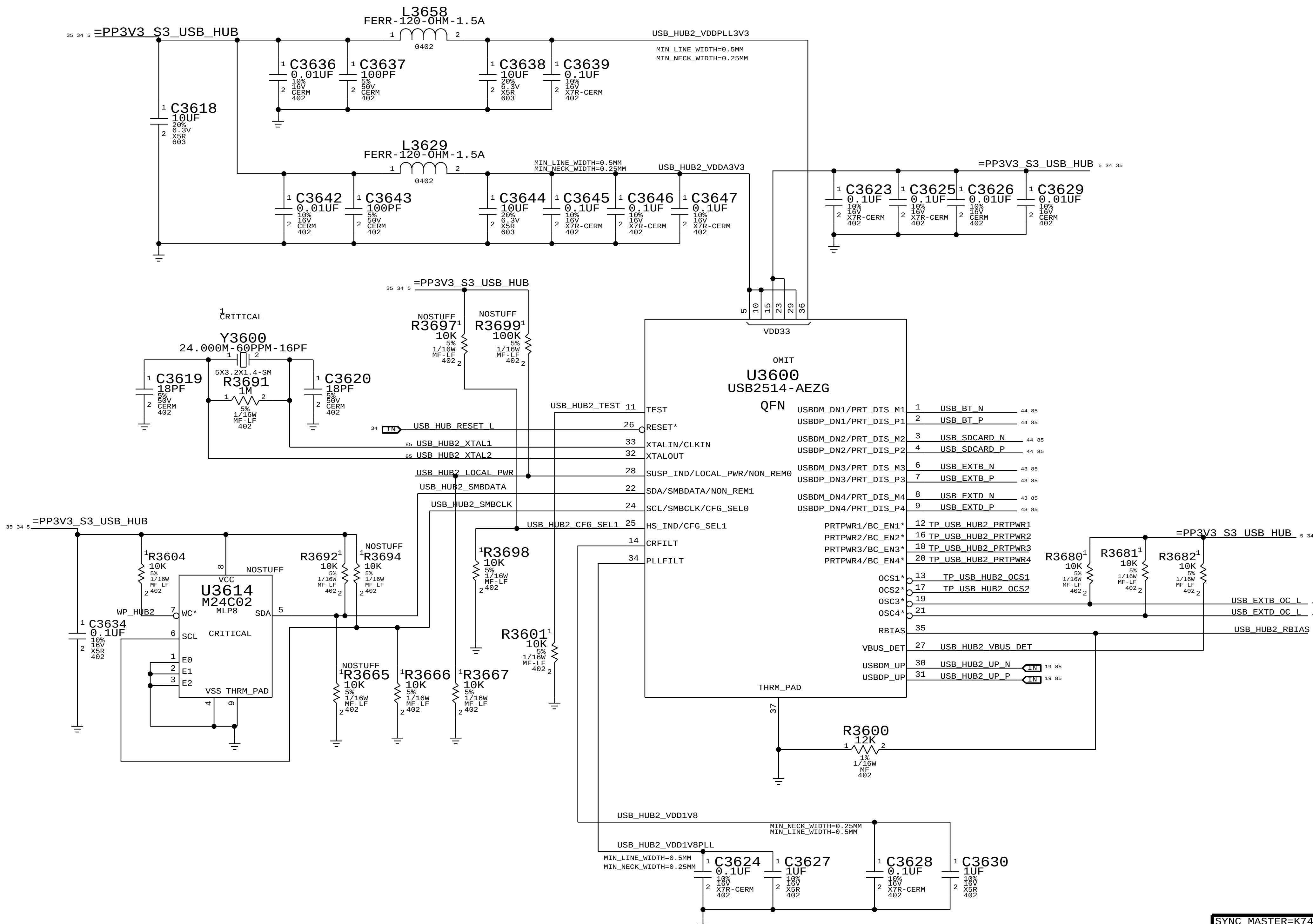
Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

BRANCH

USB HUB-2



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: USB HUB 2

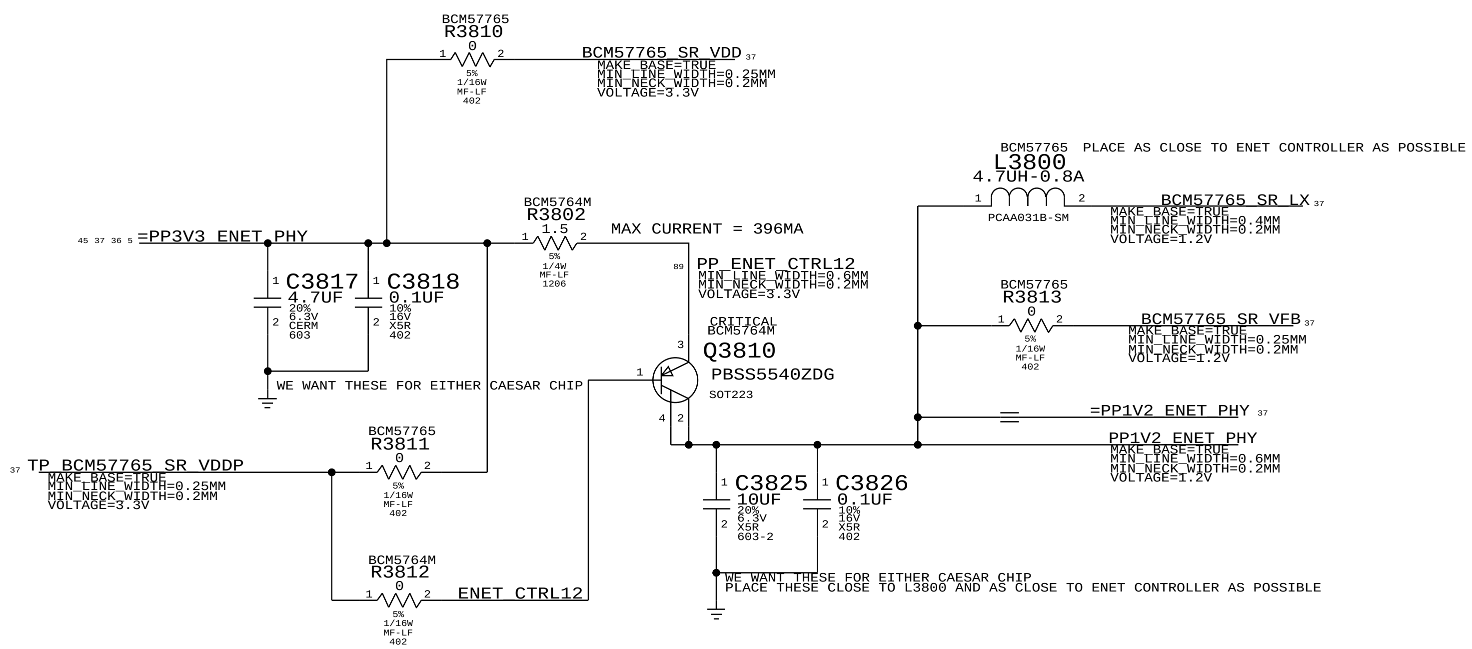
Apple Inc.

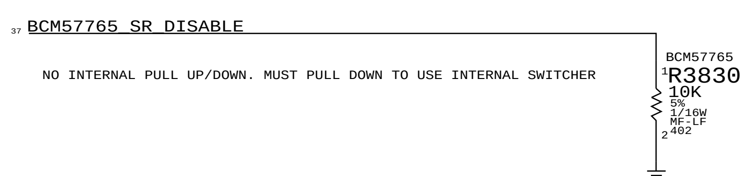
DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

BRANCH

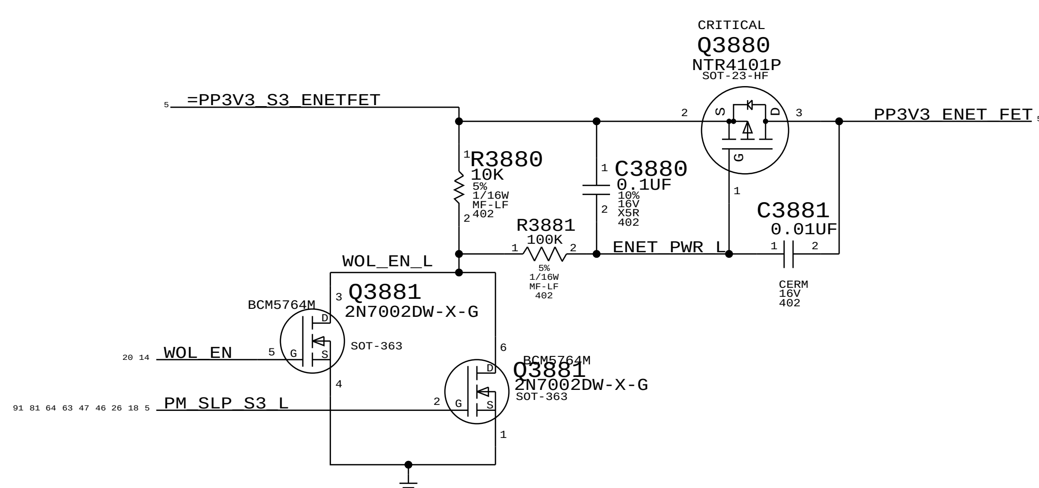
CAESAR II/IV 1V2 RAIL SUPPLY



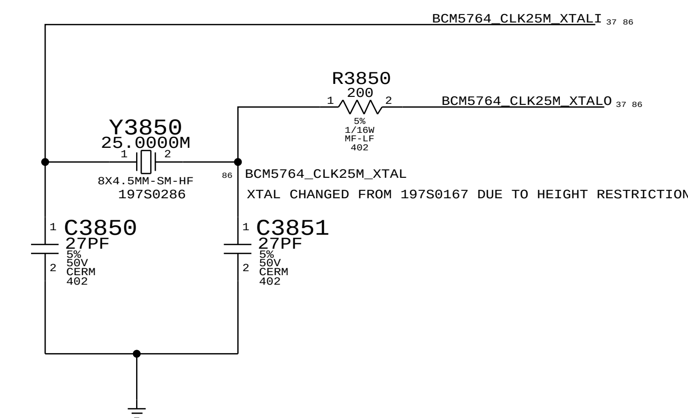


3.3V ENET FET

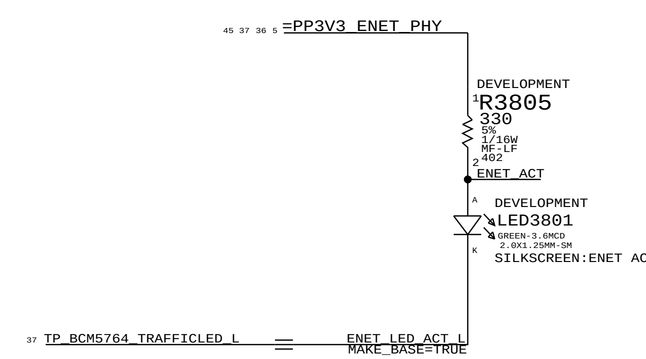
ENET_PWR_ON = "S0" || (S3 power && WOL_EN)



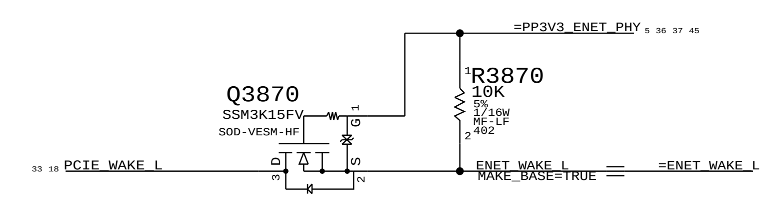
CAESAR II/IV 25MHZ XTAL



CAESAR II/IV ACTIVITY LED



CAESAR II/IV WAKE# ISOLATION



SYNC_MASTER=MASTER SYNC_DATE=N/A

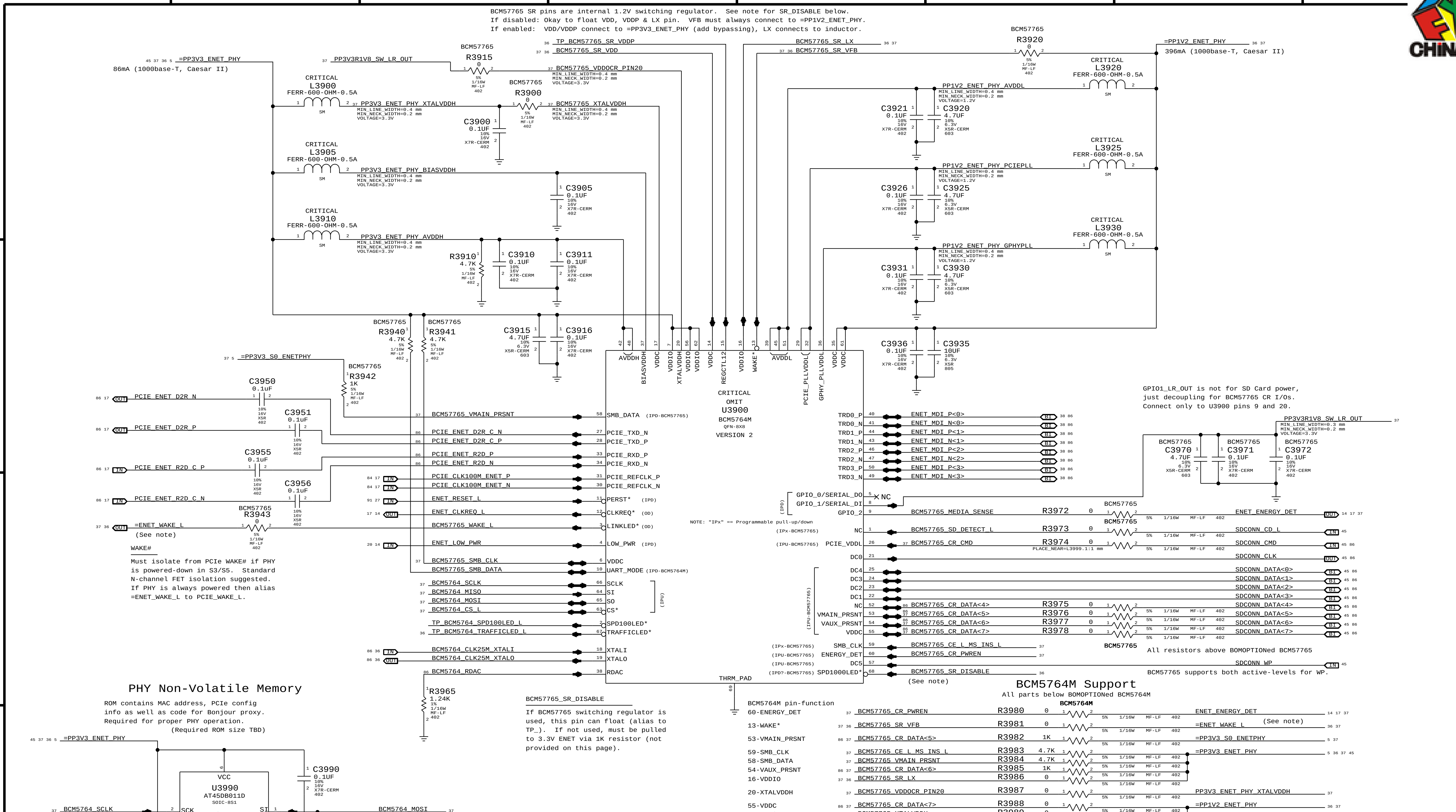
PAGE TITLE: Caesar II/IV Support

Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

BRANCH



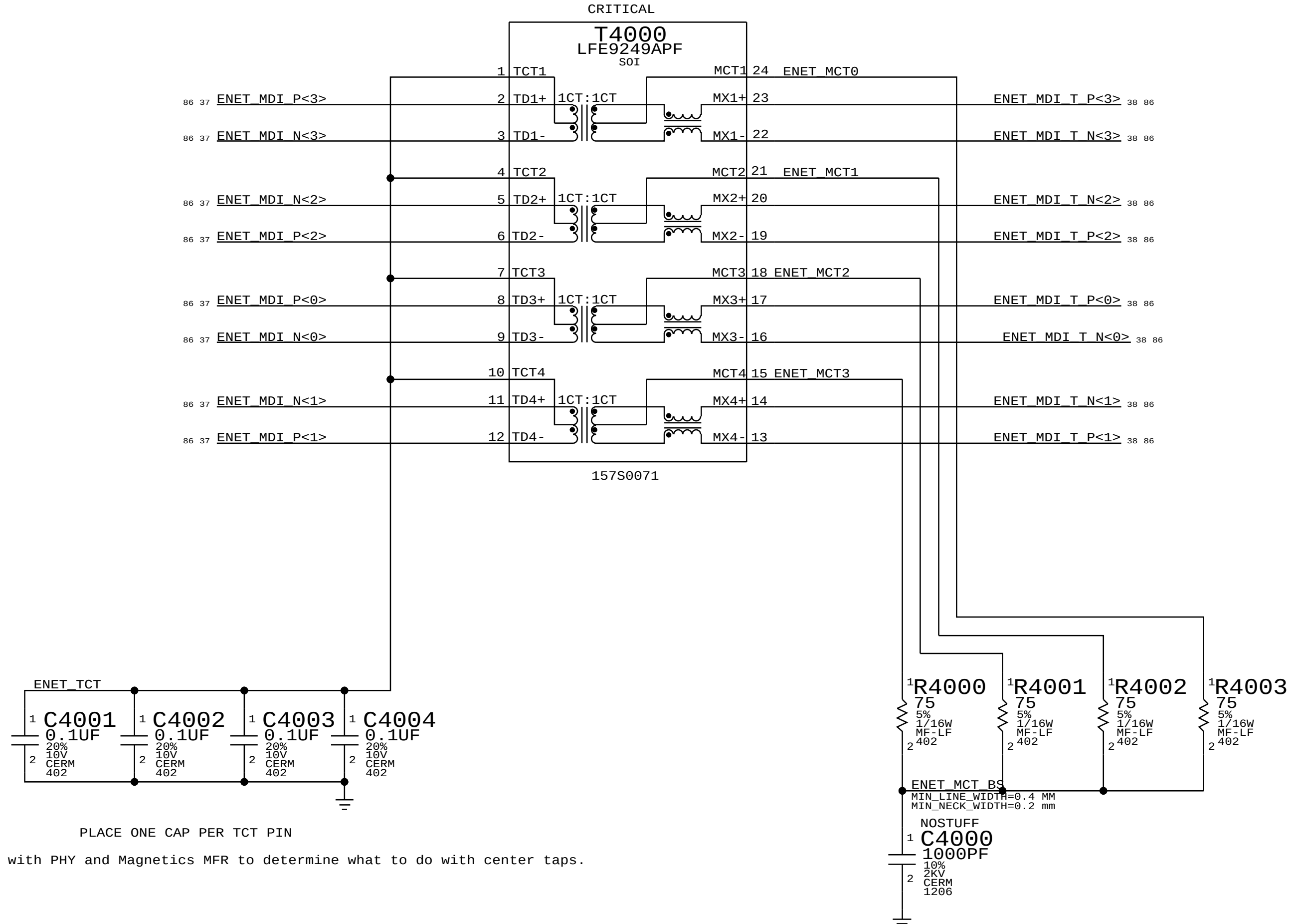
SYNC_MASTER=T27 SYNC_DATE=11/30/2009

PAGE TITLE: Ethernet PHY (Caesar II/IV)

Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

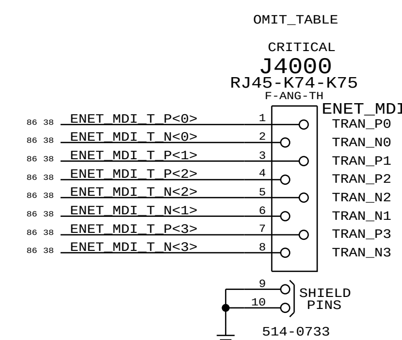
REVISION: A.0.0



PLACE ONE CAP PER TCT PIN

NOTE: Check with PHY and Magnetics MFR to determine what to do with center taps.

NOTE: BOB SMITH TERMINATION FOR EMC.



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
514-0654	1	K22/K23 PROD. RJ45	J4000	CRITICAL	METAL_IO
514-0733	1	K74/K75 RJ45, PLASTIC, PD/NI	J4000	CRITICAL	PLASTIC_IO

SYNC_MASTER=MASTER SYNC_DATE=N/A

PAGE TITLE

Ethernet Connector

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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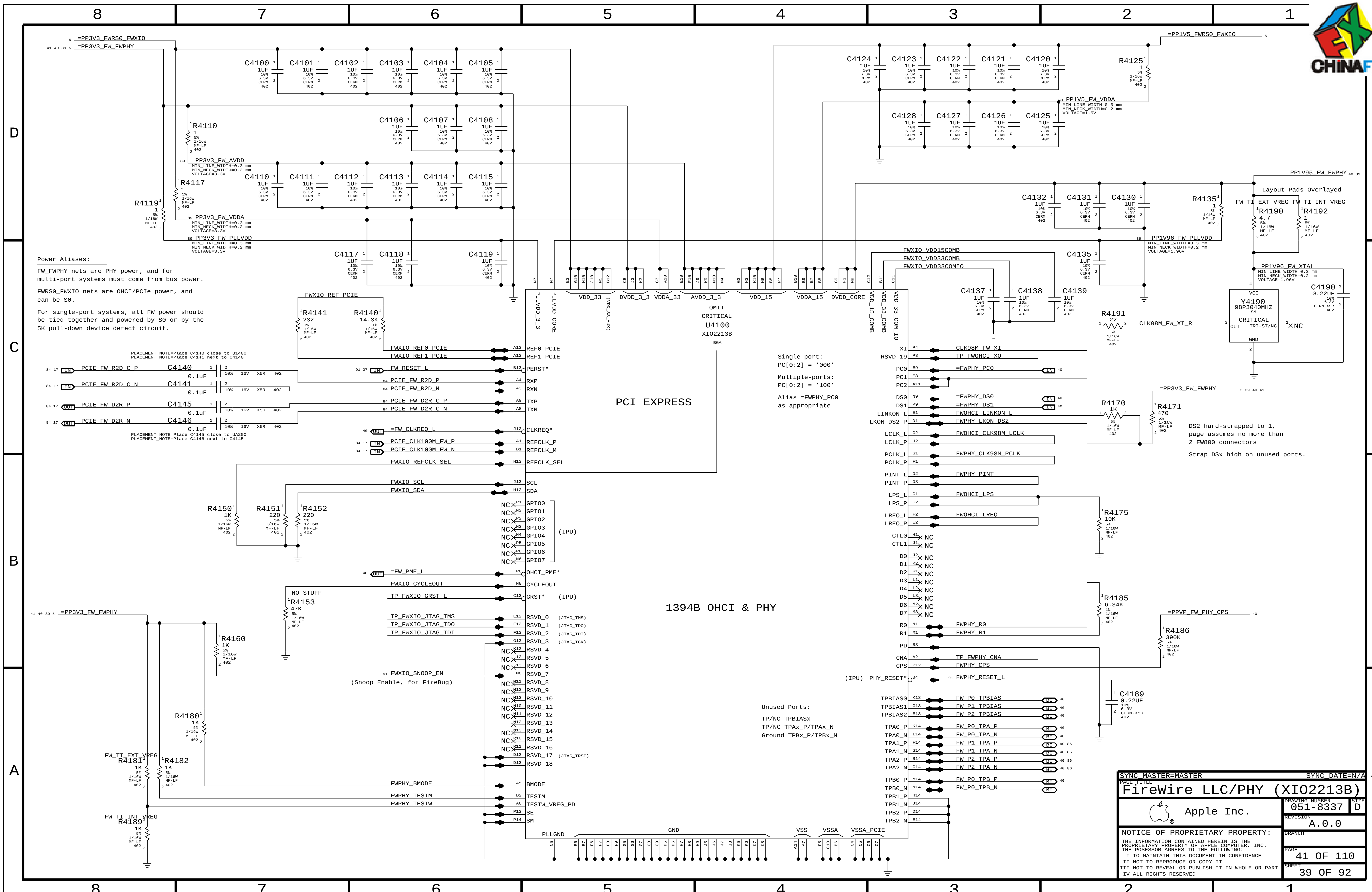


SYNC_MASTER=MASTER SYNC_DATE=N/A

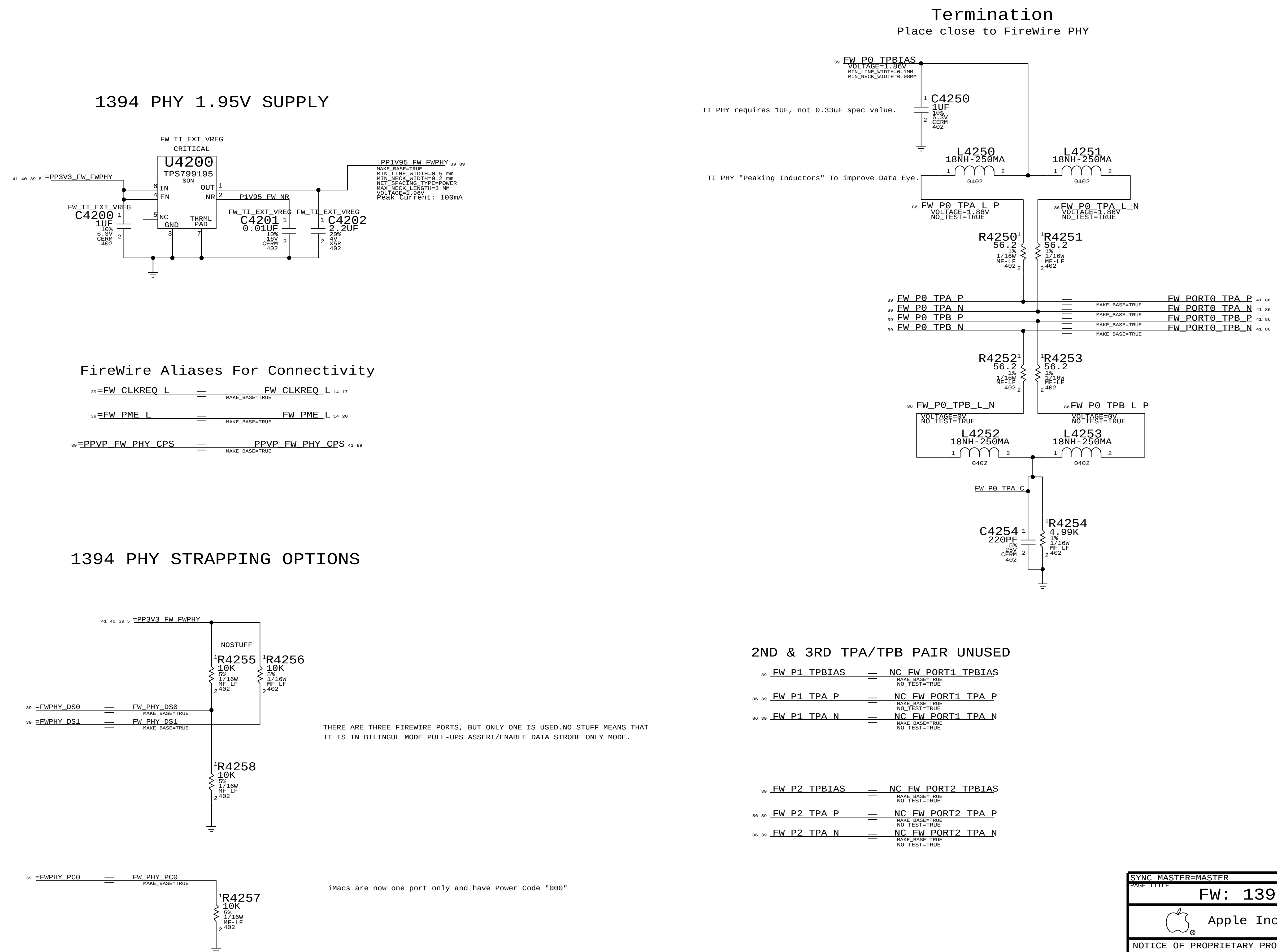
PAGE TITLE
FireWire LLC/PHY (XIO2213B)

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH



SYNC_MASTER=MASTER SYNC_DATE=N/A

PAGE TITLE: FW: 1394B MISC

Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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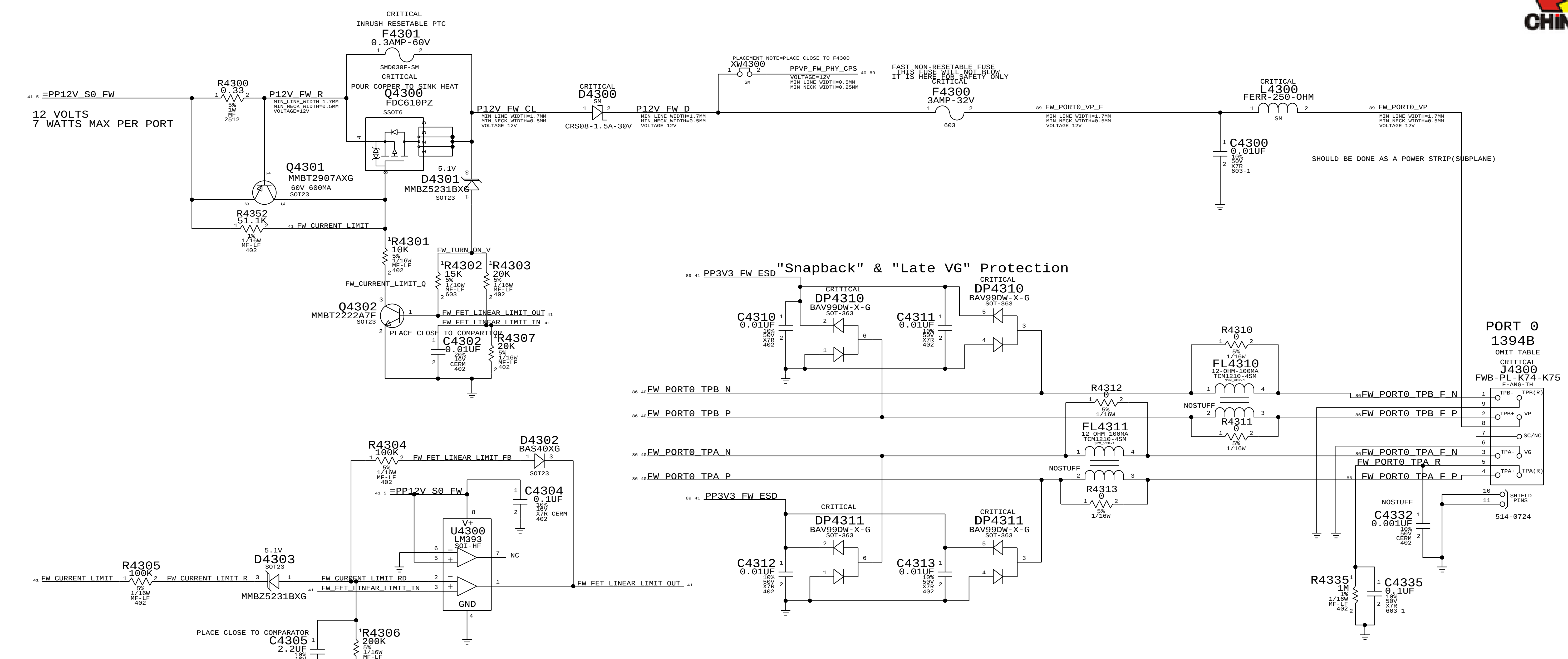
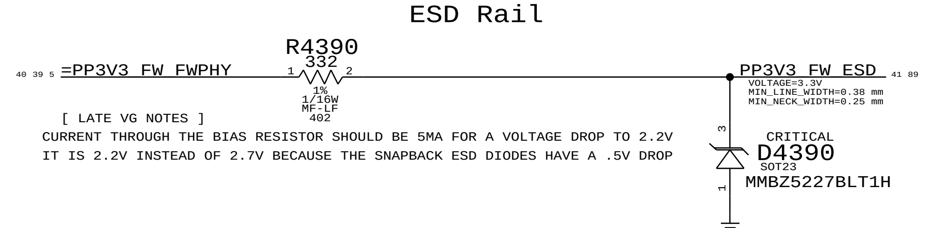


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
514-0656	1	K22/K23 PROD. FW	J4300	CRITICAL	METAL_IO
514-0724	1	K74/K75 FW, PLASTIC, PD/NI	J4300	CRITICAL	PLASTIC_IO

ESD Rail



[LATE VG NOTES]
CURRENT THROUGH THE BIAS RESISTOR SHOULD BE 5MA FOR A VOLTAGE DROP TO 2.2V
IT IS 2.2V INSTEAD OF 2.7V BECAUSE THE SNAPBACK ESD DIODES HAVE A .5V DROP

SYNC_MASTER=MASTER SYNC_DATE=11/17/2009

PAGE TITLE
FIREWIRE CONNECTOR

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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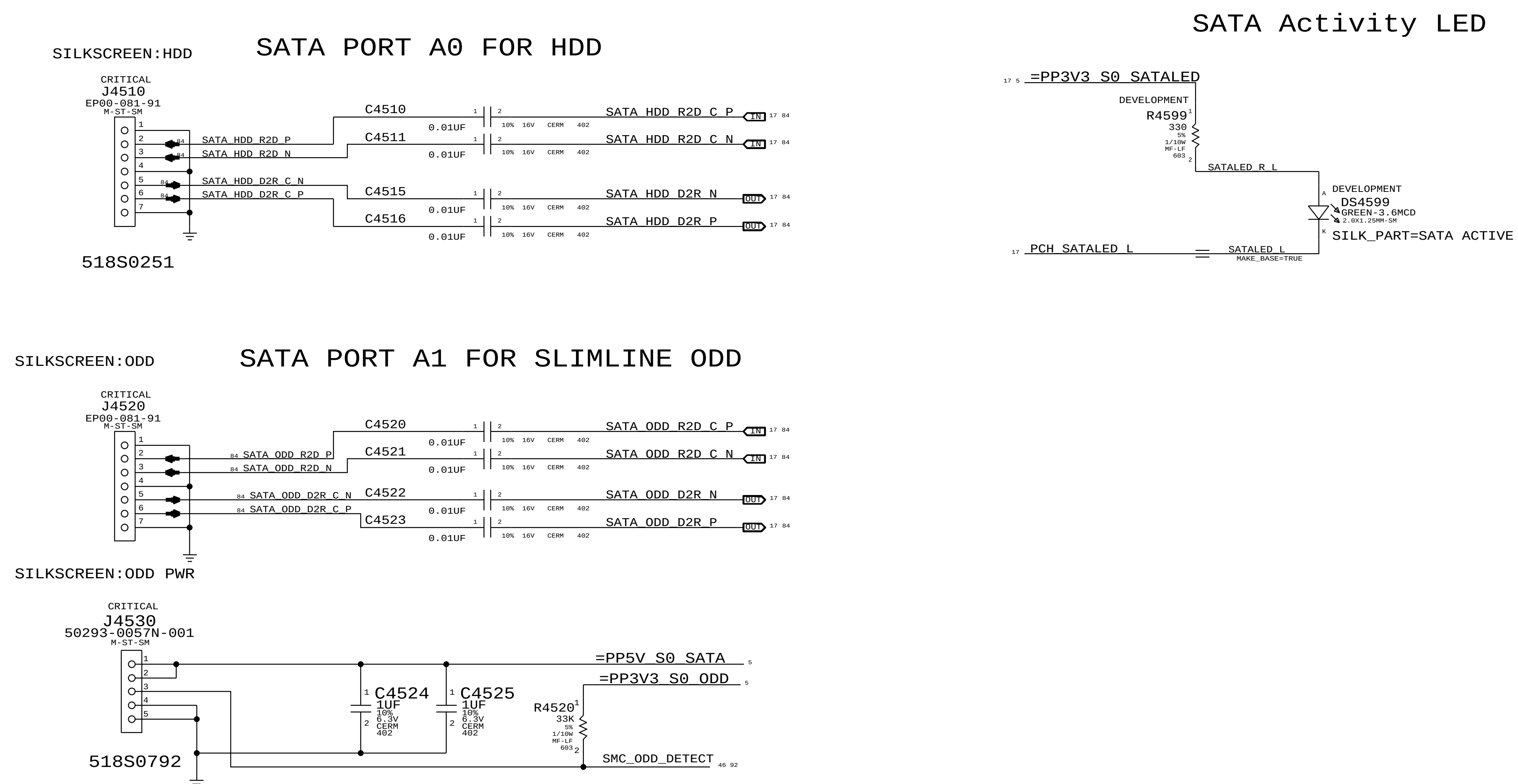

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

SATA Connectors

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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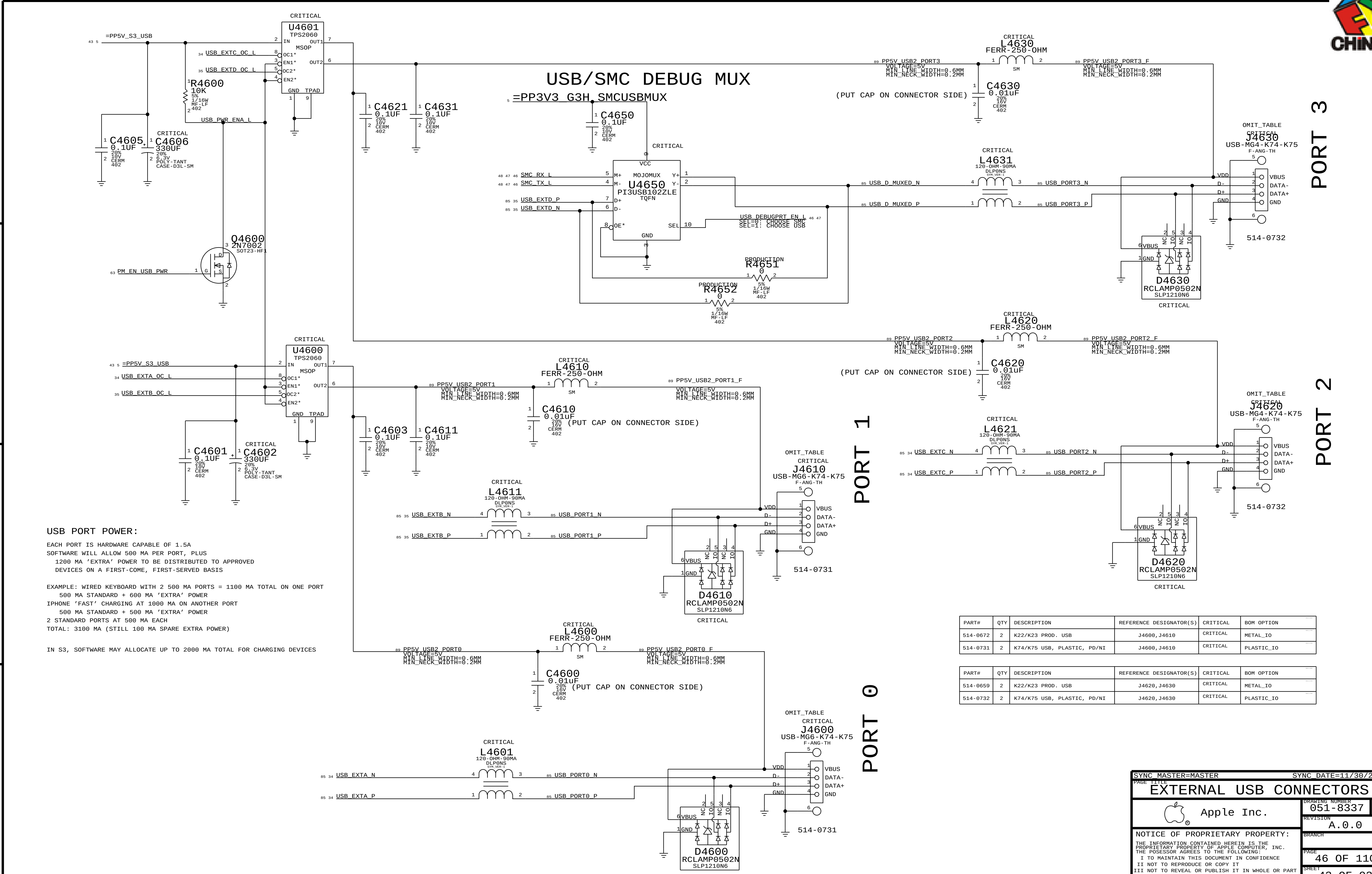

USB PORT POWER:

EACH PORT IS HARDWARE CAPABLE OF 1.5A
SOFTWARE WILL ALLOW 500 MA PER PORT, PLUS
1200 MA 'EXTRA' POWER TO BE DISTRIBUTED TO APPROVED
DEVICES ON A FIRST-COME, FIRST-SERVED BASIS

EXAMPLE: WIRED KEYBOARD WITH 2 500 MA PORTS = 1100 MA TOTAL ON ONE PORT
500 MA STANDARD + 600 MA 'EXTRA' POWER
IPHONE 'FAST' CHARGING AT 1000 MA ON ANOTHER PORT
500 MA STANDARD + 500 MA 'EXTRA' POWER
2 STANDARD PORTS AT 500 MA EACH
TOTAL: 3100 MA (STILL 100 MA SPARE EXTRA POWER)

IN S3, SOFTWARE MAY ALLOCATE UP TO 2000 MA TOTAL FOR CHARGING DEVICES

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
514-0672	2	K22/K23 PROD. USB	J4600,J4610	CRITICAL	METAL_IO
514-0731	2	K74/K75 USB, PLASTIC, PD/NI	J4600,J4610	CRITICAL	PLASTIC_IO

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
514-0659	2	K22/K23 PROD. USB	J4620,J4630	CRITICAL	METAL_IO
514-0732	2	K74/K75 USB, PLASTIC, PD/NI	J4620,J4630	CRITICAL	PLASTIC_IO

SYNC_MASTER=MASTER SYNC_DATE=11/30/2009

PAGE TITLE
EXTERNAL USB CONNECTORS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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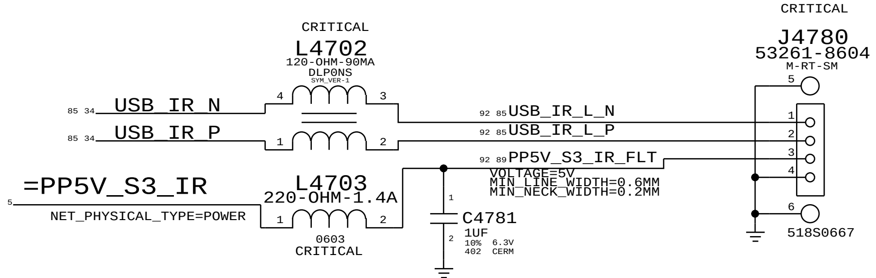
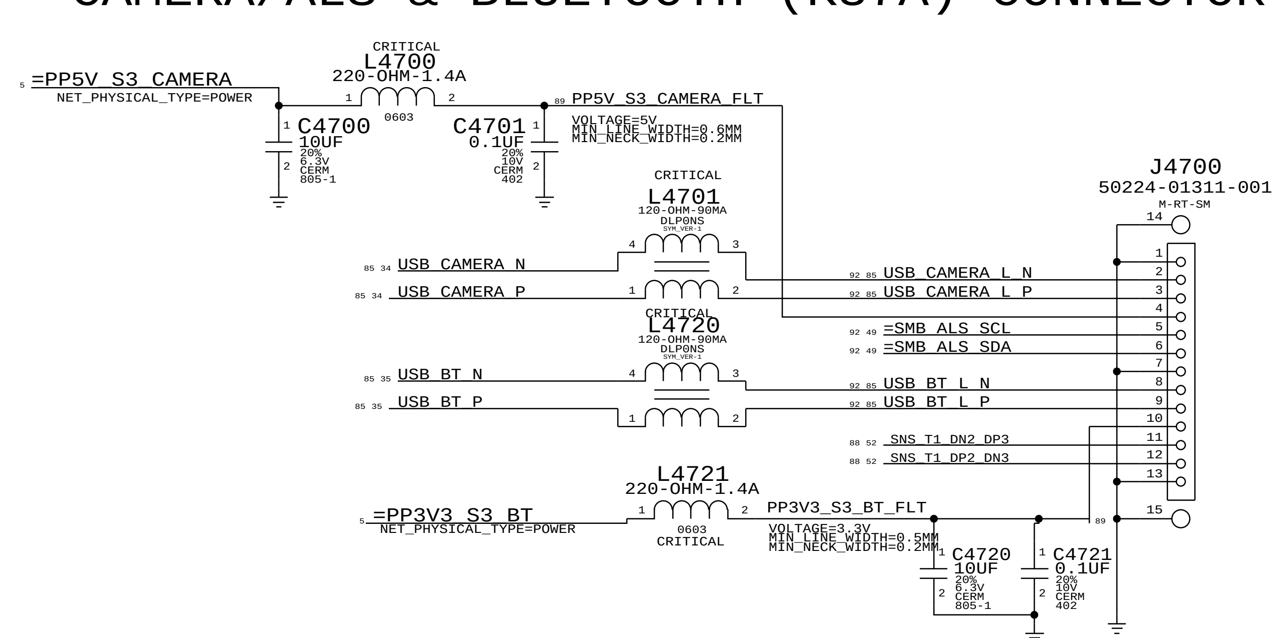
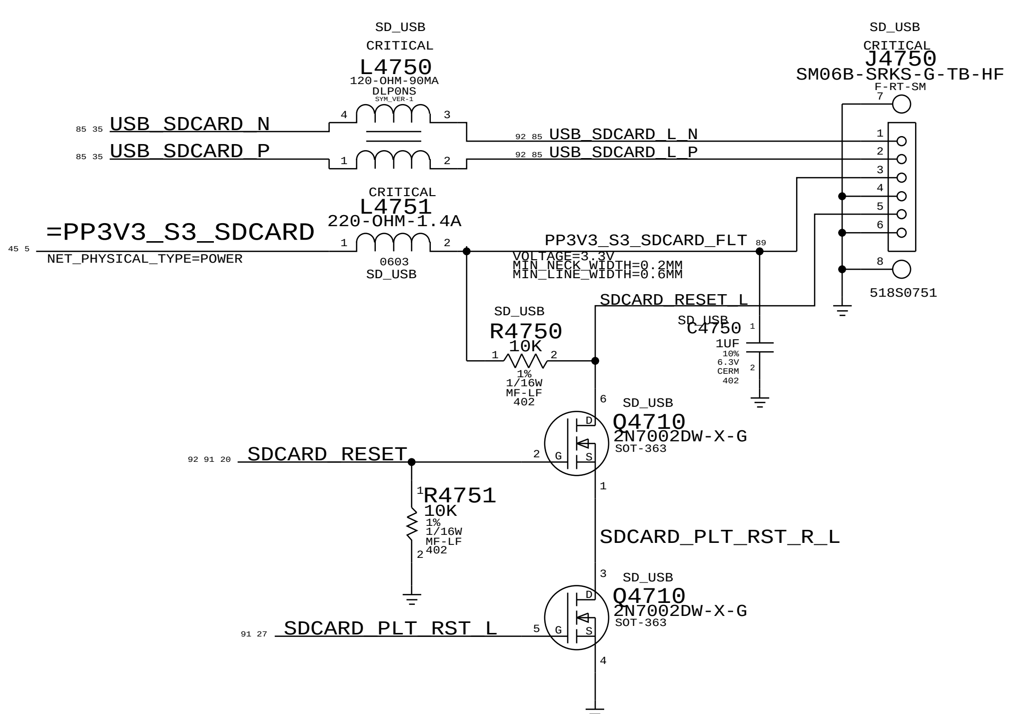
IR RECEIVER CONNECTOR



CAMERA/ALS & BLUETOOTH (K37A) CONNECTOR



LAZAURS SD CARD READER BOARD CONNECTOR
BACKUP TO CAESAR IV



SYNC_MASTER=MASTER SYNC_DATE=11/06/2009

PAGE TITLE
Internal USB Connections

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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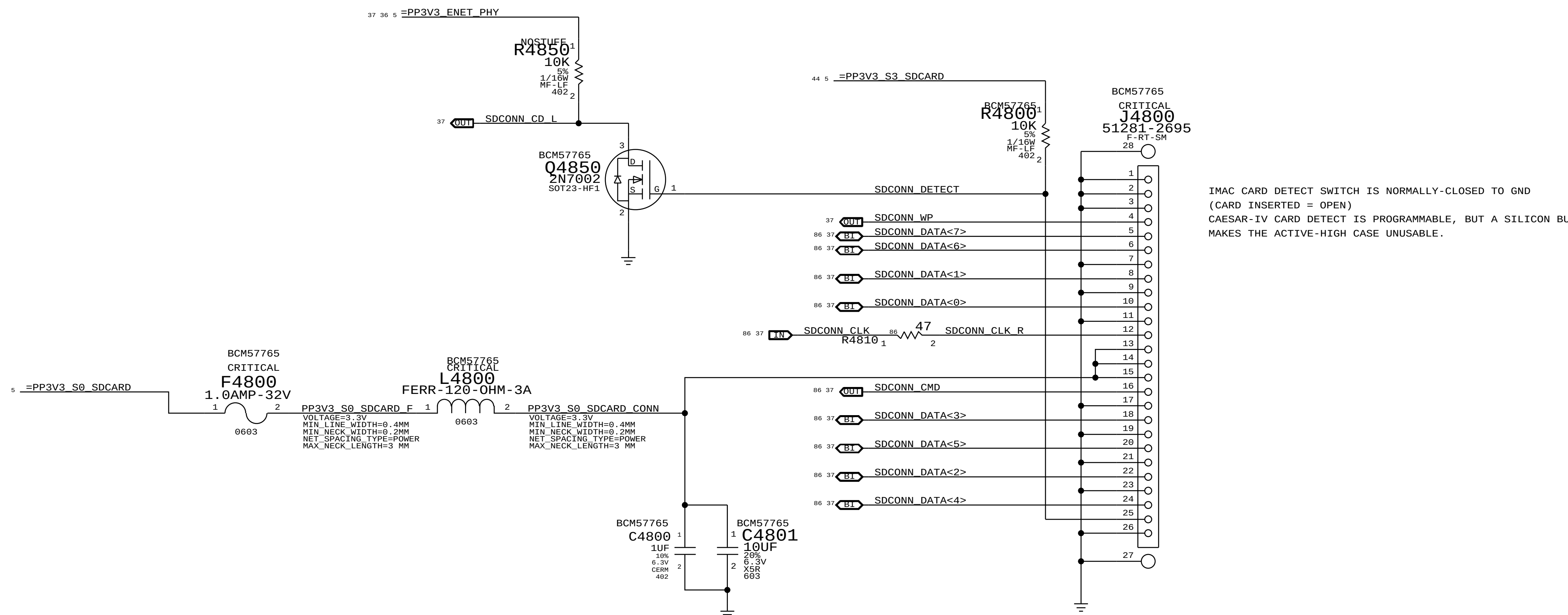


SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

SD READER CONNECTOR

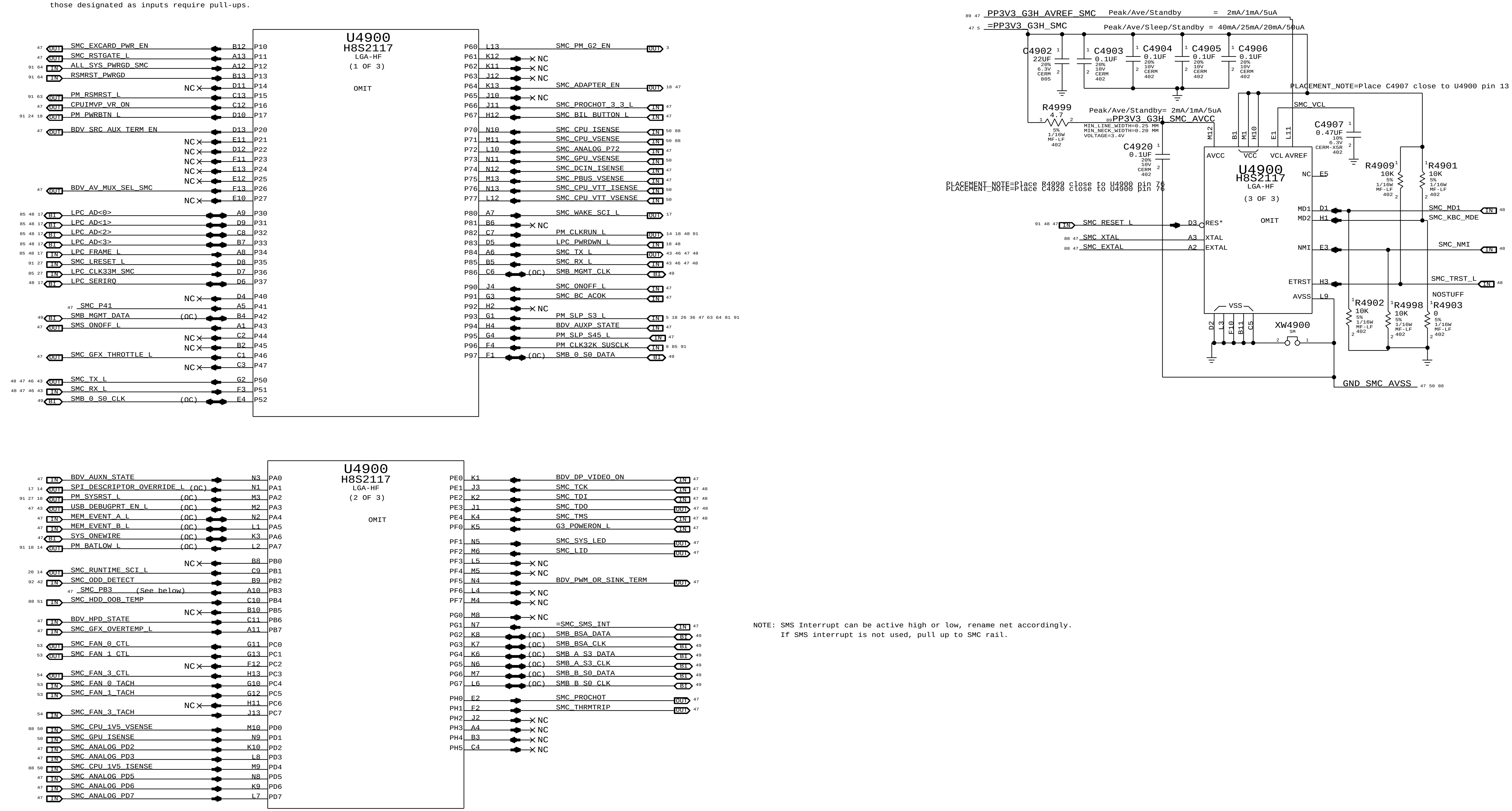
Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.



SMC_PB3:

SMC_IG_THROTTLE_L for MG systems.
Otherwise, TP/NC okay (was ISENSE_CAL_EN)

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

SMC

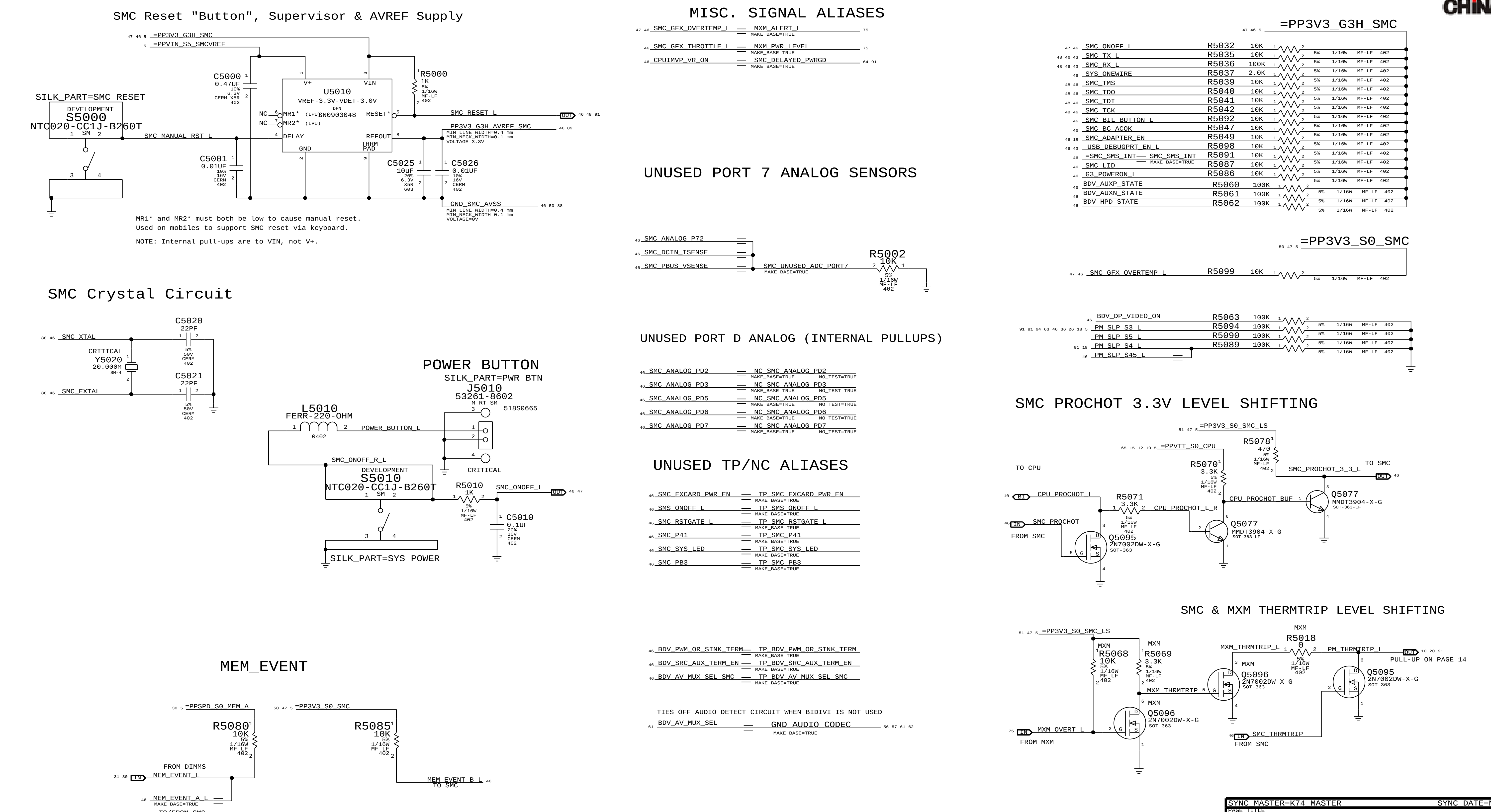
Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

8 7 6 5 4 3 2 1



SYNC_MASTER=K74_MASTER	SYNC_DATE=N/A
PAGE TITLE	SMC Support
Apple Inc.	DRAWING NUMBER 051-8337 SIZE D
	REVISION A.0.0
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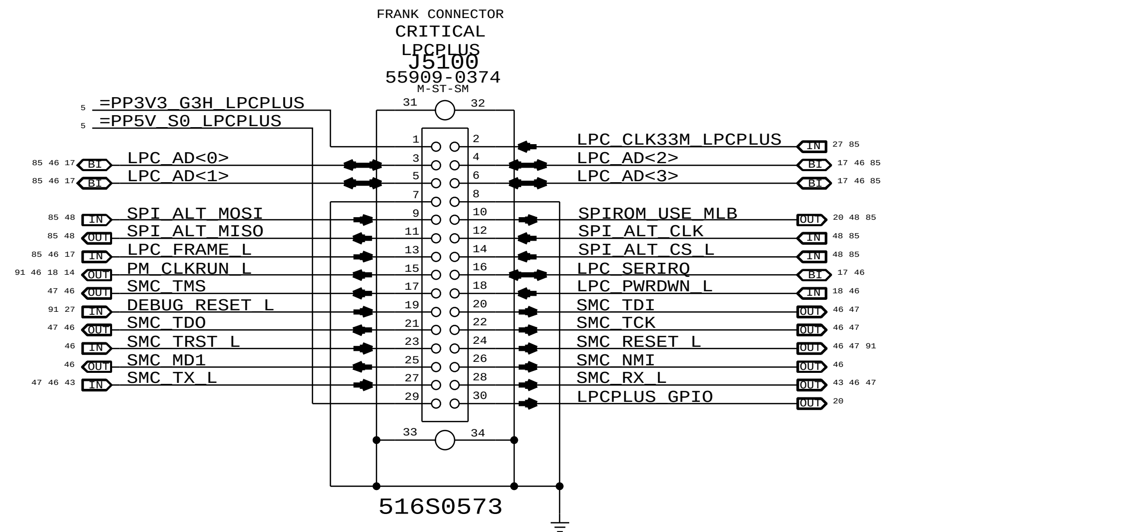
PAGE 50 OF 110

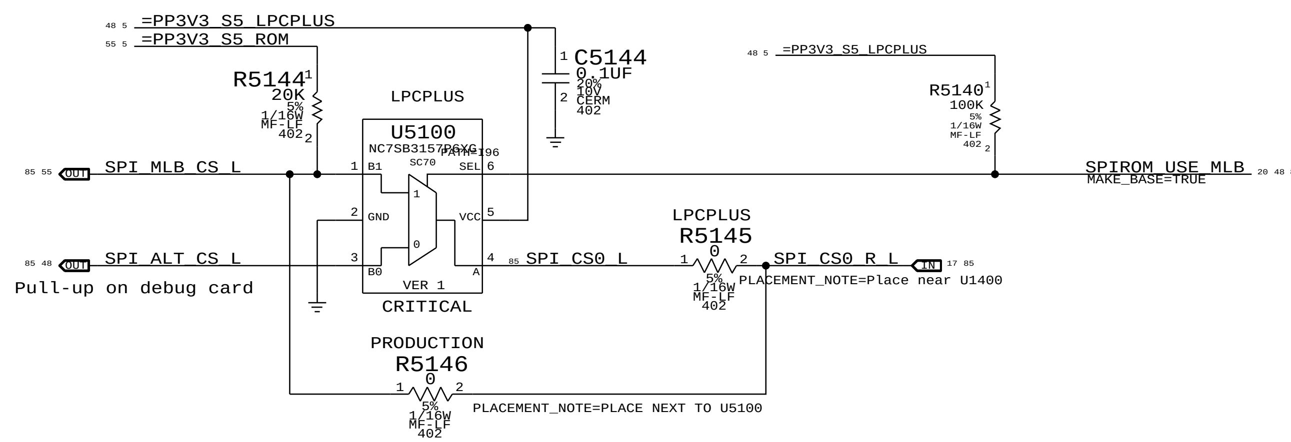
SHEET 47 OF 92

8 7 6 5 4 3 2 1

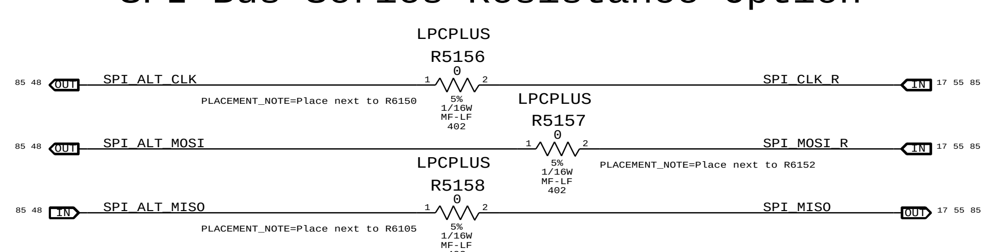
LPC+SPI Connector



Alternate SPI ROM Support



SPI Bus Series Resistance Option



SYNC_MASTER=K23F	SYNC_DATE=11/30/2009
PAGE TITLE: LPC+SPI Debug Connector	
Apple Inc.	DRAWING NUMBER 051-8337 SIZE D
	REVISION A.0.0
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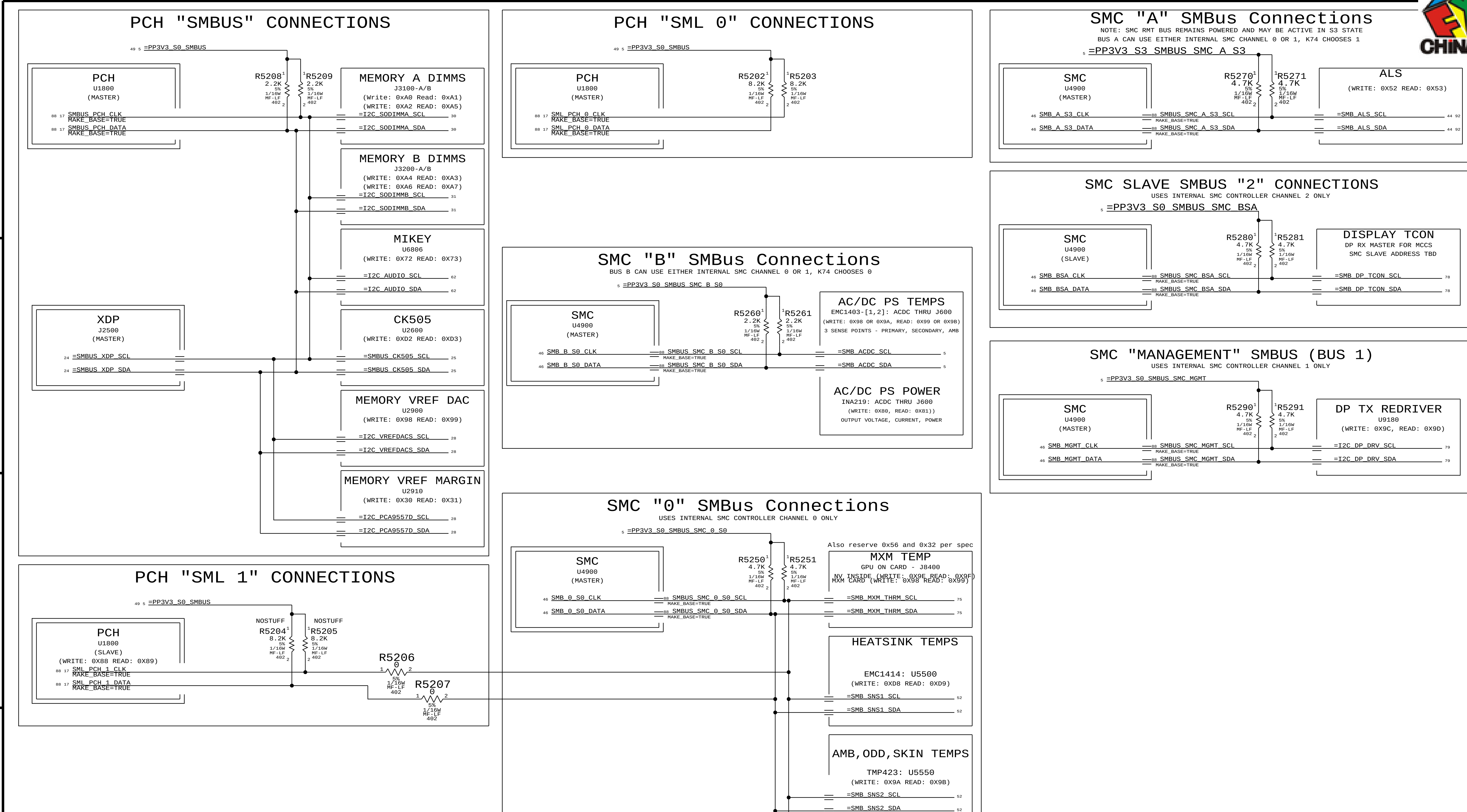
	SHEET 48 OF 92



SYNC_MASTER=DAVE SYNC_DATE=01/07/2010

PAGE TITLE

SMBus Connections

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

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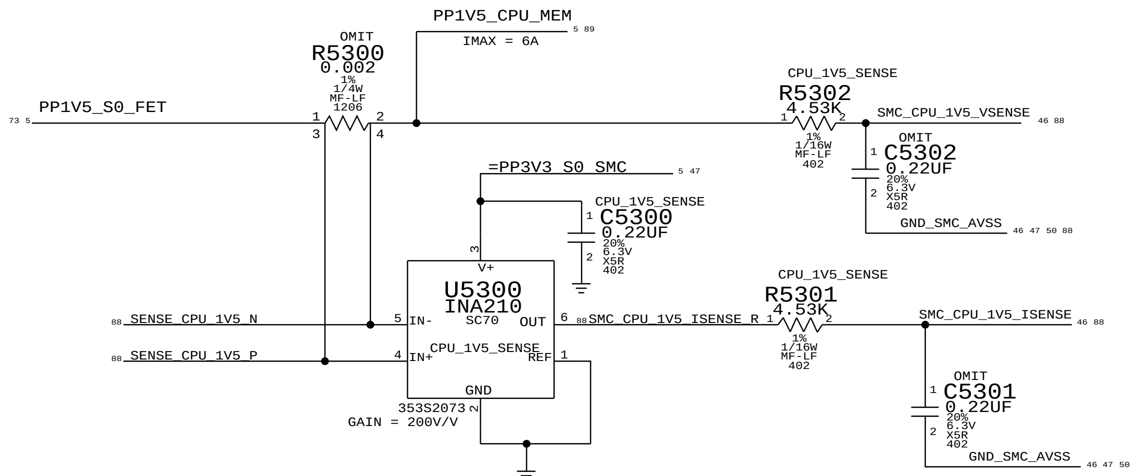
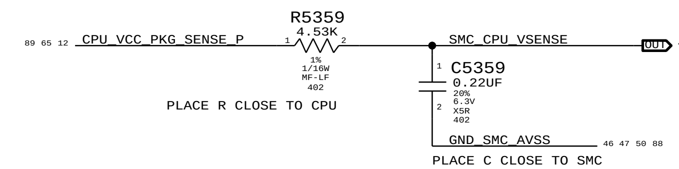
CPU 1.5V CURRENT SENSE

CPU 1.5V VOLTAGE SENSE

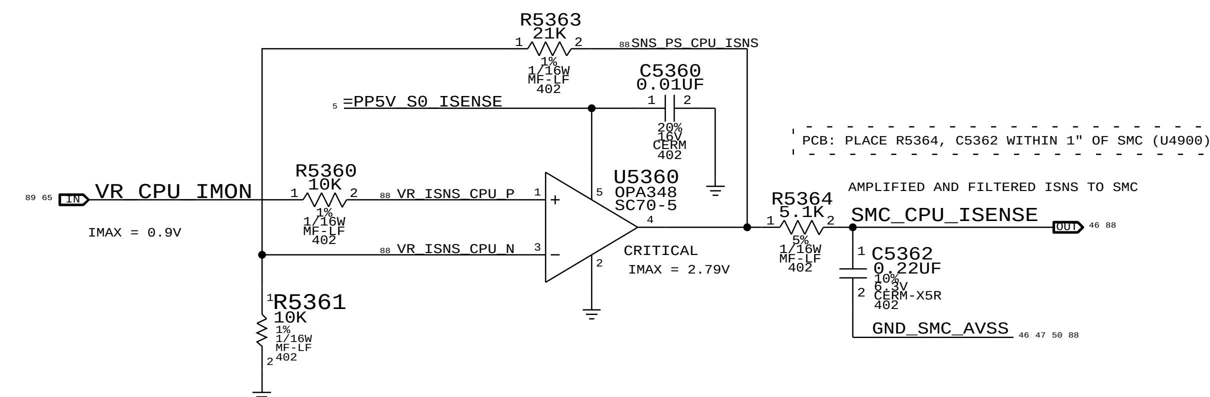


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5300	CPU_1V5_SENSE
101S0414	1	RES,0 OHM,1206,20 MILLIOHM MAX	R5300	PRODUCTION
132S0080	2	CAP,0.22UF,402	C5301,C5302	CPU_1V5_SENSE
116S0004	2	RES,0 OHM,402	C5301,C5302	PRODUCTION

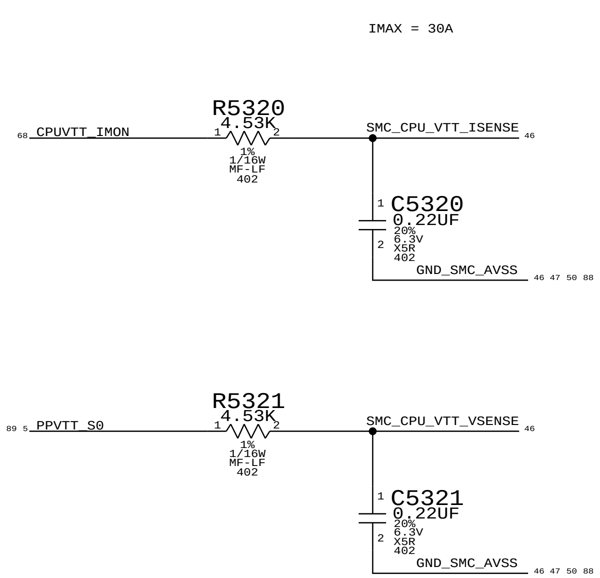
CPU Voltage Sense / Filter



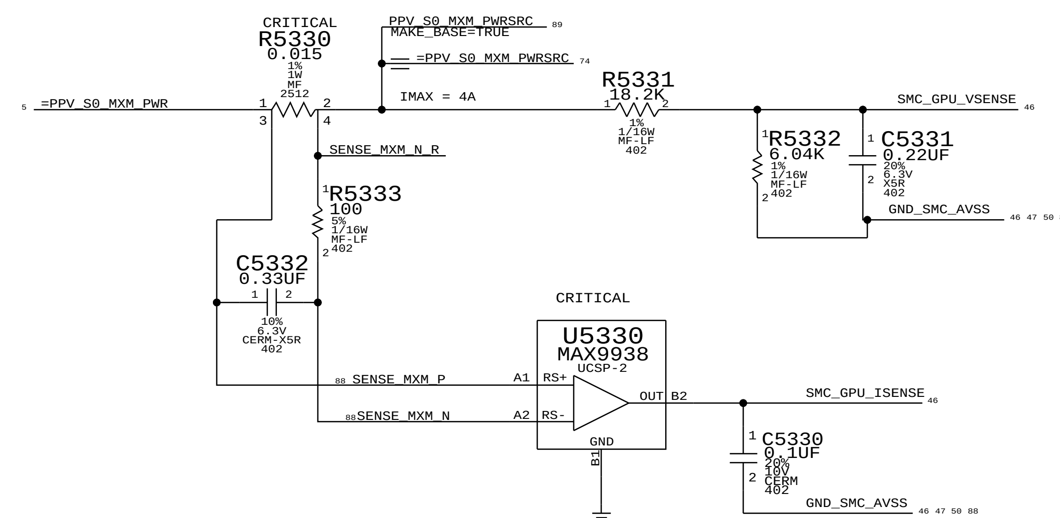
CPU CURRENT SENSE AMP & FILTER



CPU VTT CURRENT SENSE



MXM PWRSRC CURRENT & VOLTAGE SENSE



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
CPU/GPU POWER SENSE

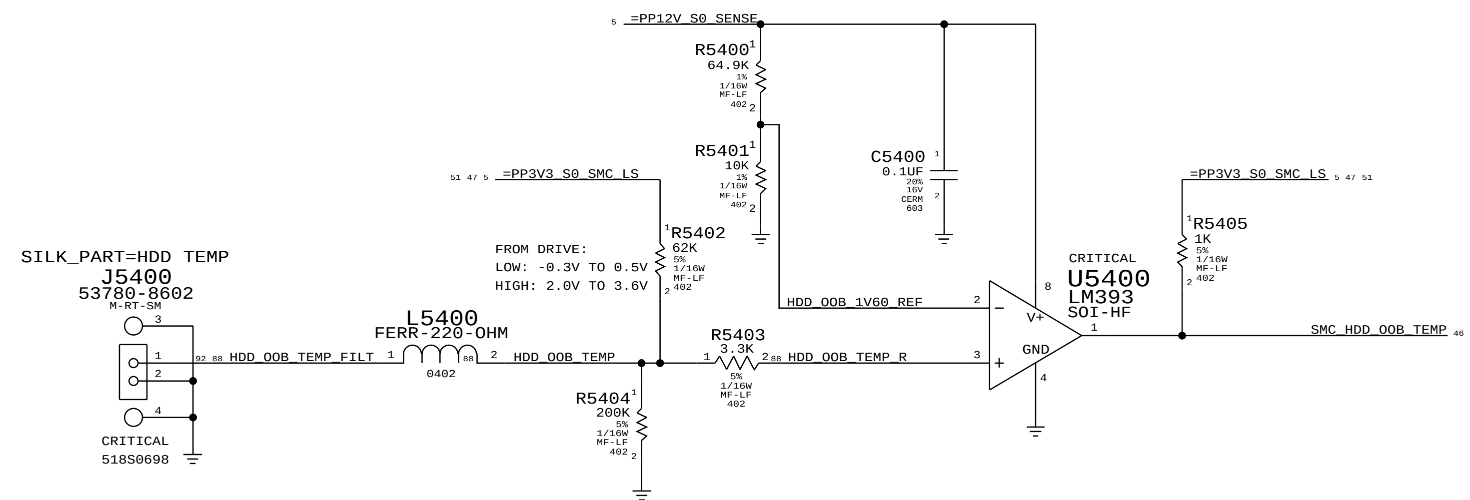
Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

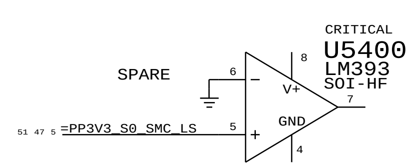
HDD OUT OF BAND TEMPERATURE SENSING LEVEL SHIFTING



Drive active = valid signal protocol
Drive asleep = HDD drives HDD_OOB_TEMP low
Drive disconnected = pulled high

Cannot pull low because some drives use this bit to determine 1.5 Gbps vs. 3.0 Gbps SATA

Must pull high to 2.5V for compatibility with all drives



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

HDD TEMP SENSE

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

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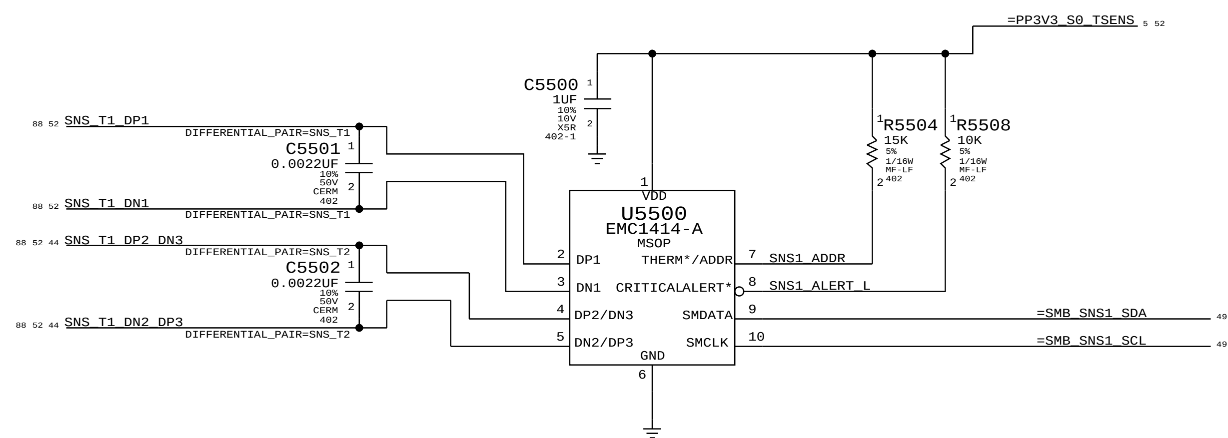
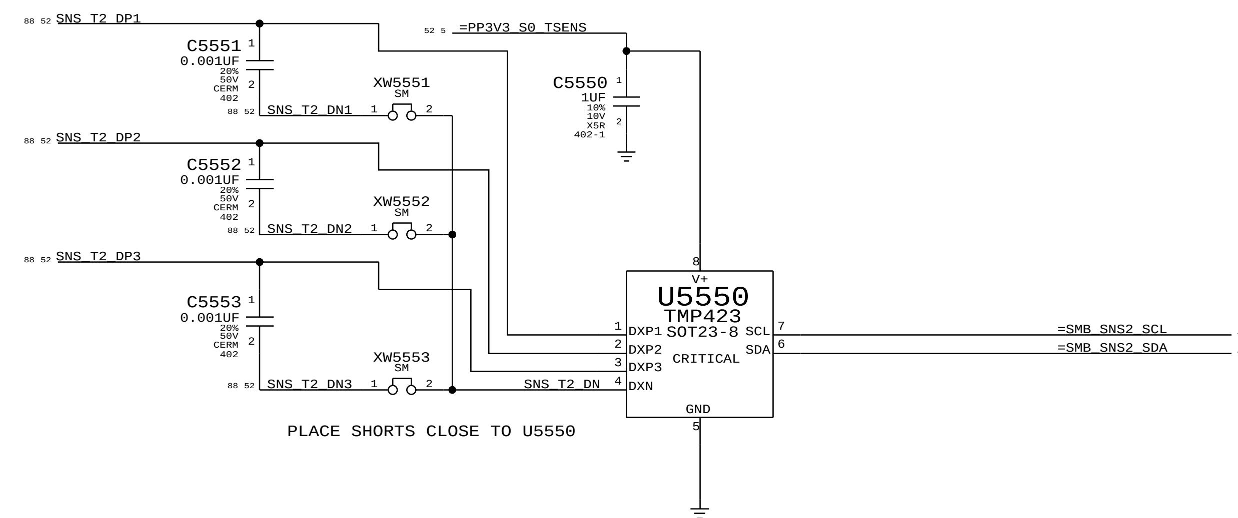
REMOTE HEATSINK SENSORS

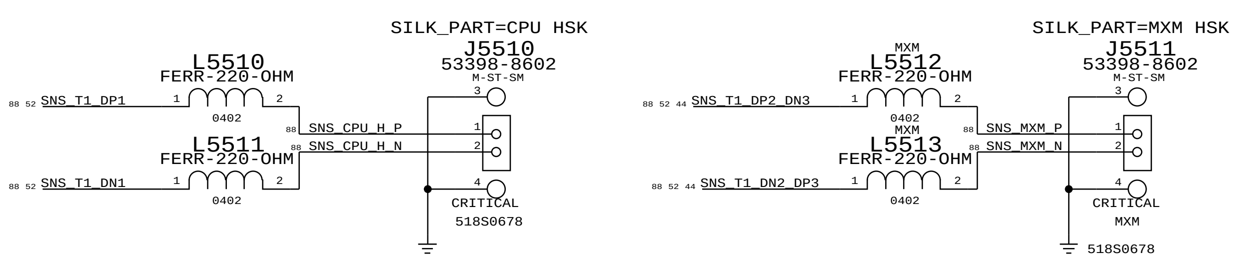


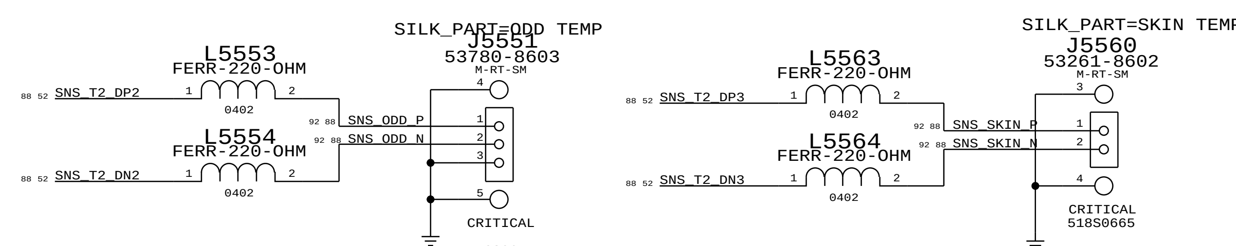
REMOTE SKIN & ODD THERMAL SENSORS



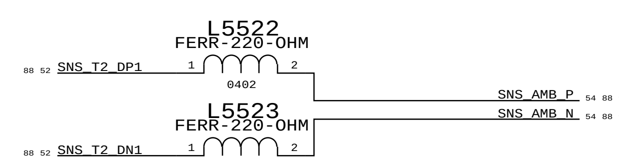
PLACE SHORTS CLOSE TO U5550

PLACE HSK SENSOR CONN. TOP SIDE NEAR MXM OR CPU





AMBIENT SENSE CONNECTOR COMBINED WITH CPU FAN



SYNC_MASTER=NICK SYNC_DATE=11/06/2009

PAGE TITLE

REMOTE TEMP/POWER SENSORS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

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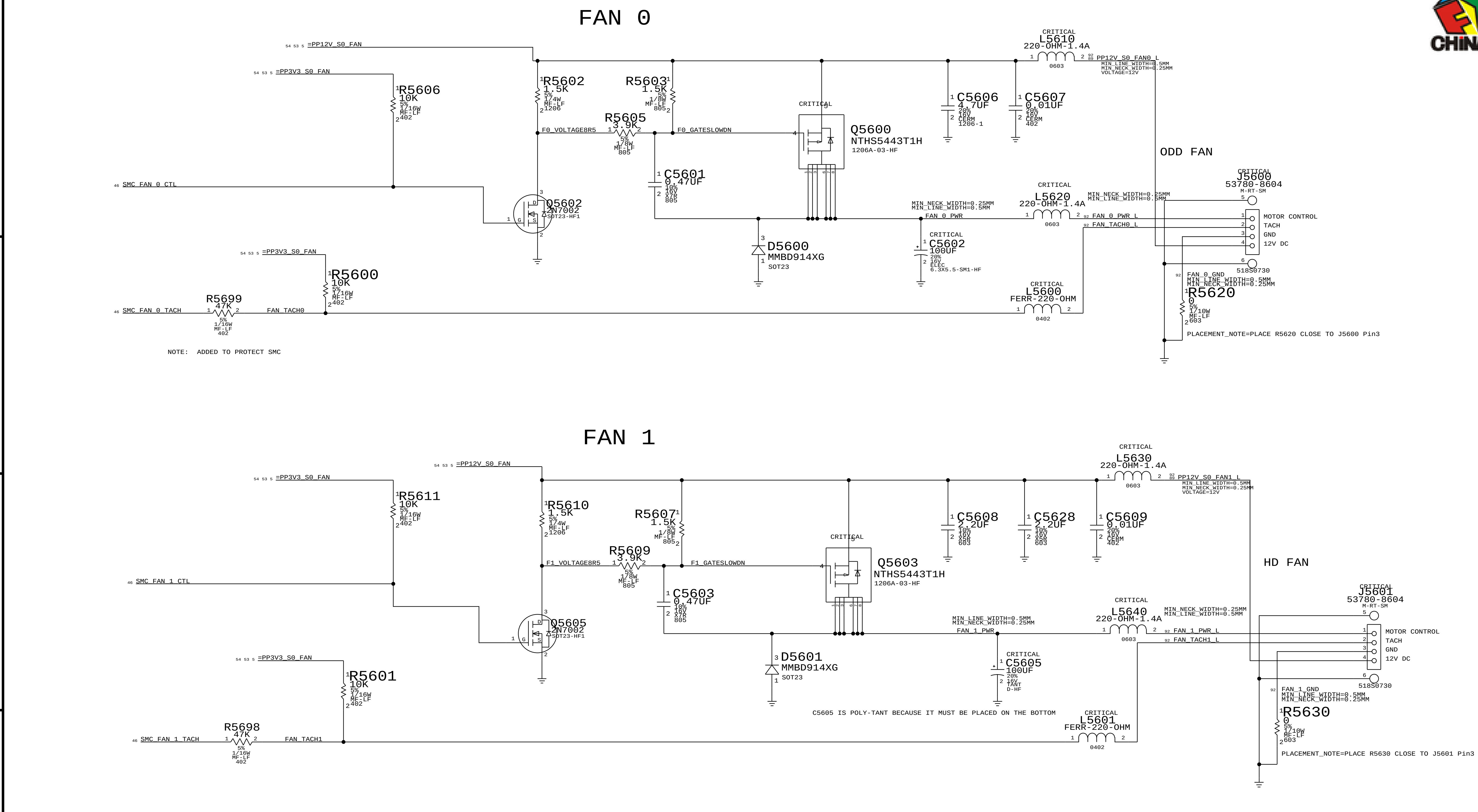


SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: HD AND OD FAN

Apple Inc.

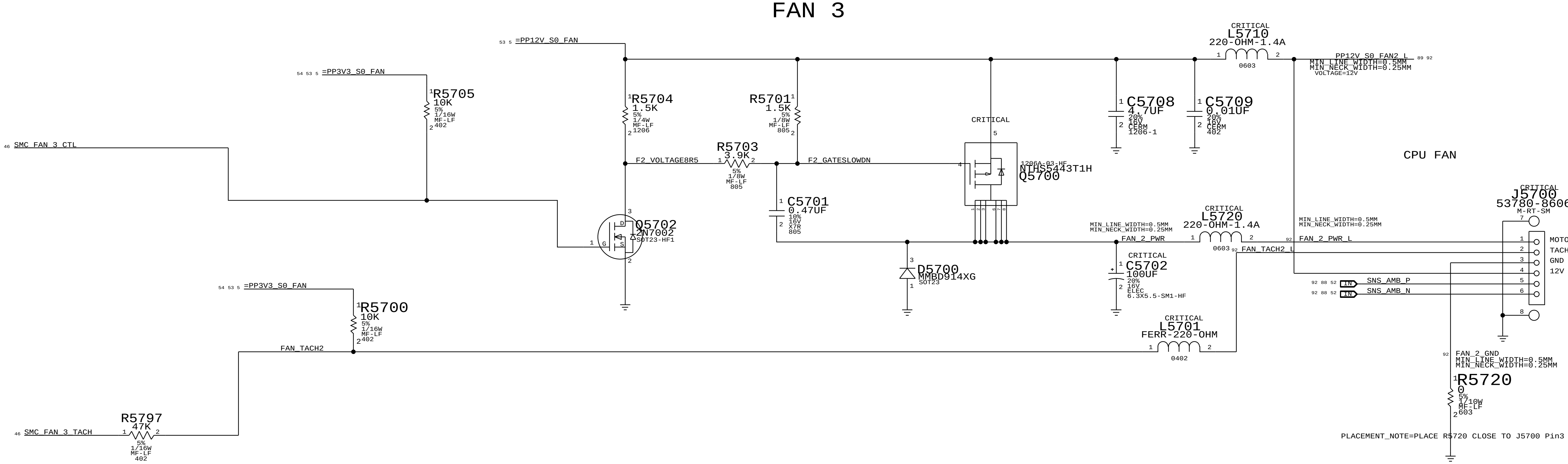
DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

BRANCH

FAN 2 UNUSED

FAN 3



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

CPU FAN & AMBIENT SENSE

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

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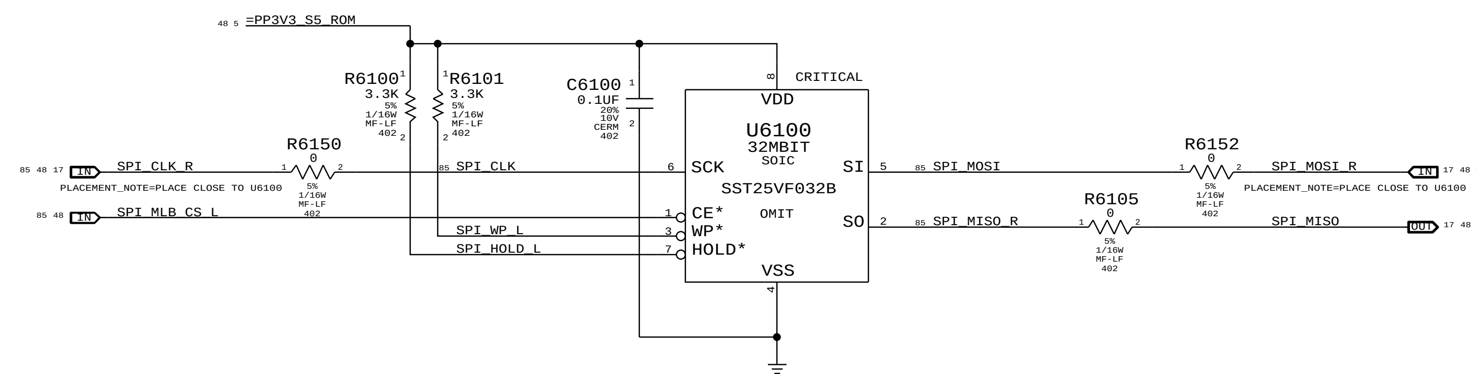

SYNC_MASTER=K23F SYNC_DATE=11/30/2009

PAGE TITLE

SPI ROM

Apple Inc.

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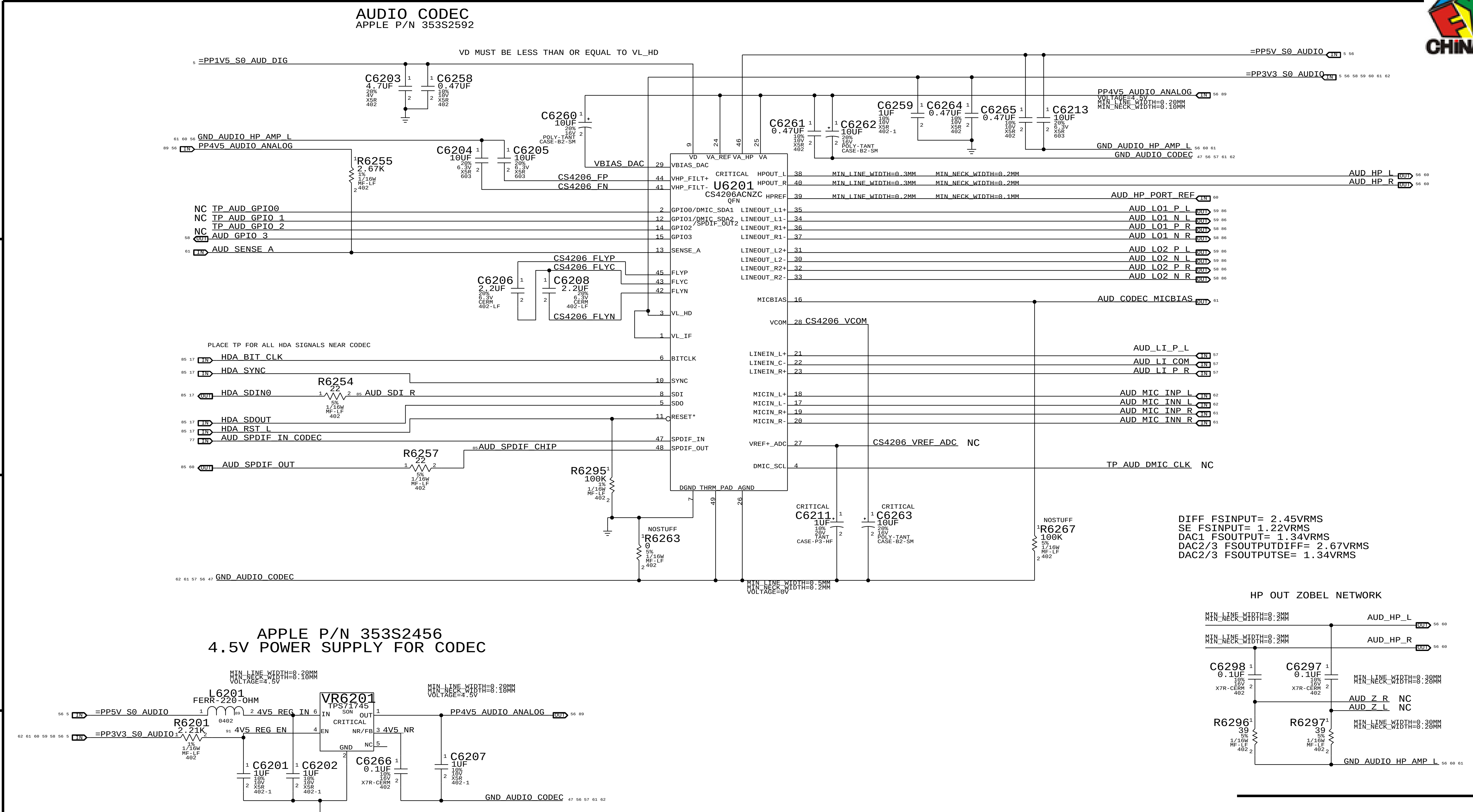
PAGE 61 OF 110

SHEET 55 OF 92



SYNC_MASTER=BREECE SYNC_DATE=02/02/2010

PAGE TITLE

AUDIO: CODEC/REGULATOR

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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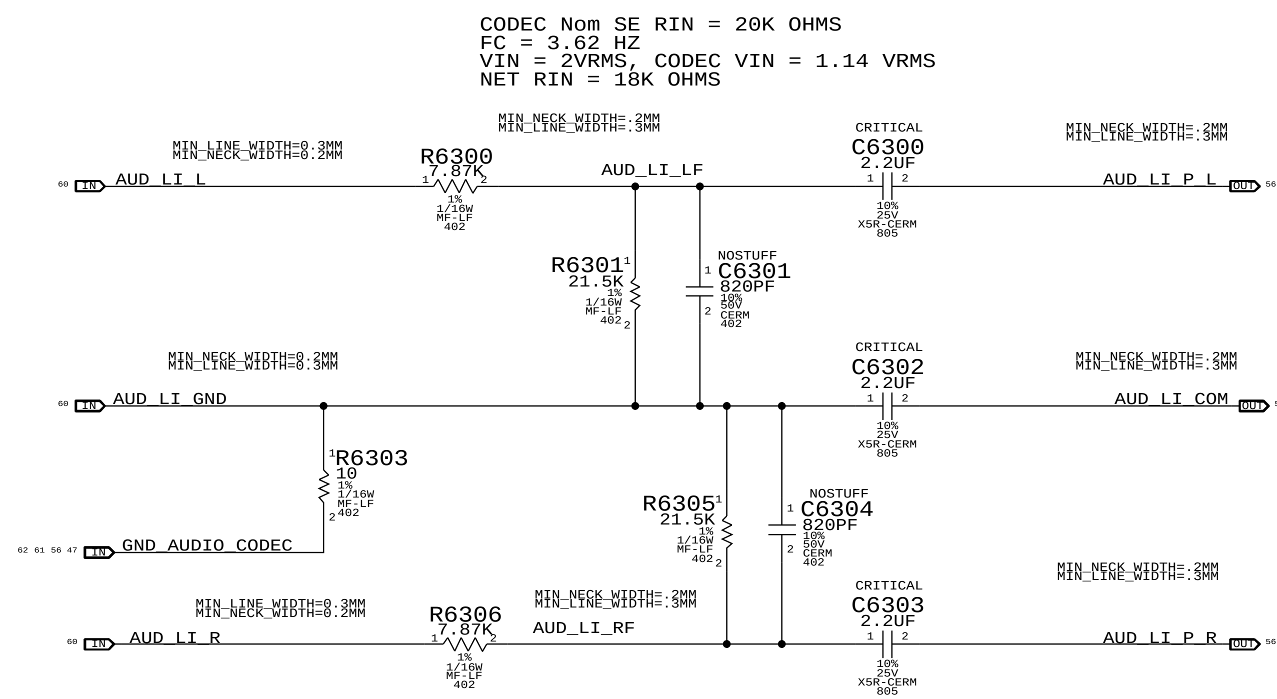


SYNC_MASTER=BREECE SYNC_DATE=02/02/2010

PAGE TITLE

AUDIO: FILTER/BUFFER

DRAWING NUMBER 051-8337 SIZE D

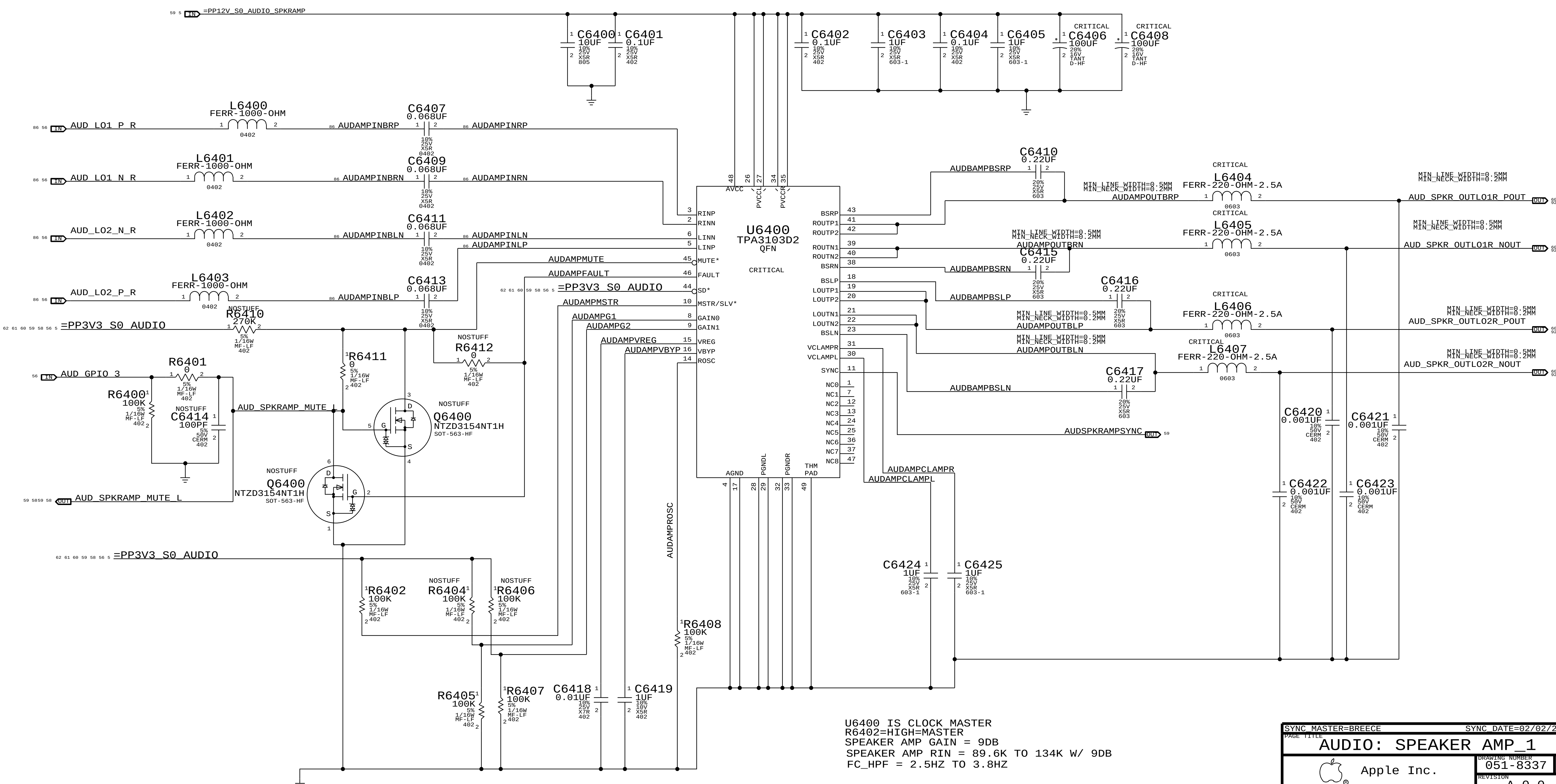
Apple Inc.

REVISION A.0.0

BRANCH

RIGHT CH. SPEAKER AMP

APPLE P/N 353S2768



U6400 IS CLOCK MASTER
R6402=HIGH=MASTER
SPEAKER AMP GAIN = 9DB
SPEAKER AMP RIN = 89.6K TO 134K W/ 9DB
FC_HPF = 2.5HZ TO 3.8HZ

SYNC_MASTER=BREECE SYNC_DATE=02/02/2010

PAGE TITLE: AUDIO: SPEAKER AMP_1

Apple Inc.

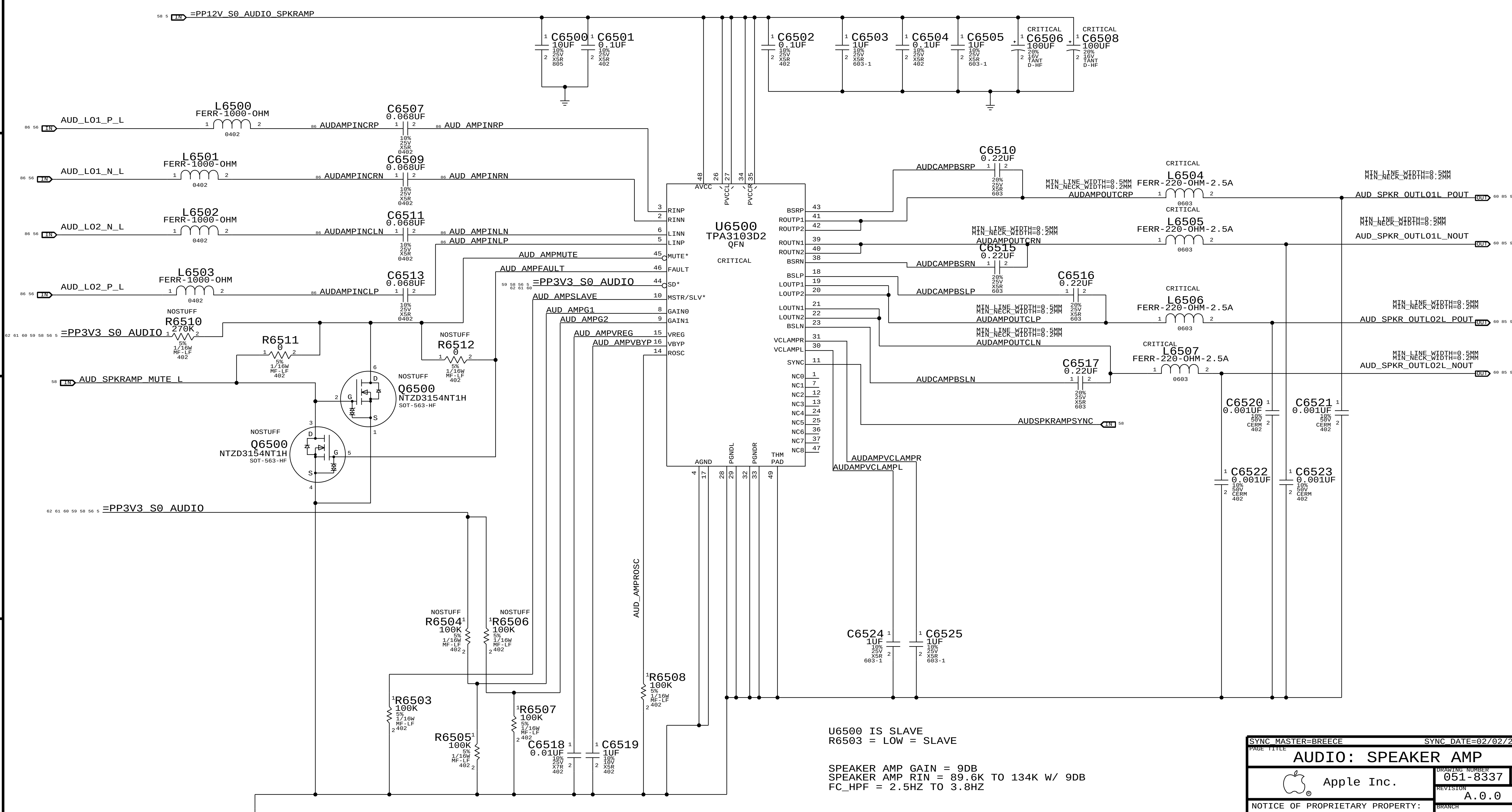
DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

BRANCH

LEFT CH. SPEAKER AMP

APPLE P/N 353S2768



U6500 IS SLAVE
R6503 = LOW = SLAVE

SPEAKER AMP GAIN = 9DB
SPEAKER AMP RIN = 89.6K TO 134K W/ 9DB
FC_HPF = 2.5HZ TO 3.8HZ

SYNC_MASTER=BREECE SYNC_DATE=02/02/2010

PAGE TITLE
AUDIO: SPEAKER AMP

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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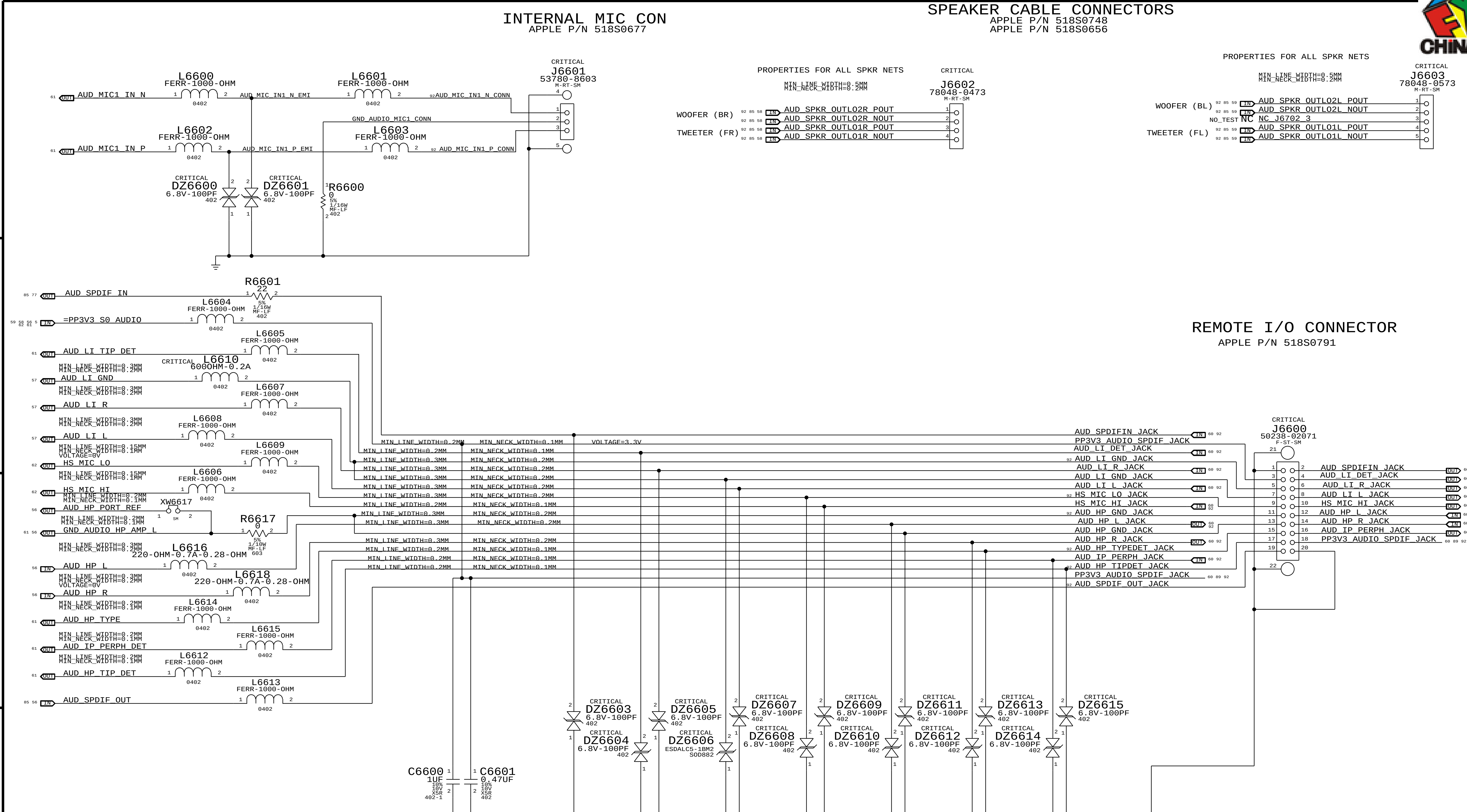

SYNC_MASTER=BREECE SYNC_DATE=02/02/2010

PAGE TITLE
Audio: MLB to I/O Conn.

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

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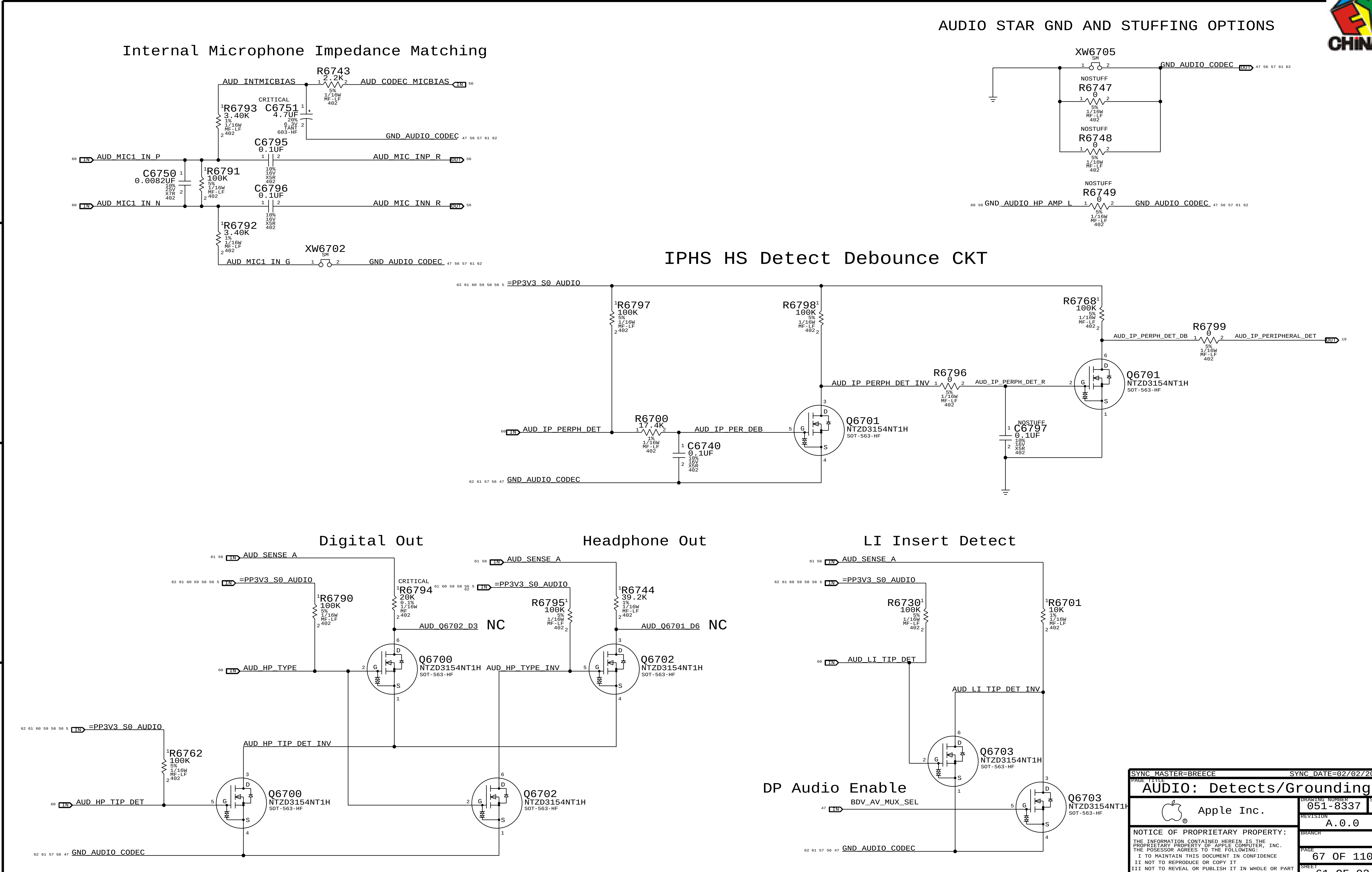
SHEET 60 OF 92



SYNC_MASTER=BREECE SYNC_DATE=02/02/2010

PAGE TITLE

AUDIO: Detects/Grounding

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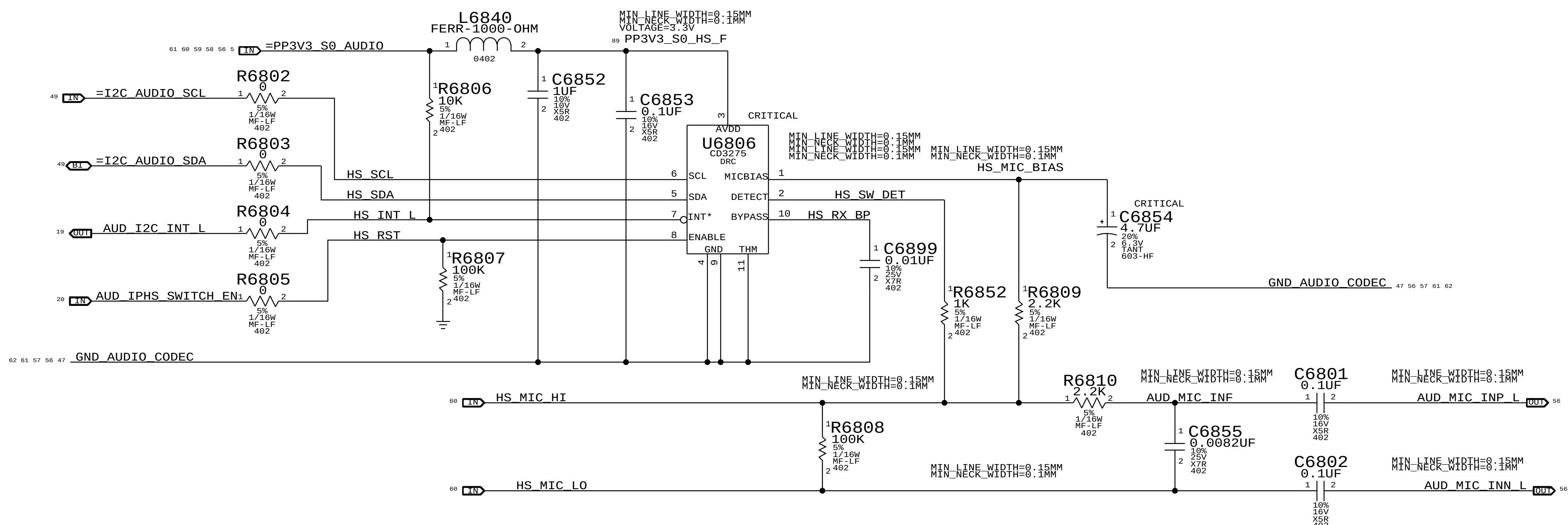
NOTICE OF PROPRIETARY PROPERTY:

FUNCTION	PIN	CONVERTER	VOLUME	ENABLE/ CNTRL TYPE	DETECT/INTERRUPT
PRIMARY	0X0B	0X04	0X04	GPIO 3	N/A
SECONDARY	0X0A	0X03	0X03	GPIO 3	N/A
HEADPHONES	0X09	0X02	0X02	N/A	0X09 (A)
LINE INPUT	0X0C	0X05	0X05	N/A	LINE IN
BUILT-IN MICROPHONE	0X0D(13,B,RIGHT)	0X06	0X06	MICBIAS 80%	N/A
HEADSET MICROPHONE	0X0D (13,V22,B,LEFT)	0X06	0X06	MIKEY	MIKEY
SPDIF OUT	0X10	0X08	N/A	N/A	0X0D (B)
SPDIF IN	0X0F	0X07	N/A	N/A	N/A
MIKEY	N/A	N/A	N/A	MCP GPIO_38	MCP GPIO_5

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256



FLP = 8.82 KHZ
FHP = 80 HZ

SYNC_MASTER=BREECE SYNC_DATE=02/02/2010

PAGE TITLE
AUDIO: Mikey

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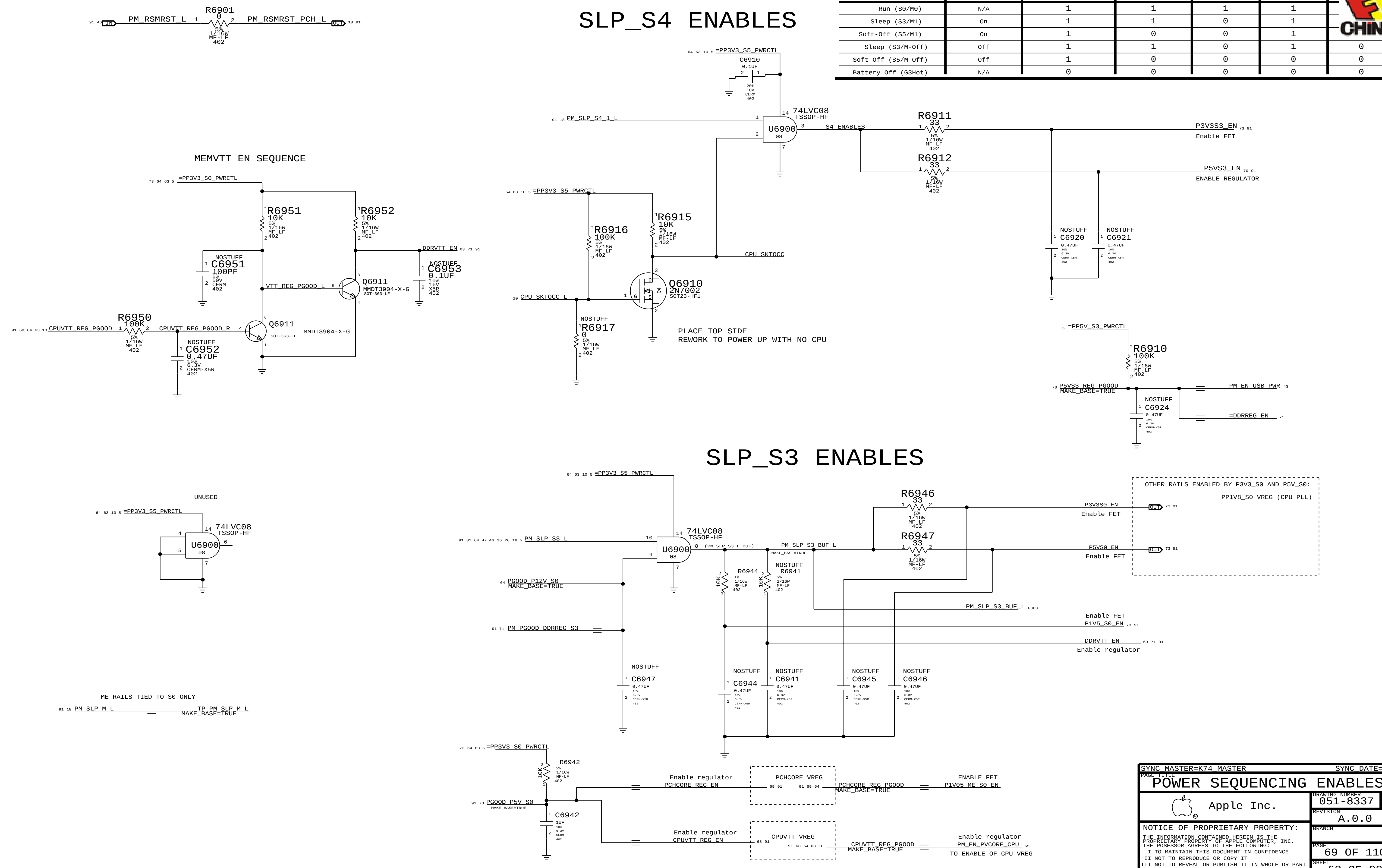
SLP_S4 ENABLES

State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	
Run (S0/M0)	N/A	1	1	1	1	
Sleep (S3/M1)	On	1	1	0	1	
Soft-Off (S5/M1)	On	1	0	0	1	
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: POWER SEQUENCING ENABLES

Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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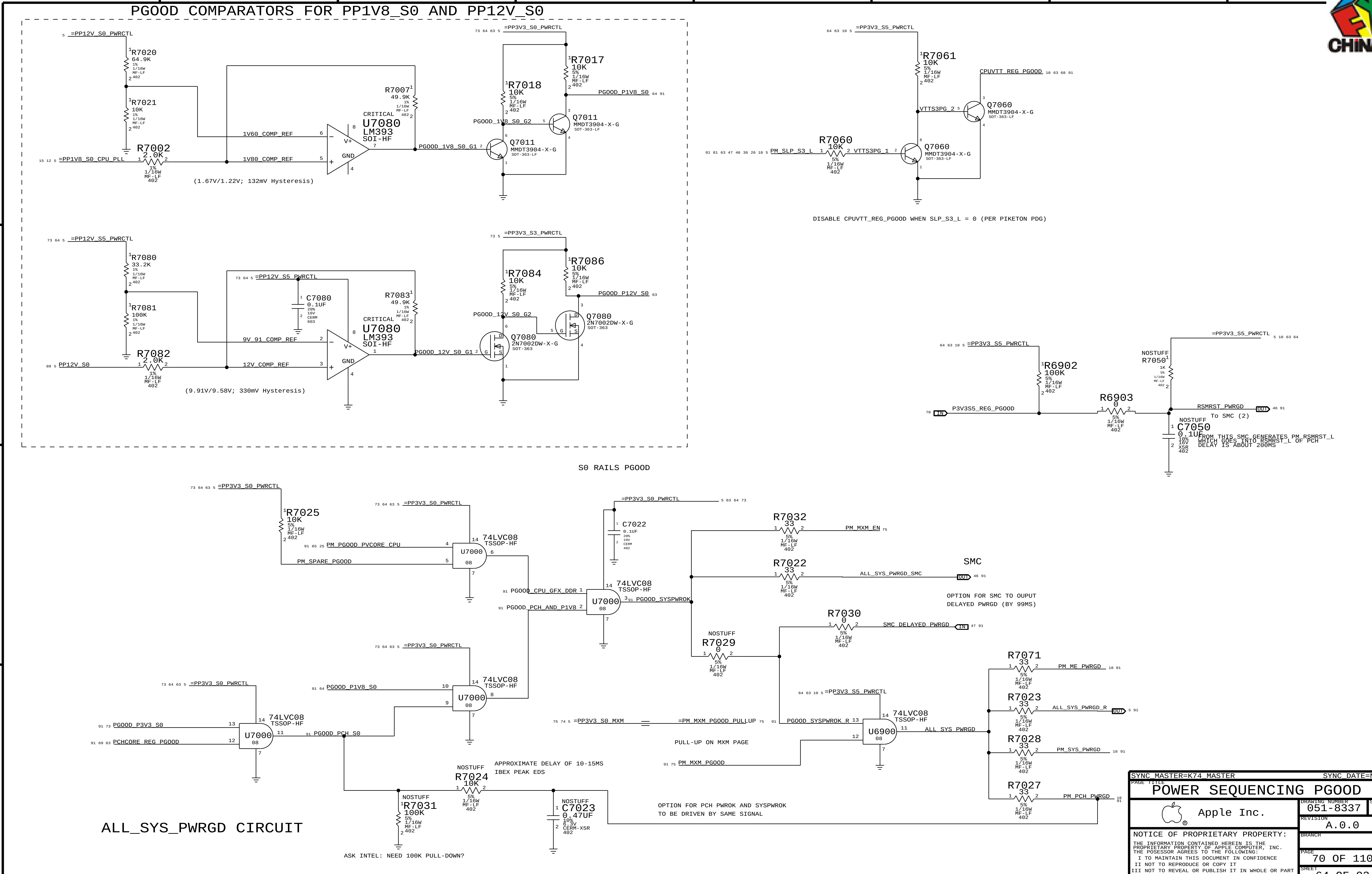

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

POWER SEQUENCING PGOOD

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

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BRANCH

NOTICE OF PROPRIETARY PROPERTY:

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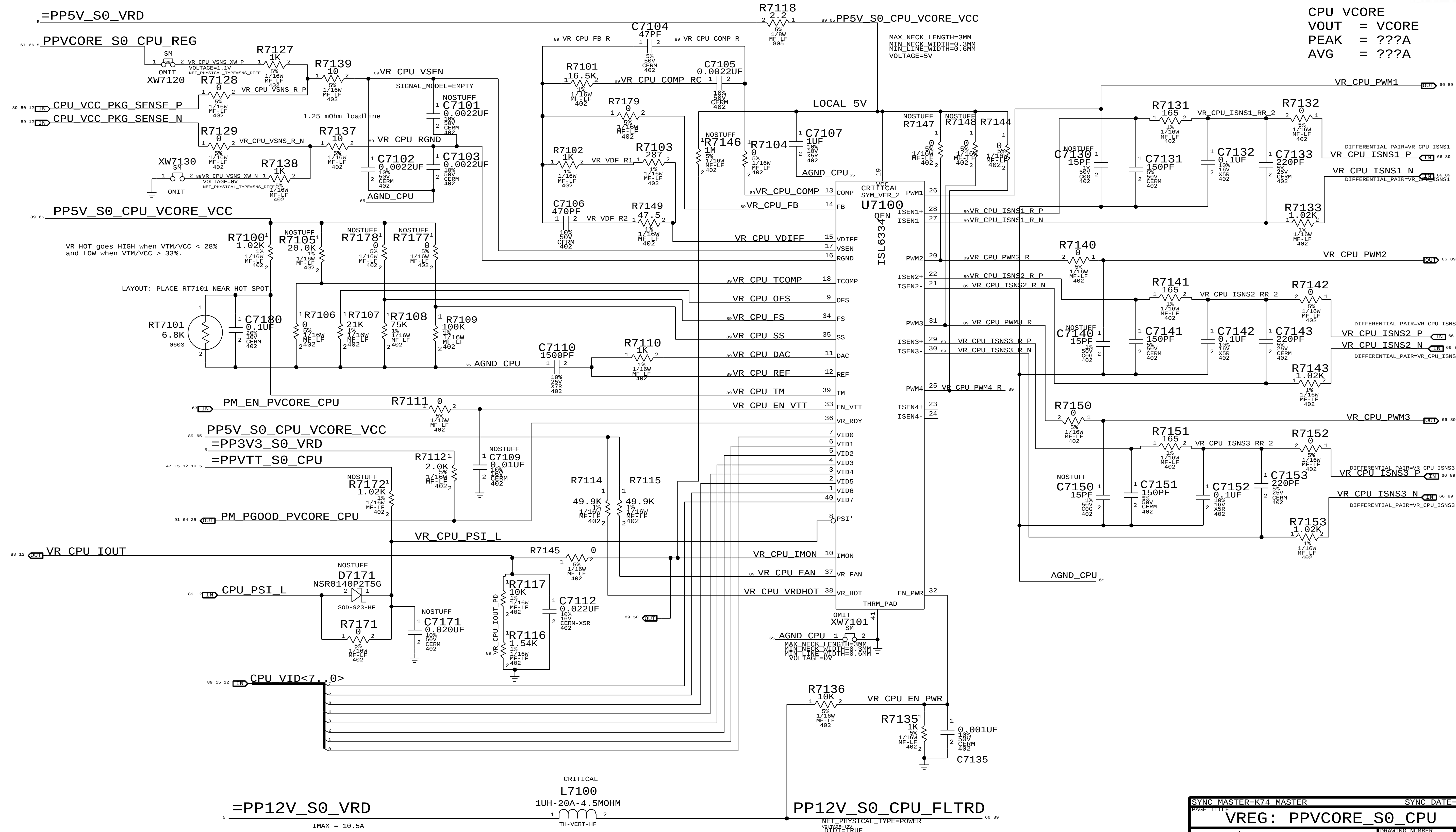
CPU CORE REG 1.1V/???A O/P= PPVCORE_S0_CPU_REG

CPU VCORE
VOUT = VCORE
PEAK = ???A
AVG = ???A



CPU CORE INPUT Filtering

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
VREG: PPVCORE_S0_CPU

Apple Inc.

DRAWING NUMBER	SIZE
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BRANCH	
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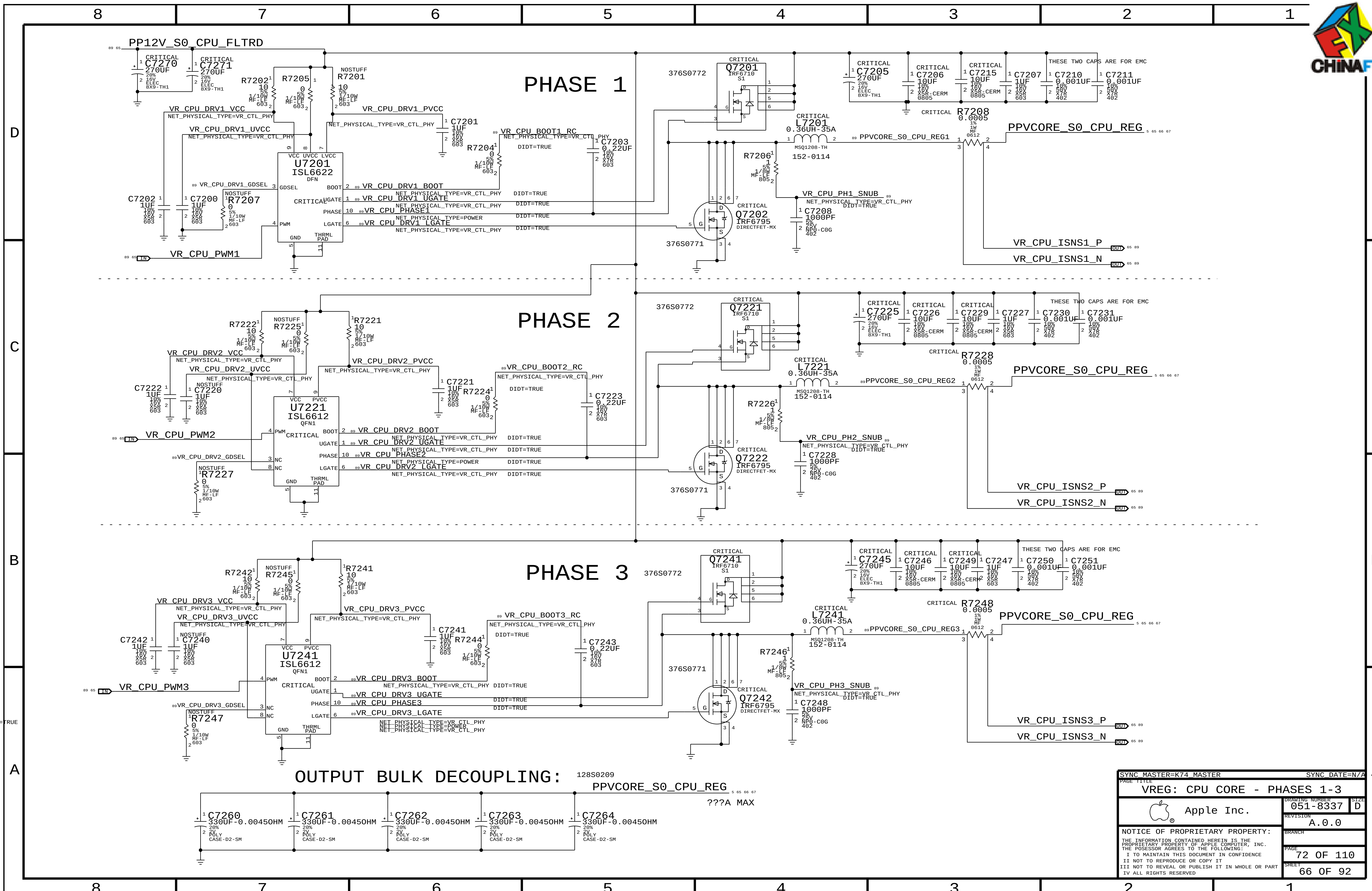
SHEET 65 OF 92	


PP12V_S0_CPU_FLTRD
PHASE 1
PHASE 2
PHASE 3
U7201
ISL6622
U7221
ISL6612
U7241
ISL6612
Q7201
IRF6710
Q7202
IRF6795
Q7221
Q7222
Q7241
Q7242
L7201
0.36UH-35A
L7221
L7241
R7208
R7228
R7248
PPVCORE_S0_CPU_REG
PPVCORE_S0_CPU_REG1
PPVCORE_S0_CPU_REG2
PPVCORE_S0_CPU_REG3
VR_CPU_PWM1
VR_CPU_PWM2
VR_CPU_PWM3
VR_CPU_DRV1_VCC
VR_CPU_DRV1_UVCC
VR_CPU_DRV1_PVCC
VR_CPU_DRV1_BOOT
VR_CPU_DRV1_UGATE
VR_CPU_PHASE1
VR_CPU_DRV1_LGATE
VR_CPU_BOOT1_RC
VR_CPU_PH1_SNUB
VR_CPU_ISNS1_P
VR_CPU_ISNS1_N
VR_CPU_ISNS2_P
VR_CPU_ISNS2_N
VR_CPU_ISNS3_P
VR_CPU_ISNS3_N
THESE TWO CAPS ARE FOR EMC
OUTPUT BULK DECOUPLING:
128S0209
???A MAX
C7260
C7261
C7262
C7263
C7264
330UF-0.0045OHM
SYNC_MASTER=K74_MASTER
SYNC_DATE=N/A
VREG: CPU CORE - PHASES 1-3
Apple Inc.
051-8337
D
A.0.0
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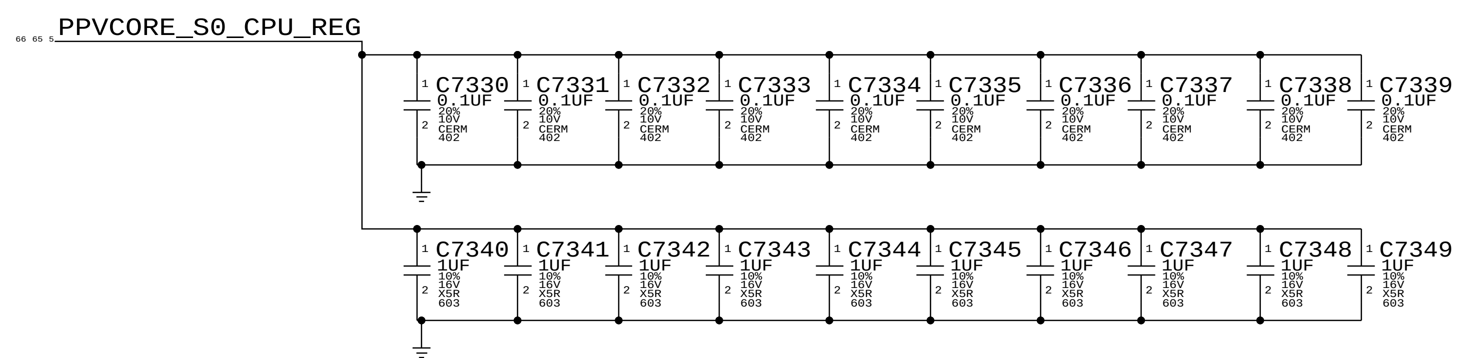

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

VREG: CPU CORE - CAPS

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DRAWING NUMBER 051-8337 SIZE D

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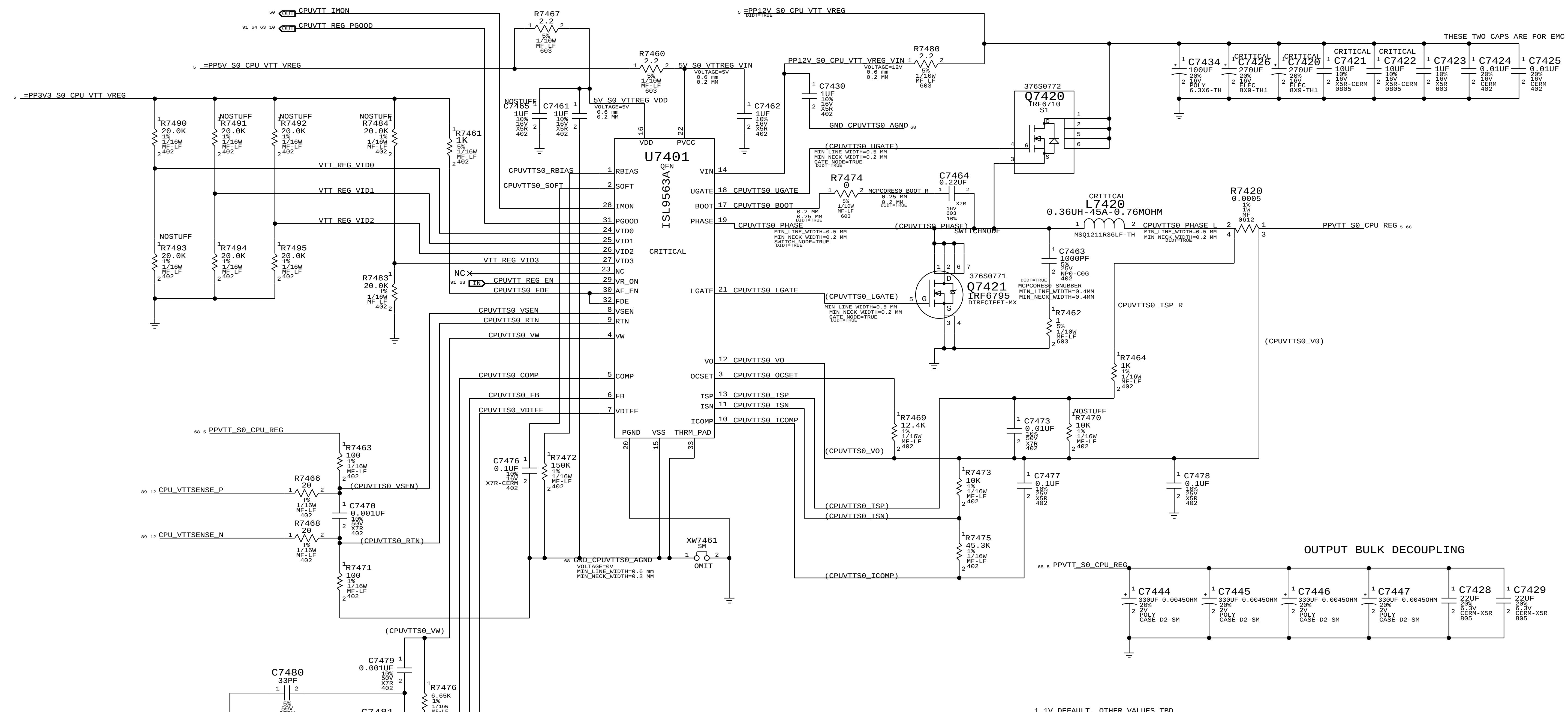
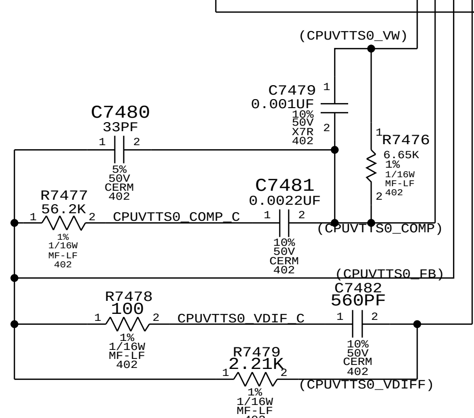
CPU VTT REG 1.1V O/P= PPVTT_S0_CPU_REG





VID<2:0>	Voltage
000	+1.100V

SYNC_MASTER=NICK SYNC_DATE=12/08/2009

PAGE TITLE: CPU VTT REGULATOR

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DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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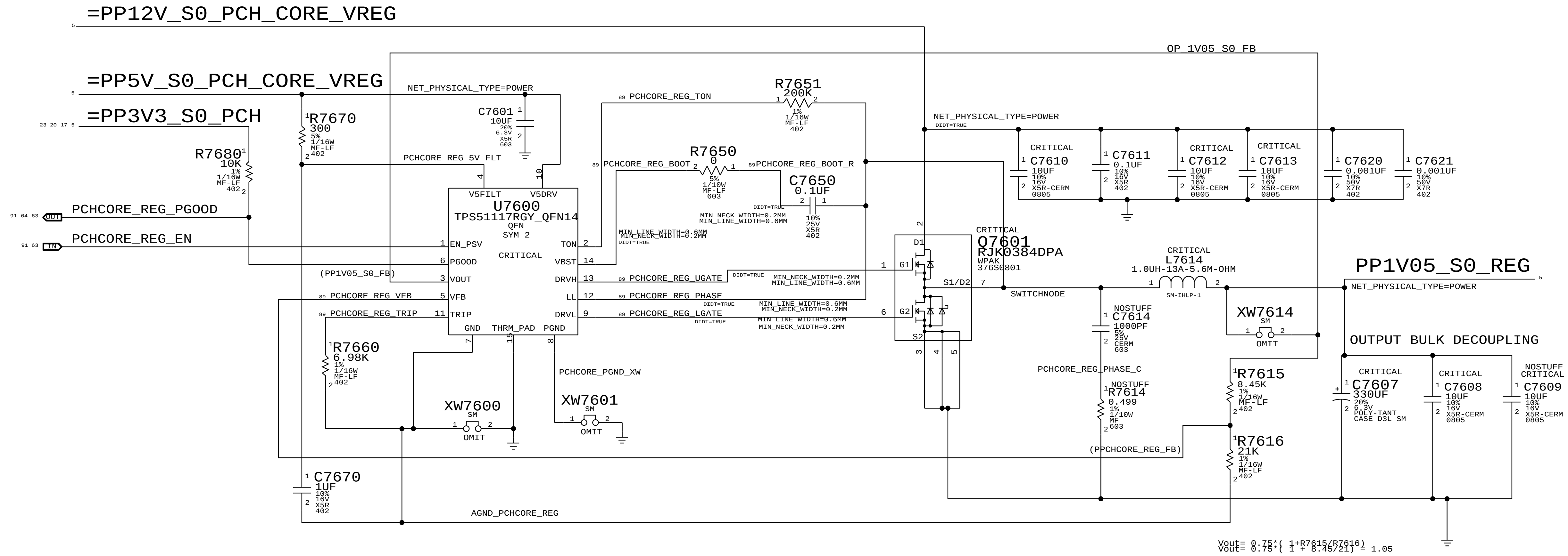
IBEX PEAK CORE REG 1.05V OUTPUT = PP1V05_S0_REG

PP1V05_S0_REG
VOUT = 1.05V
PEAK = 7.5A
AVG = 3A



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: IBEX PEAK CORE

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DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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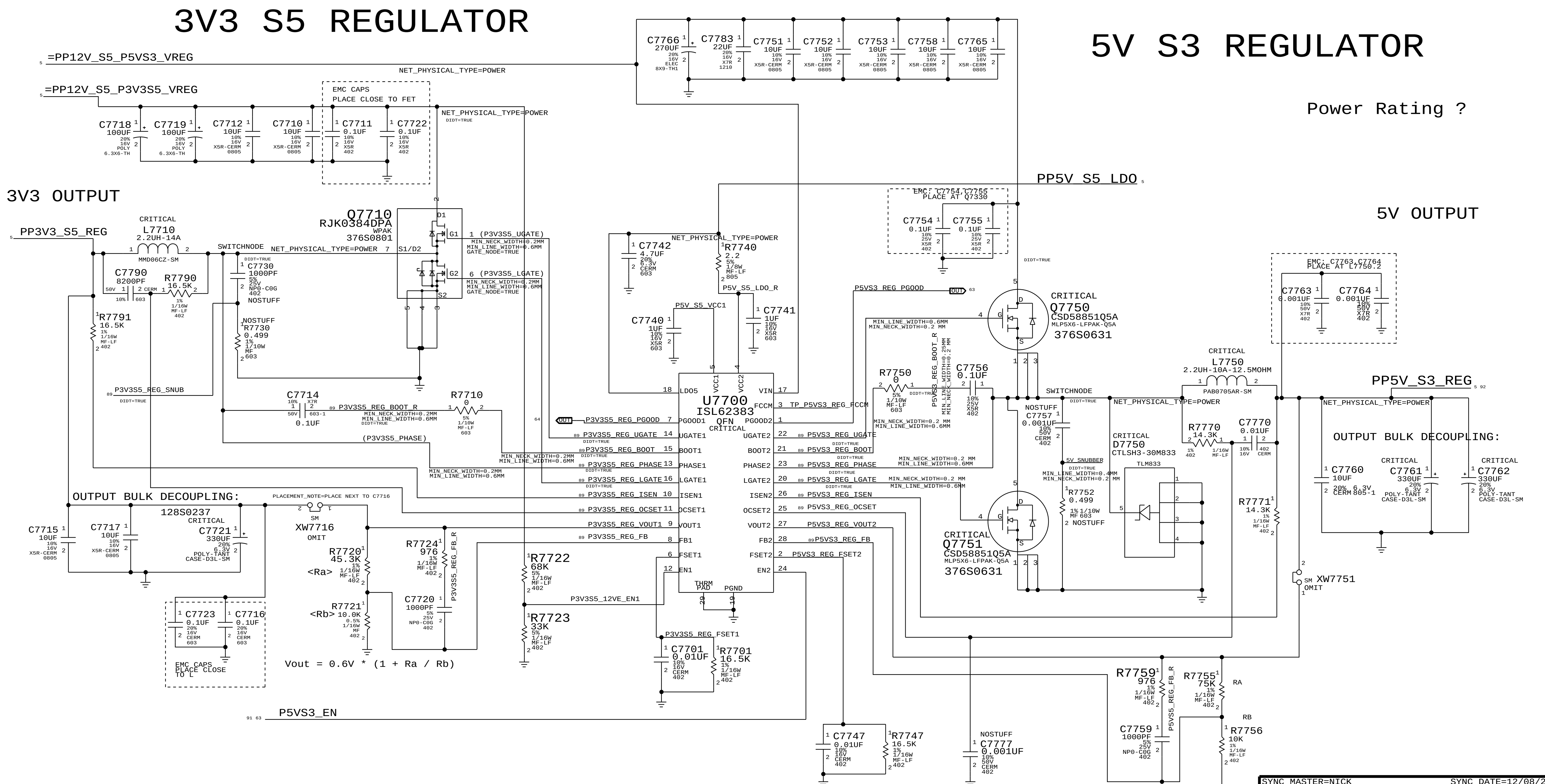
3V3 S5 REGULATOR

5V S3 REGULATOR



SYNC_MASTER=NICK	SYNC_DATE=12/08/2009
PAGE TITLE	5V_S3 / 3V3_S5 VREGS
Apple Inc.	DRAWING NUMBER 051-8337 SIZE D
	REVISION A.0.0
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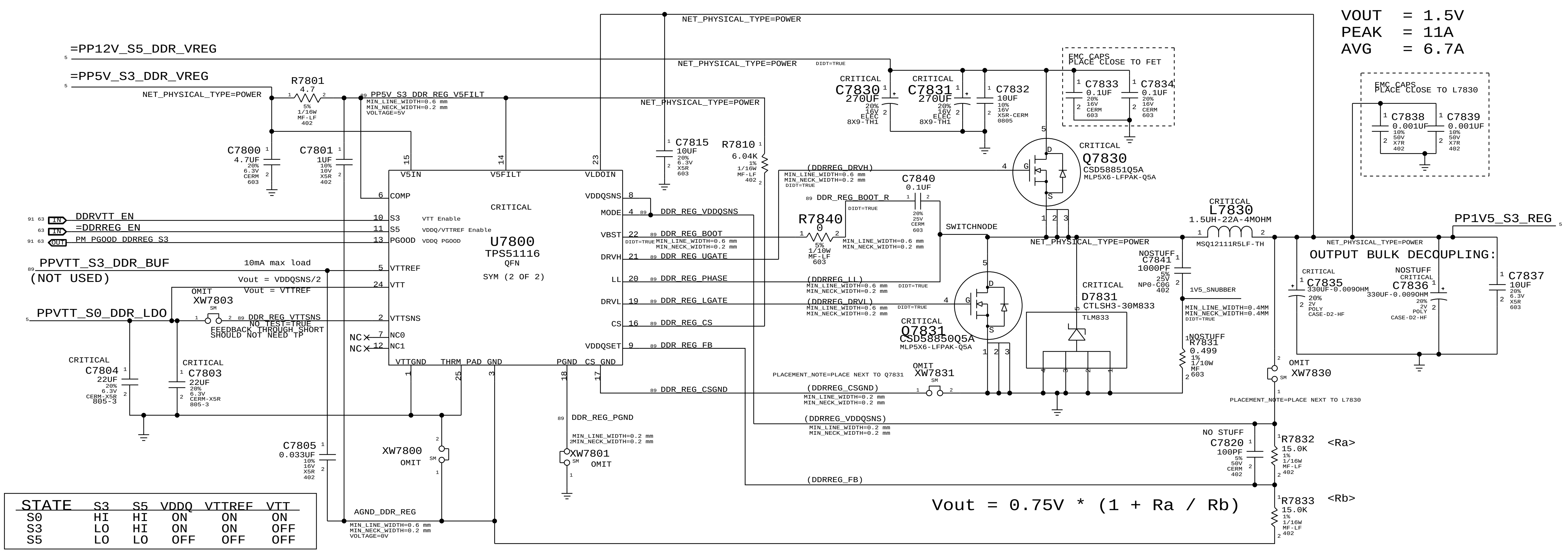
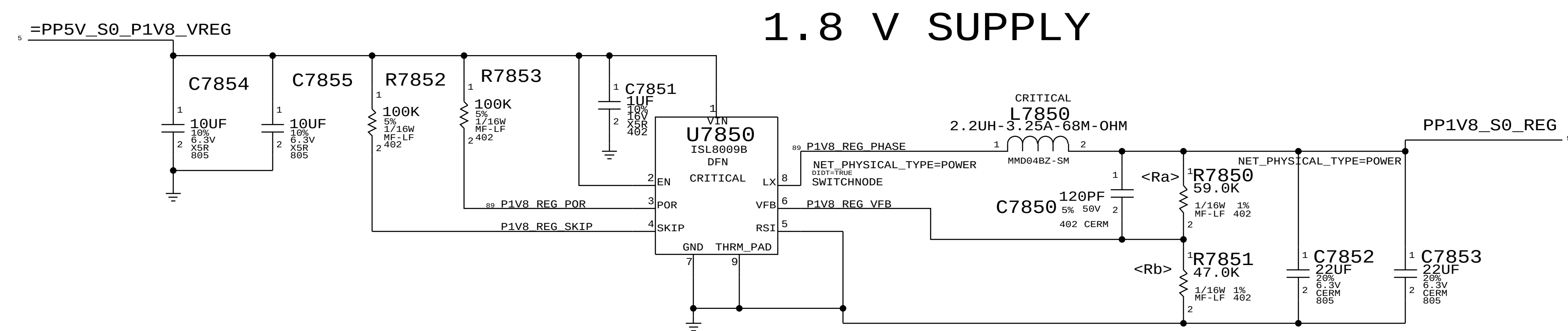
1.5 V DDR SUPPLY



STATE	S3	S5	VDDQ	VTTREF	VTT
S0	HI	HI	ON	ON	ON
S3	LO	HI	ON	ON	OFF
S5	LO	LO	OFF	OFF	OFF

1.8 V SUPPLY



Vo=0.8*(1+ Ra/Rb)

Vo=0.8*(1+ 59/47)=1.804V

SYNC_MASTER=K23F SYNC_DATE=11/30/2009

PAGE TITLE

1.5V / 1.8V VREGS

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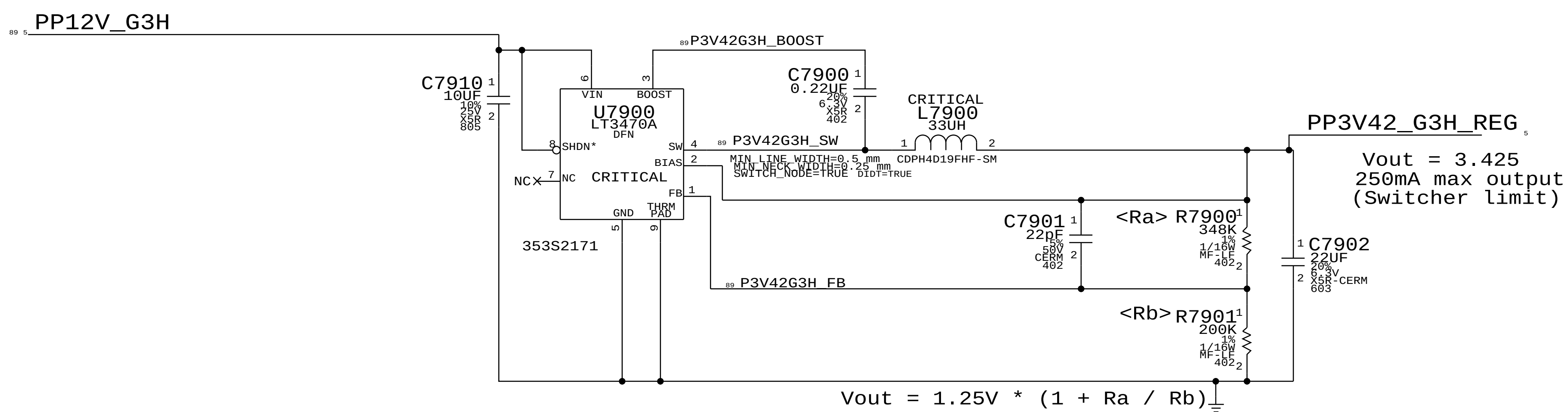
NOTICE OF PROPRIETARY PROPERTY:

3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: 3.42 G3HOT SUPPLY

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DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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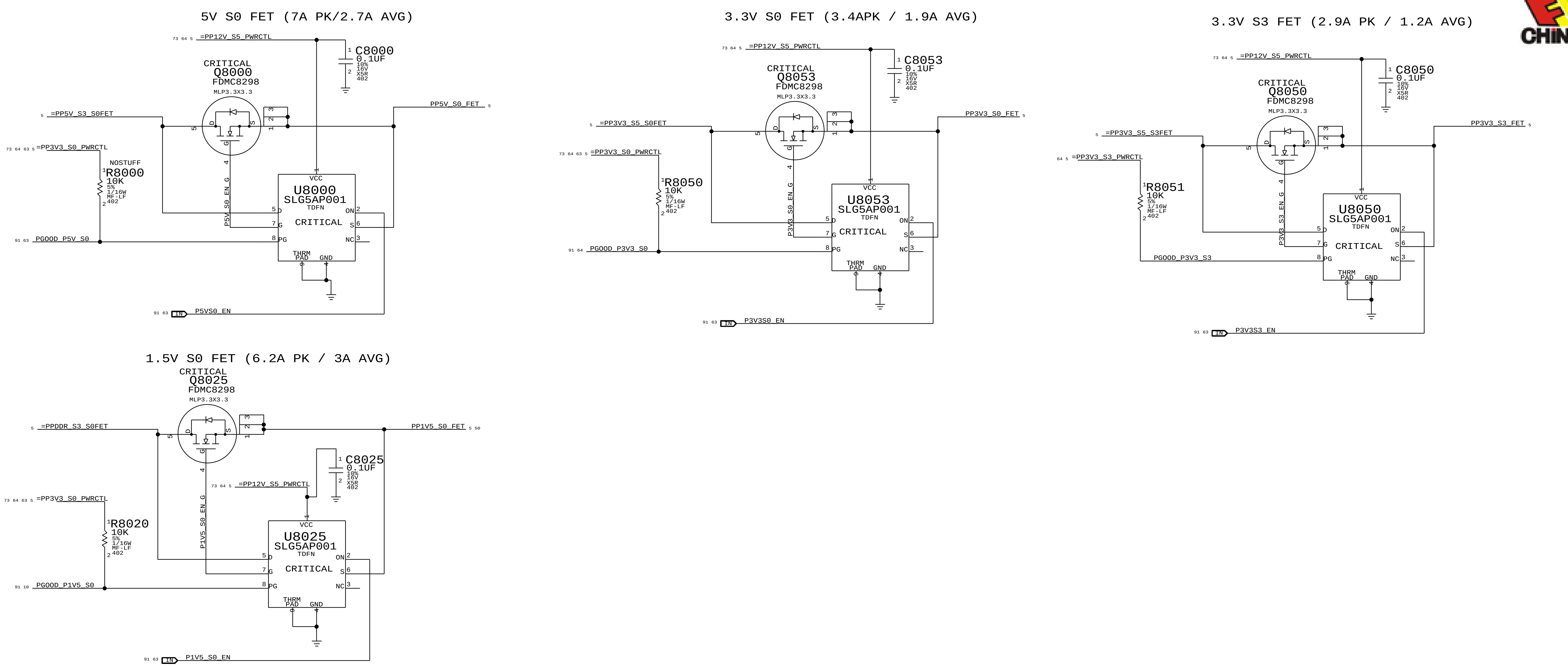
NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

S3+S0 FETS

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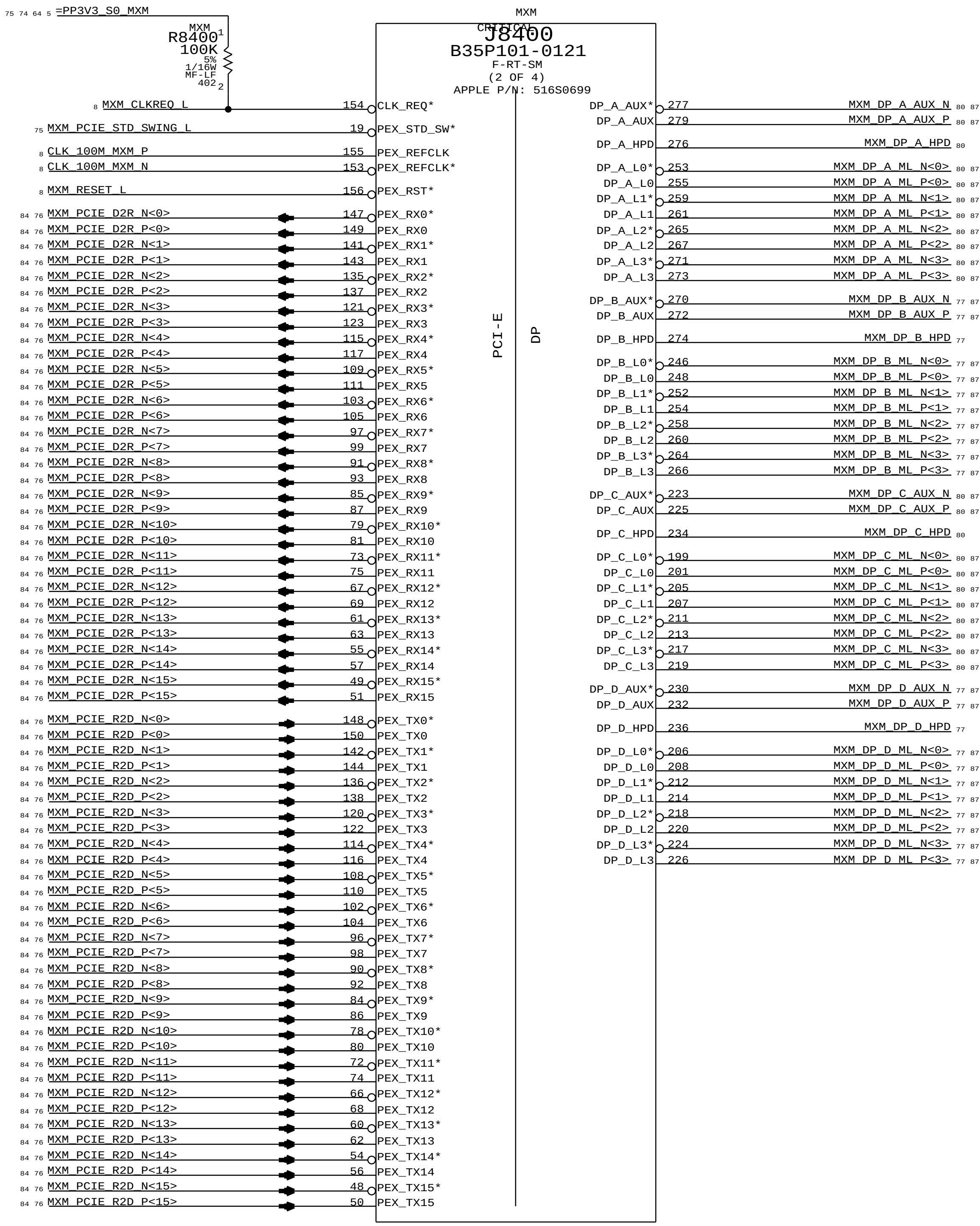
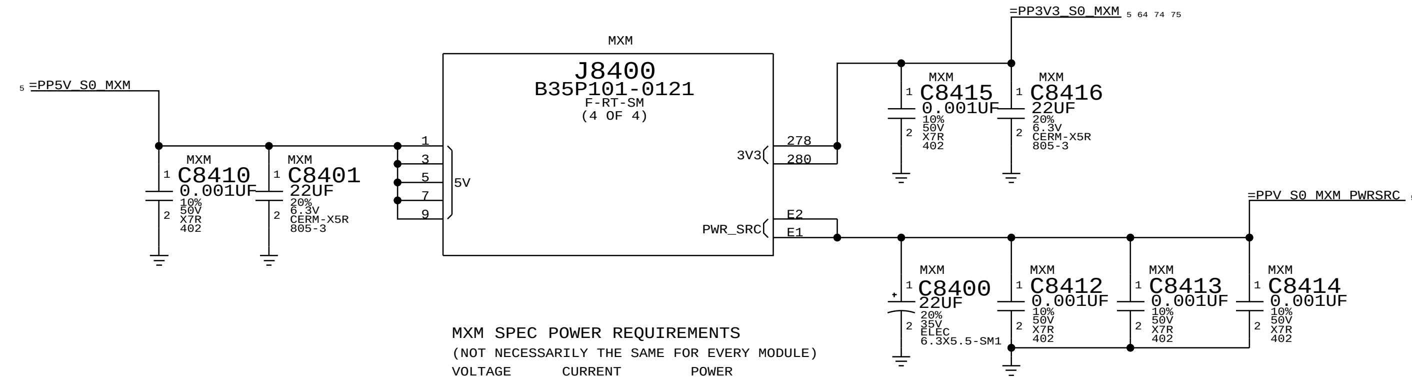
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
- MXM





MXM SPEC POWER REQUIREMENTS
(NOT NECESSARILY THE SAME FOR EVERY MODULE)

VOLTAGE	CURRENT	POWER
3V3	1.0 A	3.3 W
5V	2.5 A	12.5 W
PWR (7-20V)	UP TO 10 A	PLATFORM DEPENDENT

SYNC_MASTER=K23F SYNC_DATE=11/30/2009

PAGE TITLE
MXM PCIe, DP & Power

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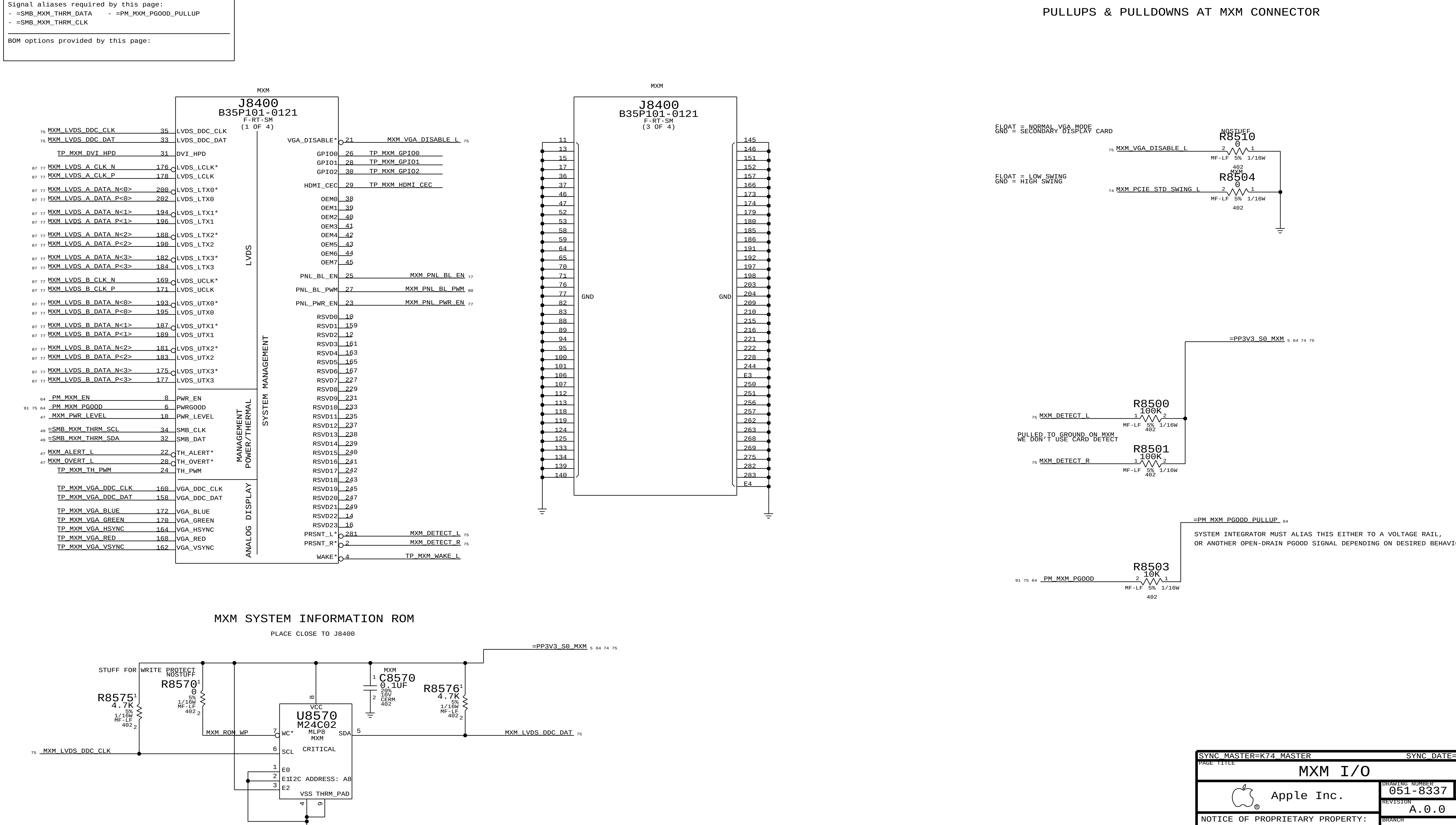
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM

Signal aliases required by this page:
- =SMB_MXM_THRM_DATA   - =PM_MXM_PGOOD_PULLUP
- =SMB_MXM_THRM_CLK

BOM options provided by this page:



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE: MXM I/O

Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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MXM TX CAPS

Input net	Ref	Value	Rating	Output net
PEG_R2D_C_P<0>	MXM C8600	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<15>
PEG_R2D_C_N<0>	MXM C8601	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<15>
PEG_R2D_C_N<1>	MXM C8602	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<14>
PEG_R2D_C_P<1>	MXM C8603	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<14>
PEG_R2D_C_N<2>	MXM C8604	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<13>
PEG_R2D_C_P<2>	MXM C8605	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<13>
PEG_R2D_C_P<3>	MXM C8606	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<12>
PEG_R2D_C_N<3>	MXM C8607	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<12>
PEG_R2D_C_N<4>	MXM C8608	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<11>
PEG_R2D_C_P<4>	MXM C8609	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<11>
PEG_R2D_C_N<5>	MXM C8610	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<10>
PEG_R2D_C_P<5>	MXM C8611	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<10>
PEG_R2D_C_P<6>	MXM C8612	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<9>
PEG_R2D_C_N<6>	MXM C8613	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<9>
PEG_R2D_C_N<7>	MXM C8614	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<8>
PEG_R2D_C_P<7>	MXM C8615	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<8>
PEG_R2D_C_P<8>	MXM C8616	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<7>
PEG_R2D_C_N<8>	MXM C8617	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<7>
PEG_R2D_C_P<9>	MXM C8618	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<6>
PEG_R2D_C_N<9>	MXM C8619	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<6>
PEG_R2D_C_N<10>	MXM C8620	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<5>
PEG_R2D_C_P<10>	MXM C8621	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<5>
PEG_R2D_C_N<11>	MXM C8622	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<4>
PEG_R2D_C_P<11>	MXM C8623	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<4>
PEG_R2D_C_P<12>	MXM C8624	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<3>
PEG_R2D_C_N<12>	MXM C8625	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<3>
PEG_R2D_C_N<13>	MXM C8626	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<2>
PEG_R2D_C_P<13>	MXM C8627	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<2>
PEG_R2D_C_P<14>	MXM C8628	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<1>
PEG_R2D_C_N<14>	MXM C8629	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<1>
PEG_R2D_C_N<15>	MXM C8630	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_P<0>
PEG_R2D_C_P<15>	MXM C8631	0.1UF	10% 16V X5R 402	MXM_PCIE_R2D_N<0>

MXM RX CAPS

Input net	Ref	Value	Rating	Output net
MXM_PCIE_D2R_P<15>	MXM C8632	0.1UF	10% 16V X5R 402	PEG_D2R_N<0>
MXM_PCIE_D2R_N<15>	MXM C8633	0.1UF	10% 16V X5R 402	PEG_D2R_P<0>
MXM_PCIE_D2R_P<14>	MXM C8634	0.1UF	10% 16V X5R 402	PEG_D2R_N<1>
MXM_PCIE_D2R_N<14>	MXM C8635	0.1UF	10% 16V X5R 402	PEG_D2R_P<1>
MXM_PCIE_D2R_P<13>	MXM C8636	0.1UF	10% 16V X5R 402	PEG_D2R_N<2>
MXM_PCIE_D2R_N<13>	MXM C8637	0.1UF	10% 16V X5R 402	PEG_D2R_P<2>
MXM_PCIE_D2R_P<12>	MXM C8638	0.1UF	10% 16V X5R 402	PEG_D2R_N<3>
MXM_PCIE_D2R_N<12>	MXM C8639	0.1UF	10% 16V X5R 402	PEG_D2R_P<3>
MXM_PCIE_D2R_P<11>	MXM C8640	0.1UF	10% 16V X5R 402	PEG_D2R_N<4>
MXM_PCIE_D2R_N<11>	MXM C8641	0.1UF	10% 16V X5R 402	PEG_D2R_P<4>
MXM_PCIE_D2R_P<10>	MXM C8642	0.1UF	10% 16V X5R 402	PEG_D2R_N<5>
MXM_PCIE_D2R_N<10>	MXM C8643	0.1UF	10% 16V X5R 402	PEG_D2R_P<5>
MXM_PCIE_D2R_P<9>	MXM C8644	0.1UF	10% 16V X5R 402	PEG_D2R_N<6>
MXM_PCIE_D2R_N<9>	MXM C8645	0.1UF	10% 16V X5R 402	PEG_D2R_P<6>
MXM_PCIE_D2R_P<8>	MXM C8646	0.1UF	10% 16V X5R 402	PEG_D2R_N<7>
MXM_PCIE_D2R_N<8>	MXM C8647	0.1UF	10% 16V X5R 402	PEG_D2R_P<7>
MXM_PCIE_D2R_P<7>	MXM C8648	0.1UF	10% 16V X5R 402	PEG_D2R_N<8>
MXM_PCIE_D2R_N<7>	MXM C8649	0.1UF	10% 16V X5R 402	PEG_D2R_P<8>
MXM_PCIE_D2R_P<6>	MXM C8650	0.1UF	10% 16V X5R 402	PEG_D2R_N<9>
MXM_PCIE_D2R_N<6>	MXM C8651	0.1UF	10% 16V X5R 402	PEG_D2R_P<9>
MXM_PCIE_D2R_P<5>	MXM C8652	0.1UF	10% 16V X5R 402	PEG_D2R_N<10>
MXM_PCIE_D2R_N<5>	MXM C8653	0.1UF	10% 16V X5R 402	PEG_D2R_P<10>
MXM_PCIE_D2R_P<4>	MXM C8654	0.1UF	10% 16V X5R 402	PEG_D2R_N<11>
MXM_PCIE_D2R_N<4>	MXM C8655	0.1UF	10% 16V X5R 402	PEG_D2R_P<11>
MXM_PCIE_D2R_P<3>	MXM C8656	0.1UF	10% 16V X5R 402	PEG_D2R_N<12>
MXM_PCIE_D2R_N<3>	MXM C8657	0.1UF	10% 16V X5R 402	PEG_D2R_P<12>
MXM_PCIE_D2R_P<2>	MXM C8658	0.1UF	10% 16V X5R 402	PEG_D2R_P<13>
MXM_PCIE_D2R_N<2>	MXM C8659	0.1UF	10% 16V X5R 402	PEG_D2R_N<13>
MXM_PCIE_D2R_P<1>	MXM C8662	0.1UF	10% 16V X5R 402	PEG_D2R_P<14>
MXM_PCIE_D2R_N<1>	MXM C8663	0.1UF	10% 16V X5R 402	PEG_D2R_N<14>
MXM_PCIE_D2R_P<0>	MXM C8660	0.1UF	10% 16V X5R 402	PEG_D2R_N<15>
MXM_PCIE_D2R_N<0>	MXM C8661	0.1UF	10% 16V X5R 402	PEG_D2R_P<15>

SYNC_MASTER=K23F SYNC_DATE=11/30/2009

PAGE TITLE: MXM PCIE CAPS

Apple Inc.

DRAWING NUMBER: 051-8337 SIZE: D

REVISION: A.0.0

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Page Notes

Power aliases required by this page:

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Unused MXM Interfaces

Pages	Net	Alias	Properties
87 75	MXM_LVDS_A_CLK_N	NC_MXM_LVDS_A_CLK_N	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_CLK_P	NC_MXM_LVDS_A_CLK_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_N<0>	NC_MXM_LVDS_A_DATA_N<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_P<0>	NC_MXM_LVDS_A_DATA_P<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_N<1>	NC_MXM_LVDS_A_DATA_N<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_P<1>	NC_MXM_LVDS_A_DATA_P<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_N<2>	NC_MXM_LVDS_A_DATA_N<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_P<2>	NC_MXM_LVDS_A_DATA_P<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_N<3>	NC_MXM_LVDS_A_DATA_N<3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_A_DATA_P<3>	NC_MXM_LVDS_A_DATA_P<3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_CLK_N	NC_MXM_LVDS_B_CLK_N	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_CLK_P	NC_MXM_LVDS_B_CLK_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_N<0>	NC_MXM_LVDS_B_DATA_N<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_P<0>	NC_MXM_LVDS_B_DATA_P<0>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_N<1>	NC_MXM_LVDS_B_DATA_N<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_P<1>	NC_MXM_LVDS_B_DATA_P<1>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_N<2>	NC_MXM_LVDS_B_DATA_N<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_P<2>	NC_MXM_LVDS_B_DATA_P<2>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_N<3>	NC_MXM_LVDS_B_DATA_N<3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 75	MXM_LVDS_B_DATA_P<3>	NC_MXM_LVDS_B_DATA_P<3>	MAKE_BASE=TRUE NO_TEST=TRUE

Unused MXM DP Interfaces

Pages	Net	Alias	Properties
87 74	MXM_DP_B_ML_P<0..3>	NC_MXM_DP_B_ML_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_DP_B_ML_N<0..3>	NC_MXM_DP_B_ML_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_DP_B_AUX_P	NC_MXM_DP_B_AUX_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_DP_B_AUX_N	NC_MXM_DP_B_AUX_N	MAKE_BASE=TRUE NO_TEST=TRUE
74	MXM_DP_B_HPD	NC_MXM_DP_B_HPD	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_DP_D_ML_P<0..3>	NC_MXM_DP_D_ML_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_DP_D_ML_N<0..3>	NC_MXM_DP_D_ML_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_DP_D_AUX_P	NC_MXM_DP_D_AUX_P	MAKE_BASE=TRUE NO_TEST=TRUE
87 74	MXM_DP_D_AUX_N	NC_MXM_DP_D_AUX_N	MAKE_BASE=TRUE NO_TEST=TRUE
74	MXM_DP_D_HPD	NC_MXM_DP_D_HPD	MAKE_BASE=TRUE NO_TEST=TRUE

UNUSED MXM CONTROL SIGNALS

Pages	Net	Alias	Properties
75	MXM_PNL_BL_EN	NC_MXM_PNL_BL_EN	MAKE_BASE=TRUE NO_TEST=TRUE
75	MXM_PNL_PWR_EN	NC_MXM_PNL_PWR_EN	MAKE_BASE=TRUE NO_TEST=TRUE

DISPLAY AUDIO MUX NOT USED - SEND SPDIF TO CODEC

Pages	Port	Net	Properties	Alias	Port	Pages
85 60	IN	AUD_SPDIF_IN	MAKE_BASE=TRUE	AUD_SPDIF_IN_CODEC	OUT	56
78	IN	DP_INT_SPDIF_AUDIO		TP_DP_INT_SPDIF_AUDIO MAKE_BASE=TRUE		

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
Display: Aliases

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DRAWING NUMBER 051-8337 SIZE D

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8 7 6 5 4 3 2 1

D C B A

Page Notes

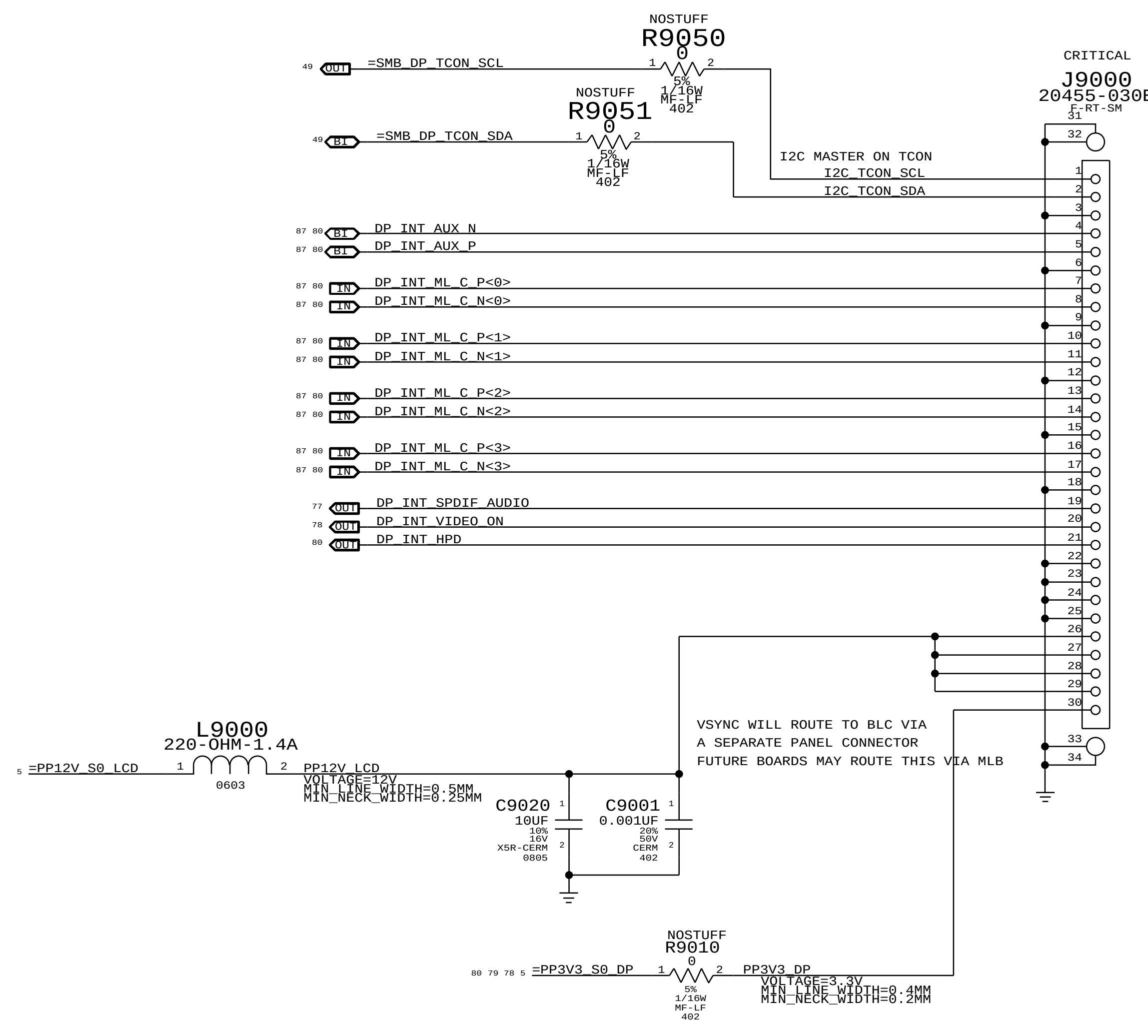
Power aliases required by this page:

- =PP12V_S0_LCD
- =PP3V3_S0_DP

Signal aliases required by this page:
- =SMB_DP_TCON_SCL, =SMB_DP_TCON_SDA

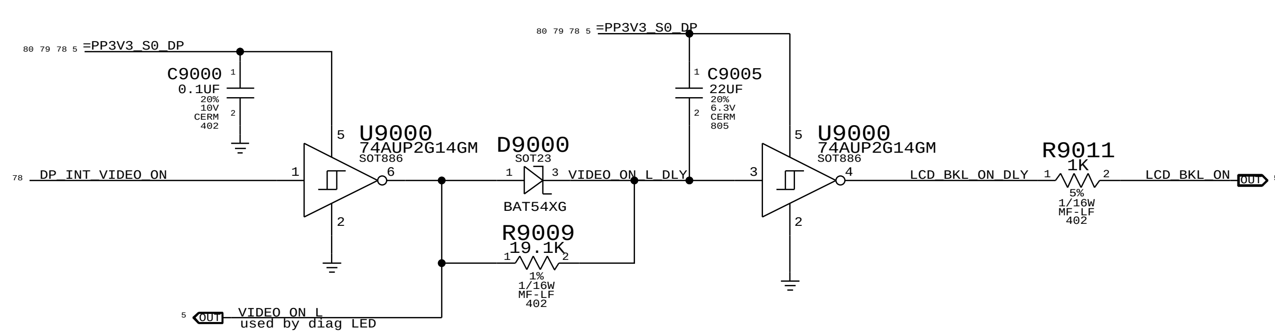
BOM options provided by this page:

INTERNAL DP INTERFACE



BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

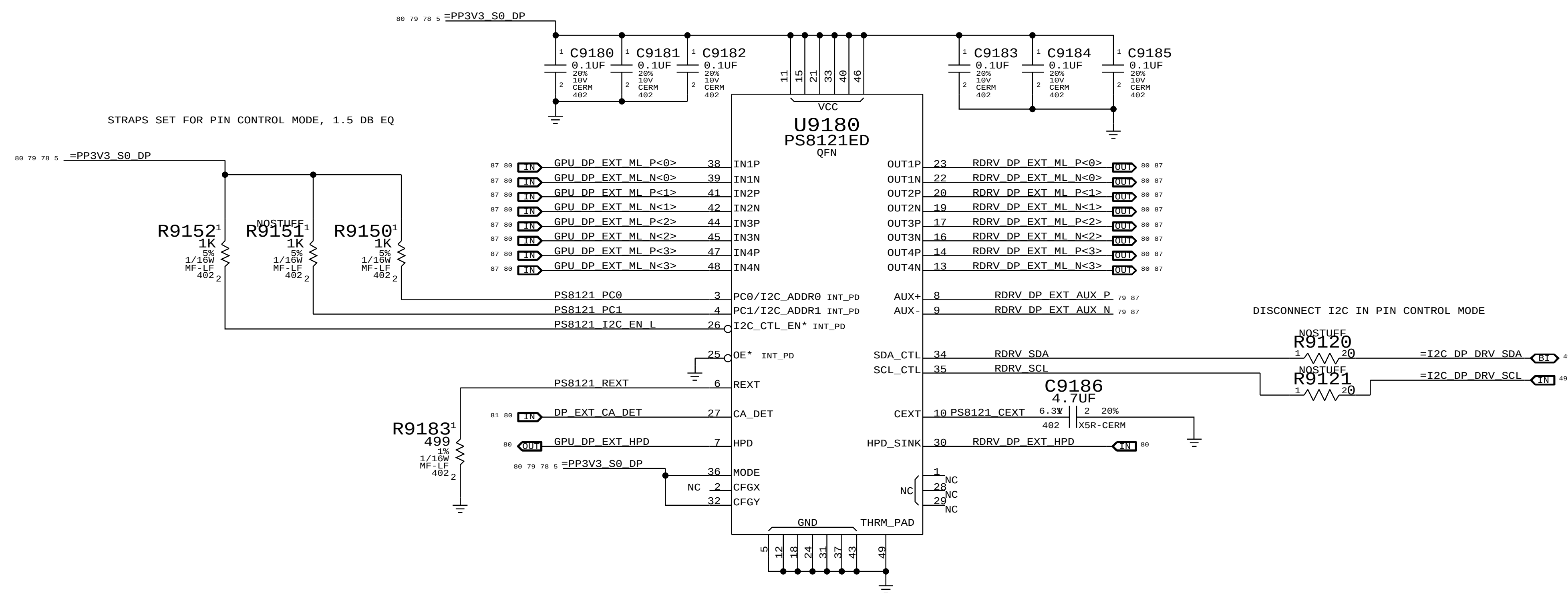


SYNC_MASTER=K74_MASTER		SYNC_DATE=N/A
PAGE TITLE: Display: Int DP Connector		
Apple Inc.	DRAWING NUMBER: 051-8337	SIZE: D
	REVISION: A.0.0	
NOTICE OF PROPRIETARY PROPERTY:	BRANCH	
	PAGE: 90 OF 110	
	SHEET: 78 OF 92	

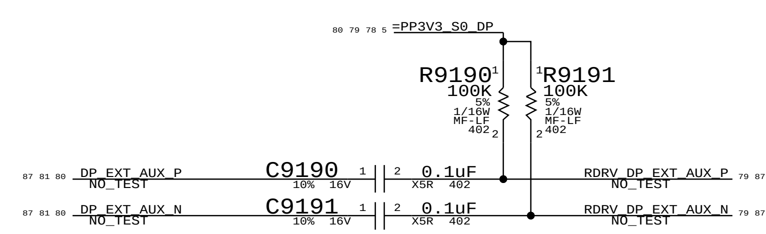
8 7 6 5 4 3 2 1

EQ & Re-Driver for DP source

STRAPS SET FOR PIN CONTROL MODE, 1.5 DB EQ



COMMON MODE BIAS FOR PS8121ED AUX INTERCEPTION



SYNC_MASTER=DAVE SYNC_DATE=01/07/2010

PAGE TITLE

DISPLAY: DP REDRIVER

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DRAWING NUMBER 051-8337 SIZE D

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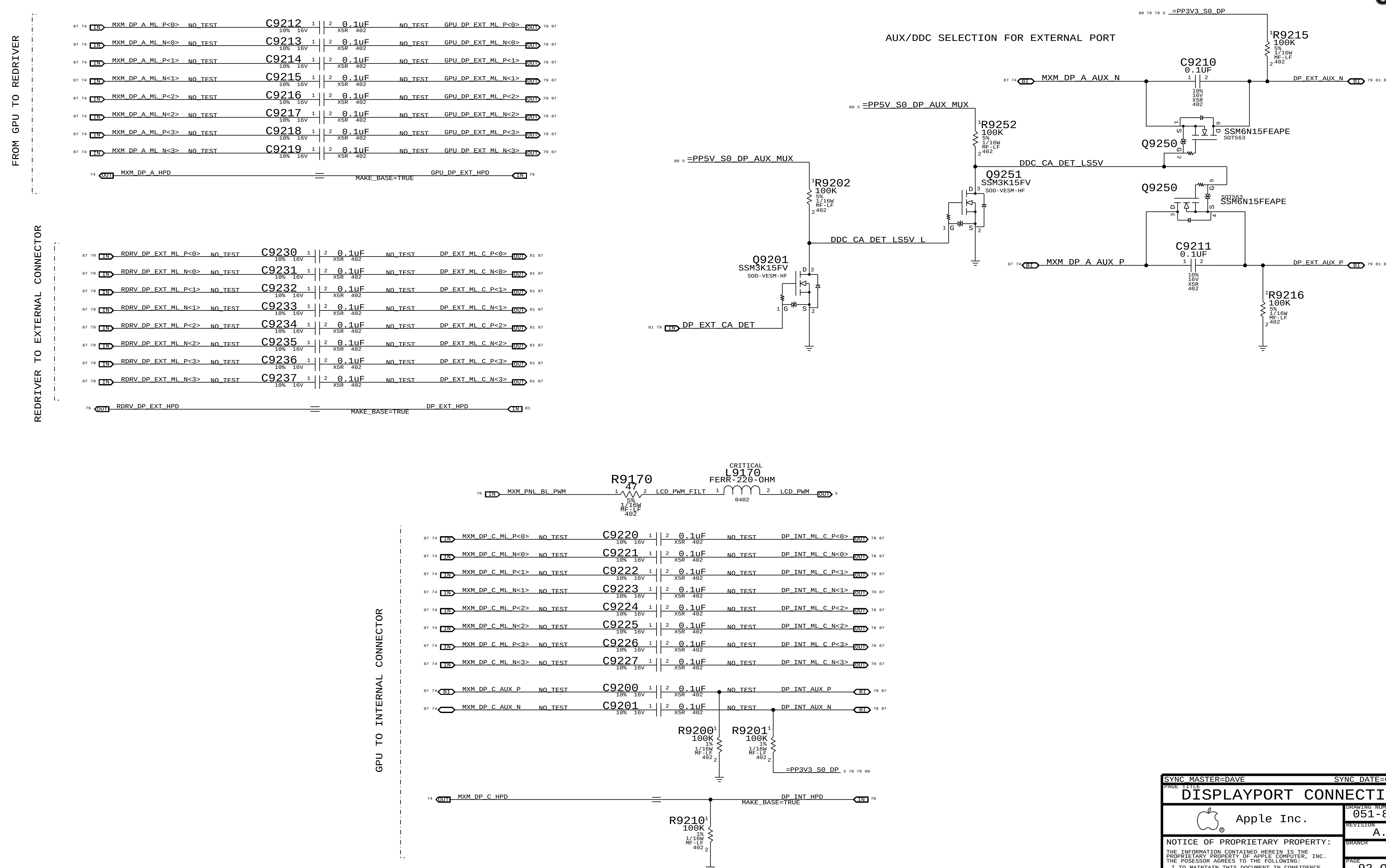

SYNC_MASTER=DAVE SYNC_DATE=01/07/2010

PAGE TITLE

DISPLAYPORT CONNECTIONS

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DRAWING NUMBER 051-8337 SIZE D

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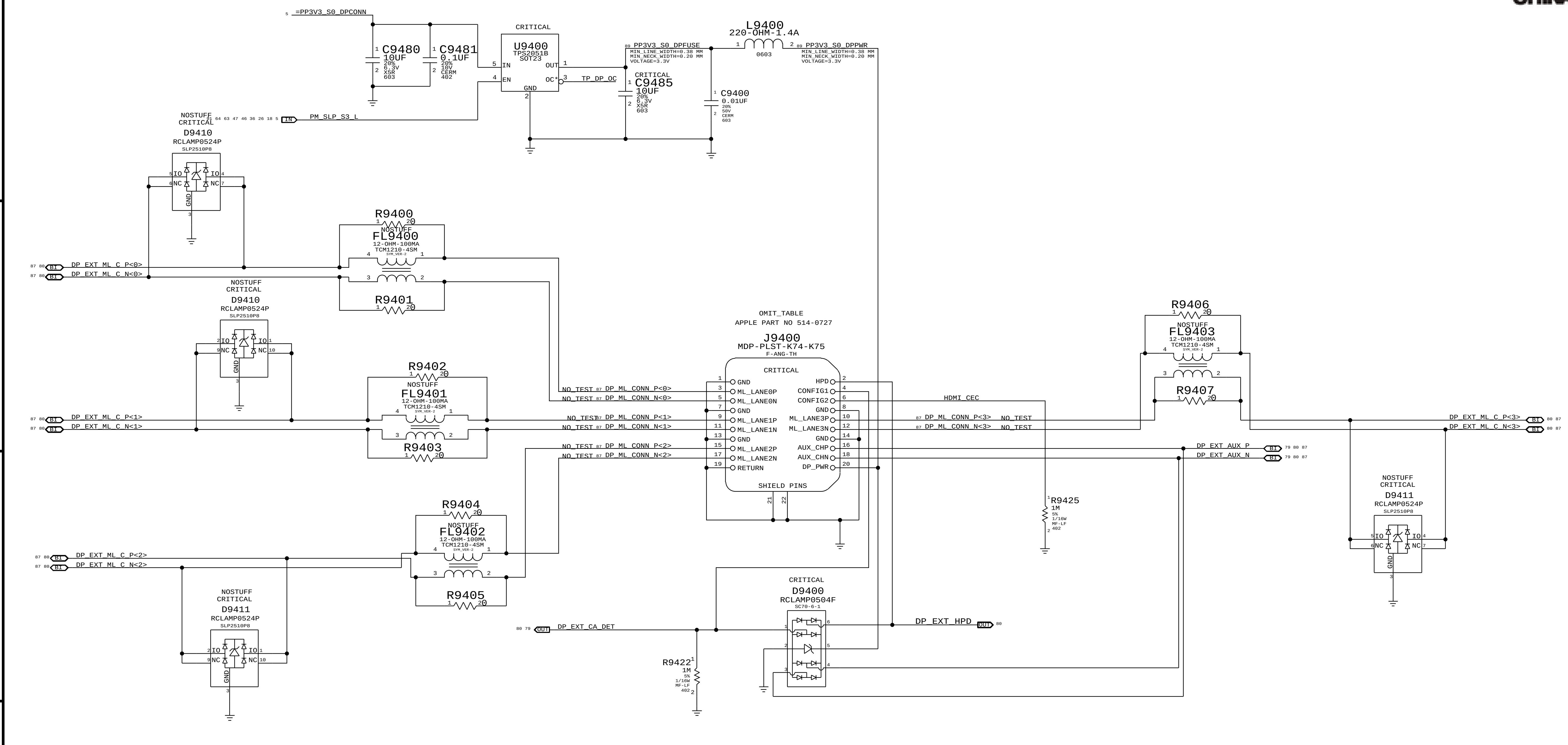
NOTICE OF PROPRIETARY PROPERTY:



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
514-0686	1	K22/K23 PROD. MDP	J9400	CRITICAL	METAL_IO
514-0727	1	K74/K75 MDP, PLASTIC, PD/NI	J9400	CRITICAL	PLASTIC_IO

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

Display: Ext DP Connector

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DRAWING NUMBER 051-8337 SIZE D

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PAGE 94 OF 110

SHEET 81 OF 92

K74/K75 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,BOTTOM	NO_TYPE,BGA_P1MM	MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
35_OHM_SE	TOP,BOTTOM	Y	0.21 MM	0.085 MM	=STANDARD		
35_OHM_SE	*	Y	0.19 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD		
39_OHM_SE	*	Y	0.16 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD		
45_OHM_SE	*	Y	0.12 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3,ISL6	Y	0.155 MM	0.085 MM	=STANDARD	0.135 MM	0.1 MM
70_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL6	Y	0.115 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL6	Y	0.099 MM	0.085 MM	12 MM	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3,ISL6	Y	0.081 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.085 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_CTL	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
RCOMP	*	Y	0.254 MM	0.200 MM	3.0 MM	=STANDARD	

=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	BGA_P1MM	POWER_CTL
POWER	*	POWER_WIDTH
VR_CTL_PHY	BGA_P1MM	DEFAULT
VR_CTL_PHY	*	POWER_CTL

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?
6:1_SPACING	*	0.6 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.150 MM	?
2X_DIELECTRIC	TOP,BOTTOM	0.160 MM	?
3X_DIELECTRIC	*	0.220 MM	?
3X_DIELECTRIC	TOP,BOTTOM	0.240 MM	?
4X_DIELECTRIC	*	0.300 MM	?
4X_DIELECTRIC	TOP,BOTTOM	0.320 MM	?
5X_DIELECTRIC	*	0.380 MM	?
5X_DIELECTRIC	TOP,BOTTOM	0.400 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	*	*	STANDARD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.8 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	0.2 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P1MM
CLK_LPC	*	BGA_P1MM	BGA_P1MM
CLK_PCI	*	BGA_P1MM	BGA_P1MM

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
MEM_35S	*	=35_OHM_SE	=35_OHM_SE	=35_OHM_SE	=35_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DQ_ODD2DQ_ODD	*	=3:1_SPACING	?
MEM_DQ_ODD2MEM	*	=3:1_SPACING	?
MEM_DQ_EVEN2DQ_EVEN	*	=3:1_SPACING	?
MEM_DQ_EVEN2MEM	*	=3:1_SPACING	?
MEM_DQ_EVEN2DQ_ODD	*	=5:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_2OTHER	*	=3:1_SPACING	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DQ_ODD	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM
MEM_CLK	MEM_DQ_EVEN	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_ODD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_EVEN	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_ODD	MEM_CLK	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_CTRL	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_CMD	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_DQ_ODD	*	MEM_DQ_ODD2DQ_ODD
MEM_DQ_ODD	MEM_DQS	*	MEM_DQ_ODD2MEM
MEM_DQ_ODD	MEM_DQ_EVEN	*	MEM_DQ_EVEN2DQ_ODD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_EVEN	MEM_CLK	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_CTRL	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_CMD	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_DQ_EVEN	*	MEM_DQ_EVEN2DQ_EVEN
MEM_DQ_EVEN	MEM_DQS	*	MEM_DQ_EVEN2MEM
MEM_DQ_EVEN	MEM_DQ_ODD	*	MEM_DQ_EVEN2DQ_ODD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQ_ODD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQ_EVEN	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DQ_ODD	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM
MEM_CMD	MEM_DQ_EVEN	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_DQ_ODD	*	*	MEM_2OTHER
MEM_DQS	*	*	MEM_2OTHER
MEM_DQ_EVEN	*	*	MEM_2OTHER

Need to support MEM_*-style wildcards!

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_POWER_WIDTH	*	Y	0.500 MM	0.175 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_POWER_PHY	*	MEM_POWER_WIDTH

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_POWER	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_RCOMP_PHY	*	Y	0.175 MM	0.175 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_RCOMP	*	0.2 MM	?

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
	MEM_70D	MEM_CLK	MEM_A_CLK_P<3..0>	11 32
	MEM_70D	MEM_CLK	MEM_A_CLK_N<3..0>	11 32
	MEM_39S	MEM_CTRL	MEM_A_CKE<3..0>	11 30
	MEM_39S	MEM_CTRL	MEM_A_CS_L<3..0>	11 30
	MEM_39S	MEM_CTRL	MEM_A_ODT<3..0>	11 30
	MEM_35S	MEM_CMD	MEM_A_A<15..0>	11 30
	MEM_35S	MEM_CMD	MEM_A_BA<2..0>	11 30
	MEM_35S	MEM_CMD	MEM_A_RAS_L	11 30
	MEM_35S	MEM_CMD	MEM_A_CAS_L	11 30
	MEM_35S	MEM_CMD	MEM_A_WE_L	11 30
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<7..0>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<0>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<15..8>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<1>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<23..16>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<2>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<31..24>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<3>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<39..32>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<4>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<47..40>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<5>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DQ<55..48>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_A_DM<6>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DQ<63..56>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_A_DM<7>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<0>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<0>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<1>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<1>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<2>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<2>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<3>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<3>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<4>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<4>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<5>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<5>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<6>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<6>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_P<7>	11 32
	MEM_70D	MEM_DQS	MEM_A_DQS_N<7>	11 32
	MEM_70D	MEM_CLK	MEM_B_CLK_P<3..0>	11 32
	MEM_70D	MEM_CLK	MEM_B_CLK_N<3..0>	11 32
	MEM_39S	MEM_CTRL	MEM_B_CKE<3..0>	11 31
	MEM_39S	MEM_CTRL	MEM_B_CS_L<3..0>	11 31
	MEM_39S	MEM_CTRL	MEM_B_ODT<3..0>	11 31
	MEM_35S	MEM_CMD	MEM_B_A<15..0>	11 31
	MEM_35S	MEM_CMD	MEM_B_BA<2..0>	11 31
	MEM_35S	MEM_CMD	MEM_B_RAS_L	11 31
	MEM_35S	MEM_CMD	MEM_B_CAS_L	11 31
	MEM_35S	MEM_CMD	MEM_B_WE_L	11 31
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<7..0>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<0>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<15..8>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<1>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<23..16>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<2>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<31..24>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<3>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<39..32>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<4>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<47..40>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<5>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DQ<55..48>	11 32
	MEM_45S	MEM_DQ_EVEN	MEM_B_DM<6>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DQ<63..56>	11 32
	MEM_45S	MEM_DQ_ODD	MEM_B_DM<7>	11 32

MEMORY POWER PROPERTIES

VOLTAGE	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_A	11 28
	MEM_POWER_PHY	MEM_POWER	CPU_DIMM_VREF_B	11 28

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
	MEM_70D	MEM_DQS	MEM_B_DQS_P<0>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<0>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<1>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<1>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<2>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<2>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<3>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<3>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<4>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<4>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<5>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<5>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<6>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<6>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_P<7>	11 32
	MEM_70D	MEM_DQS	MEM_B_DQS_N<7>	11 32
I160	MEM_RCOMP_PHY	MEM_RCOMP	CPU_SM_RCOMP0	10
I159	MEM_RCOMP_PHY	MEM_RCOMP	CPU_SM_RCOMP1	10
I161	MEM_RCOMP_PHY	MEM_RCOMP	CPU_SM_RCOMP2	10

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
Memory Constraints

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=4X_DIELECTRIC	?
CLK_PCIE	*	0.5 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

CPU

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_RCOMP_PHY	*	Y	0.254 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_ITP	*	0.2 MM	?
CPU_RCOMP	*	0.2 MM	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=5X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=5X_DIELECTRIC	?

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
CPU ITP				
	CPU_50S	CPU_ITP	XDP_BPM_L<7..0>	10 24
	CPU_50S	CPU_ITP	CPU_CFG<17..0>	9 14 24
	CPU_50S	CPU_ITP	XDP_OBSDATA_A<3..0>	24
CPU_MISC				
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_PEG_COMP	9
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_PEG_RBIAS	9
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_COMP3	10
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_COMP2	10
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_COMP1	10
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_COMP0	10

ANY OTHER LYNNFIELD CONSTRAINTS NOT COVERED ON PAGES 101 AND 107 SHOULD GO ON THIS PAGE TOO

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
PCIE GRAPHICS				
	PCIE_85D	PCIE	PEG_R2D_C_P<15..0>	8 76
	PCIE_85D	PCIE	PEG_R2D_C_N<15..0>	8 76
	PCIE_85D	PCIE	PEG_D2R_P<15..0>	8 76
	PCIE_85D	PCIE	PEG_D2R_N<15..0>	8 76
	PCIE_85D	PCIE	MXM_PCIE_R2D_P<15..0>	74 76
	PCIE_85D	PCIE	MXM_PCIE_R2D_N<15..0>	74 76
	PCIE_85D	PCIE	MXM_PCIE_D2R_P<15..0>	74 76
	PCIE_85D	PCIE	MXM_PCIE_D2R_N<15..0>	74 76
PCIE I/O				
	PCIE_85D	PCIE	PCIE_MINI_R2D_P	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_N	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_P	17 33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_N	17 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_P	17 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_N	17 33
	PCIE_85D	PCIE	PCIE_FW_R2D_P	39
	PCIE_85D	PCIE	PCIE_FW_R2D_N	39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_P	17 39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_N	17 39
	PCIE_85D	PCIE	PCIE_FW_D2R_P	17 39
	PCIE_85D	PCIE	PCIE_FW_D2R_N	17 39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_P	39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_N	39
DMI				
	PCIE_85D	PCIE	DMI_S2N_P<3..0>	9 18
	PCIE_85D	PCIE	DMI_S2N_N<3..0>	9 18
	PCIE_85D	PCIE	DMI_N2S_P<3..0>	9 18
	PCIE_85D	PCIE	DMI_N2S_N<3..0>	9 18
FDI				
PCIE REF CLOCKS				
	CLK_PCIE_100D	CLK_PCIE	GPU_CLK100M_PCIE_P	8
	CLK_PCIE_100D	CLK_PCIE	GPU_CLK100M_PCIE_N	8
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_P	17 33
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_N	17 33
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_P	17 39
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_N	17 39
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_P	17 37
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_N	17 37
SATA				
	SATA_85D	SATA	SATA_HDD_R2D_C_P	17 42
	SATA_85D	SATA	SATA_HDD_R2D_C_N	17 42
	SATA_85D	SATA	SATA_HDD_R2D_P	42
	SATA_85D	SATA	SATA_HDD_R2D_N	42
	SATA_85D	SATA	SATA_HDD_D2R_P	17 42
	SATA_85D	SATA	SATA_HDD_D2R_N	17 42
	SATA_85D	SATA	SATA_HDD_D2R_C_P	42
	SATA_85D	SATA	SATA_HDD_D2R_C_N	42
	SATA_85D	SATA	SATA_ODD_R2D_C_P	17 42
	SATA_85D	SATA	SATA_ODD_R2D_C_N	17 42
	SATA_85D	SATA	SATA_ODD_R2D_P	42
	SATA_85D	SATA	SATA_ODD_R2D_N	42
	SATA_85D	SATA	SATA_ODD_D2R_P	17 42
	SATA_85D	SATA	SATA_ODD_D2R_N	17 42
	SATA_85D	SATA	SATA_ODD_D2R_C_P	42
	SATA_85D	SATA	SATA_ODD_D2R_C_N	42
CLOCKS				
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_CPU_P	10 20
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_CPU_N	10 20
	CLK_PCIE_100D	CLK_PCIE	GFX_CLK120M_DPLLSS_P	10 17
	CLK_PCIE_100D	CLK_PCIE	GFX_CLK120M_DPLLSS_N	10 17
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_ITP_P	10 24
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_ITP_N	10 24
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_CPU_P	10 17
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_CPU_N	10 17
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_PCH_P	17 25
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_PCH_N	17 25
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_PCH_P	17 25
	CLK_PCIE_100D	CLK_PCIE	FSB_CLK133M_PCH_N	17 25
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK96M_DOT_P	17 25
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK96M_DOT_N	
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK100M_SATA_P	17 25
	CLK_PCIE_100D	CLK_PCIE	PCH_CLK100M_SATA_N	17 25

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
PCIE/DMI/FDI/SATA CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

PCH CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	0.2 MM	?
COMP_PCH	*	0.2 MM	?

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	0.2 MM	?

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	0.15 MM	?
CLK_LPC	*	0.2 MM	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

ANDARD=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE		

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	0.2 MM	?

XTAL Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
	PCI_55S	PCI	PCI_REQ0_L	19
	PCI_55S	PCI	PCI_REQ1_L	19
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT	19 27
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN	17 27
	LPC_55S	LPC	LPC_AD<3..0>	17 46 48
	LPC_55S	LPC	LPC_FRAME_L	17 46 48
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	19 27
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	27 46
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	27 48
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK_R	8 18 91
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK	8 46 91
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	19 27
	USB_90D	USB	USB_EXTA_P	34 43
	USB_90D	USB	USB_EXTA_N	34 43
	USB_90D	USB	USB_PORT0_P	43
	USB_90D	USB	USB_PORT0_N	43
	USB_90D	USB	USB_EXTB_P	35 43
	USB_90D	USB	USB_EXTB_N	35 43
	USB_90D	USB	USB_PORT1_P	43
	USB_90D	USB	USB_PORT1_N	43
	USB_90D	USB	USB_EXTC_P	34 43
	USB_90D	USB	USB_EXTC_N	34 43
	USB_90D	USB	USB_PORT2_P	43
	USB_90D	USB	USB_PORT2_N	43
	USB_90D	USB	USB_EXTD_P	35 43
	USB_90D	USB	USB_EXTD_N	35 43
	USB_90D	USB	USB_D_MUXED_P	43
	USB_90D	USB	USB_D_MUXED_N	43
	USB_90D	USB	USB_PORT3_P	43
	USB_90D	USB	USB_PORT3_N	43
	USB_90D	USB	USB_CAMERA_P	34 44
	USB_90D	USB	USB_CAMERA_N	34 44
	USB_90D	USB	USB_CAMERA_L_P	44 92
	USB_90D	USB	USB_CAMERA_L_N	44 92
	USB_90D	USB	USB_BT_P	35 44
	USB_90D	USB	USB_BT_N	35 44
	USB_90D	USB	USB_BT_L_P	44 92
	USB_90D	USB	USB_BT_L_N	44 92
	USB_90D	USB	USB_IR_P	34 44
	USB_90D	USB	USB_IR_N	34 44
	USB_90D	USB	USB_IR_L_P	44 92
	USB_90D	USB	USB_IR_L_N	44 92
	USB_90D	USB	USB_SDCARD_P	35 44
	USB_90D	USB	USB_SDCARD_N	35 44
	USB_90D	USB	USB_SDCARD_L_P	44 92
	USB_90D	USB	USB_SDCARD_L_N	44 92
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	17 27
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	17 27
	CLK_XTAL	XTAL	CK505_XTAL_IN	25
	CLK_XTAL	XTAL	CK505_XTAL_OUT	25
	CLK_PCH_55S	CLK_PCH	PCH_CLK14P3M_REFCLK	17 25
	USB_90D	USB	USB_HUB1_UP_P	19 34
	USB_90D	USB	USB_HUB1_UP_N	19 34
	USB_90D	USB	USB_HUB2_UP_P	19 35
	USB_90D	USB	USB_HUB2_UP_N	19 35

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
	SPI_55S	SPI	SPI_CLK_R	17 48 55
	SPI_55S	SPI	SPI_CLK	55
	SPI_55S	SPI	SPI_MOSI_R	17 48 55
	SPI_55S	SPI	SPI_MOSI	55
	SPI_55S	SPI	SPI_MISO	17 48 55
	SPI_55S	SPI	SPI_MISO_R	55
	SPI_55S	SPI	SPI_CS0_R_L	17 48
	SPI_55S	SPI	SPI_CS0_L	48
	SPI_55S	SPI	SPI_MLB_CS_L	48 55
	SPI_55S	SPI	SPI_ALT_CS_L	48
	SPI_55S	SPI	SPIROM_USE_MLB	20 48
	SPI_55S	SPI	SPI_ALT_MOSI	48
	SPI_55S	SPI	SPI_ALT_MISO	48
	SPI_55S	SPI	SPI_ALT_CLK	48
	HDA_55S	HDA	HDA_BIT_CLK	17 56
	HDA_55S	HDA	HDA_BIT_CLK_R	17
	HDA_55S	HDA	HDA_RST_L	17 56
	HDA_55S	HDA	HDA_RST_R_L	17
	HDA_55S	HDA	HDA_SDOUT	17 56
	HDA_55S	HDA	HDA_SDOUT_R	17
	HDA_55S	HDA	HDA_SYNC	17 56
	HDA_55S	HDA	HDA_SYNC_R	17
	HDA_55S	HDA	HDA_SDIN0	17 56
	HDA_55S	HDA	AUD_SDI_R	56
		HDA	AUD_SPDIF_IN	60 77
		HDA	AUD_SPDIF_OUT	56 60
		HDA	AUD_SPDIF_CHIP	56
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_NOUT	59 60 92
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_POUT	59 60 92
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_NOUT	58 60 92
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_POUT	58 60 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_NOUT	59 60 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_POUT	59 60 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_NOUT	58 60 92
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_POUT	58 60 92
	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT	17 27
	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN	17 27
	PCH_55S	COMP_PCH	PCH_USB_RBIAS	19
	PCH_55S	COMP_PCH	PCH_SATAICOMP	17
	PCH_55S	COMP_PCH	PCH_XCLK_RCOMP	17
	PCH_55S	COMP_PCH	PCH_DMI_COMP	18
	CLK_XTAL	XTAL	USB_HUB1_XTAL1	34
	CLK_XTAL	XTAL	USB_HUB1_XTAL2	34
	PCH_55S	COMP_PCH	USB_HUB1_RBIAS	34
	CLK_XTAL	XTAL	USB_HUB2_XTAL1	35
	CLK_XTAL	XTAL	USB_HUB2_XTAL2	35
	PCH_55S	COMP_PCH	USB_HUB2_RBIAS	35

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
IBEX PEAK CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D
REVISION A.0.0
BRANCH

8	7	6	5	4	3	2	1

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	0.3 MM	?
ENET_SE	*	=STANDARD	?

SOURCE:BROADCOM 5764M-DS04-RDS. PAGE 38

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF	*	0.6 MM	?

CAESAR IV (SD) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD	*	=3:1_SPACING	?

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

AUDIO CONSTRAINTS

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	=3:1_SPACING	?

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUD_DIFF	*	1:1_DIFFPAIR

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	ENET_50S	ENET_SE	BCM5764_RDAC	37
	ENET_50S	BUF0_CLK	BCM5764_CLK25M_XTALI	36 37
	ENET_50S	BUF0_CLK	BCM5764_CLK25M_XTALO	36 37
	ENET_50S	BUF0_CLK	BCM5764_CLK25M_XTAL	36
	ENET_100D	ENET_DIFF	ENET_MDI_P<3..0>	37 38
	ENET_100D	ENET_DIFF	ENET_MDI_N<3..0>	37 38
	ENET_100D	ENET_DIFF	ENET_MDI_T_P<3..0>	38
	ENET_100D	ENET_DIFF	ENET_MDI_T_N<3..0>	38
	ENET_100D	ENET_MII	PCIE_ENET_R2D_P	37
	ENET_100D	ENET_MII	PCIE_ENET_R2D_N	37
	ENET_100D	ENET_MII	PCIE_ENET_D2R_P	17 37
	ENET_100D	ENET_MII	PCIE_ENET_D2R_N	17 37
	ENET_100D	ENET_MII	PCIE_ENET_R2D_C_P	17 37
	ENET_100D	ENET_MII	PCIE_ENET_R2D_C_N	17 37
	ENET_100D	ENET_MII	PCIE_ENET_D2R_C_P	37
	ENET_100D	ENET_MII	PCIE_ENET_D2R_C_N	37
	SD_50S	SD	SDCONN_CMD	37 45
	SD_50S	SD	SDCONN_CLK	37 45
	SD_50S	SD	SDCONN_CLK_R	45
	SD_50S	SD	SDCONN_DATA<7..0>	37 45
	SD_50S	SD	BCM57765_CR_DATA<7..4>	37

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	FW_110D	FW_TP	FW_PORT0_TPA_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPA_N	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_N	40 41
	FW_110D	FW_TP	FW_PORT0_TPA_F_P	41
	FW_110D	FW_TP	FW_PORT0_TPA_F_N	41
	FW_110D	FW_TP	FW_PORT0_TPB_F_P	41
	FW_110D	FW_TP	FW_PORT0_TPB_F_N	41
PORT 1 & 2 NOT USED				
	FW_110D	FW_TP	FW_P0_TPA_L_P	40
	FW_110D	FW_TP	FW_P0_TPA_L_N	40
	FW_110D	FW_TP	FW_P0_TPB_L_P	40
	FW_110D	FW_TP	FW_P0_TPB_L_N	40
UNUSED FW NETS PHYSICAL PROPERTIES				
	FW_110D	FW_TP	FW_P1_TPA_P	39 40
	FW_110D	FW_TP	FW_P1_TPA_N	39 40
	FW_110D	FW_TP	FW_P2_TPA_P	39 40
	FW_110D	FW_TP	FW_P2_TPA_N	39 40

AUDIO NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	AUD_DIFF	AUDIO	AUDAMPINBLN	58
	AUD_DIFF	AUDIO	AUDAMPINBLP	58
	AUD_DIFF	AUDIO	AUDAMPINLN	58
	AUD_DIFF	AUDIO	AUDAMPINLP	58
	AUD_DIFF	AUDIO	AUD_LO2_N_R	56 58
	AUD_DIFF	AUDIO	AUD_LO2_P_R	56 58
	AUD_DIFF	AUDIO	AUDAMPINBRN	58
	AUD_DIFF	AUDIO	AUDAMPINBRP	58
	AUD_DIFF	AUDIO	AUDAMPINRN	58
	AUD_DIFF	AUDIO	AUDAMPINRP	58
	AUD_DIFF	AUDIO	AUD_LO1_N_R	56 58
	AUD_DIFF	AUDIO	AUD_LO1_P_R	56 58
	AUD_DIFF	AUDIO	AUDAMPINCLN	59
	AUD_DIFF	AUDIO	AUDAMPINCLP	59
	AUD_DIFF	AUDIO	AUD_AMPINLN	59
	AUD_DIFF	AUDIO	AUD_AMPINLP	59
	AUD_DIFF	AUDIO	AUD_LO2_N_L	56 59
	AUD_DIFF	AUDIO	AUD_LO2_P_L	56 59
	AUD_DIFF	AUDIO	AUDAMPINCRN	59
	AUD_DIFF	AUDIO	AUDAMPINCRP	59
	AUD_DIFF	AUDIO	AUD_AMPINRN	59
	AUD_DIFF	AUDIO	AUD_AMPINRP	59
	AUD_DIFF	AUDIO	AUD_LO1_N_L	56 59
	AUD_DIFF	AUDIO	AUD_LO1_P_L	56 59

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
ENET/SD/FW/AUD CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

8	7	6	5	4	3	2	1

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

PAIRS SHOULD BE WITHIN 100 MILS OF CLOCK LENGTH.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DIsplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.

NET_TYPE

ELECTRICAL_CONSTRAINT_SET ASSINGED IN CONT. MGR.	PHYSICAL	SPACING		
I113	DP_85D	DISPLAYPORT	MXM_DP_C_ML_P<3..0>	74 80
I114	DP_85D	DISPLAYPORT	MXM_DP_C_ML_N<3..0>	74 80
	DP_85D	DISPLAYPORT	DP_INT_ML_C_P<3..0>	78 80
	DP_85D	DISPLAYPORT	DP_INT_ML_C_N<3..0>	78 80
	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_P	74 80
	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_N	74 80
	DP_85D	DISPLAYPORT	DP_INT_AUX_P	78 80
	DP_85D	DISPLAYPORT	DP_INT_AUX_N	78 80
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_P<3..0>	74 80
	DP_85D	DISPLAYPORT	MXM_DP_A_ML_N<3..0>	74 80
	DP_85D	DISPLAYPORT	GPU_DP_EXT_ML_P<3..0>	79 80
	DP_85D	DISPLAYPORT	GPU_DP_EXT_ML_N<3..0>	79 80
	DP_85D	DISPLAYPORT	RDRV_DP_EXT_ML_P<3..0>	79 80
	DP_85D	DISPLAYPORT	RDRV_DP_EXT_ML_N<3..0>	79 80
	DP_85D	DISPLAYPORT	DP_EXT_ML_C_P<3..0>	80 81
	DP_85D	DISPLAYPORT	DP_EXT_ML_C_N<3..0>	80 81
	DP_85D	DISPLAYPORT	DP_ML_CONN_P<3..0>	81
	DP_85D	DISPLAYPORT	DP_ML_CONN_N<3..0>	81
	DP_85D	DISPLAYPORT	MXM_DP_A_AUX_P	74 80
	DP_85D	DISPLAYPORT	MXM_DP_A_AUX_N	74 80
	DP_85D	DISPLAYPORT	RDRV_DP_EXT_AUX_P	79
	DP_85D	DISPLAYPORT	RDRV_DP_EXT_AUX_N	79
	DP_85D	DISPLAYPORT	DP_EXT_AUX_P	79 80 81
	DP_85D	DISPLAYPORT	DP_EXT_AUX_N	79 80 81

UNUSED VIDEO NET PHYSICAL CONSTRAINTS

	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_P	74 77
	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_N	74 77
	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_P	74 77
	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_N	74 77
	DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_P	75 77
	DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_N	75 77
	DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_P	75 77
	DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_N	75 77
	DP_85D	DISPLAYPORT	MXM_DP_B_ML_P<3..0>	74 77
	DP_85D	DISPLAYPORT	MXM_DP_B_ML_N<3..0>	74 77
	DP_85D	DISPLAYPORT	MXM_DP_D_ML_P<3..0>	74 77
	DP_85D	DISPLAYPORT	MXM_DP_D_ML_N<3..0>	74 77
	DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_P<3..0>	75 77
	DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_N<3..0>	75 77
	DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_P<3..0>	75 77
	DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_N<3..0>	75 77

SYNC_MASTER=DAVE SYNC_DATE=01/07/2010

PAGE TITLE

GRAPHICS CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

REVISION A.0.0

BRANCH

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
THERMAL	*	*	4:1_SPACING
THERMAL	POWER	*	PWR_P2MM
THERMAL	GND	*	GND_P2MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
THERM_DIFF	*	1:1_DIFFPAIR
SNS_DIFF	*	1:1_DIFFPAIR

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	SMB_55S	SMB	SMBUS_SMC_A_S3_SCL	49
	SMB_55S	SMB	SMBUS_SMC_A_S3_SDA	49
	SMB_55S	SMB	SMBUS_SMC_B_S0_SCL	49
	SMB_55S	SMB	SMBUS_SMC_B_S0_SDA	49
	SMB_55S	SMB	SMBUS_SMC_0_S0_SCL	49
	SMB_55S	SMB	SMBUS_SMC_0_S0_SDA	49
	SMB_55S	SMB	SMBUS_SMC_BSA_SCL	49
	SMB_55S	SMB	SMBUS_SMC_BSA_SDA	49
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	49 88
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	49 88
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	49 88
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	49 88
	SMB_55S	SMB	SMBUS_PCH_CLK	17 49
	SMB_55S	SMB	SMBUS_PCH_DATA	17 49
	SMB_55S	SMB	SML_PCH_0_CLK	17 49
	SMB_55S	SMB	SML_PCH_0_DATA	17 49
	SMB_55S	SMB	SML_PCH_1_CLK	17 49
	SMB_55S	SMB	SML_PCH_1_DATA	17 49
	CLK_XTAL	XTAL	SMC_EXTAL	46 47
	CLK_XTAL	XTAL	SMC_XTAL	46 47

SMC THERMAL NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	THERM_DIFF	THERMAL	SNS_T1_DP1	52
	THERM_DIFF	THERMAL	SNS_T1_DN1	52
	THERM_DIFF	THERMAL	SNS_T1_DP2_DN3	44 52
	THERM_DIFF	THERMAL	SNS_T1_DN2_DP3	44 52
	THERM_DIFF	THERMAL	SNS_T2_DP1	52
	THERM_DIFF	THERMAL	SNS_T2_DN1	52
	THERM_DIFF	THERMAL	SNS_T2_DP2	52
	THERM_DIFF	THERMAL	SNS_T2_DN2	52
	THERM_DIFF	THERMAL	SNS_T2_DP3	52
	THERM_DIFF	THERMAL	SNS_T2_DN3	52
	THERM_DIFF	THERMAL	SNS_ODD_P	52 92
	THERM_DIFF	THERMAL	SNS_ODD_N	52 92
	THERM_DIFF	THERMAL	SNS_CPU_H_P	52
	THERM_DIFF	THERMAL	SNS_CPU_H_N	52
	THERM_DIFF	THERMAL	SNS_SKIN_P	52 92
	THERM_DIFF	THERMAL	SNS_SKIN_N	52 92
	THERM_DIFF	THERMAL	SNS_AMB_P	52 54 92
	THERM_DIFF	THERMAL	SNS_AMB_N	52 54 92
	THERM_DIFF	THERMAL	SNS_MXM_P	52
	THERM_DIFF	THERMAL	SNS_MXM_N	52
		THERMAL	HDD_OOB_TEMP_FILT	51 92
		THERMAL	HDD_OOB_TEMP	51
		THERMAL	HDD_OOB_TEMP_R	51
		THERMAL	SMC_HDD_OOB_TEMP	46 51

SMC VOLTAGE/CURRENT NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	THERM_DIFF	THERMAL	SENSE_MXM_P	50
	THERM_DIFF	THERMAL	SENSE_MXM_N	50
	THERM_DIFF	THERMAL	SENSE_VTT_R_P	
	THERM_DIFF	THERMAL	SENSE_VTT_R_N	
	THERM_DIFF	THERMAL	SENSE_CPU_1V5_P	50
	THERM_DIFF	THERMAL	SENSE_CPU_1V5_N	50
	THERM_DIFF	THERMAL	SENSE_CPU_VTT_P	
	THERM_DIFF	THERMAL	SENSE_CPU_VTT_N	
		THERMAL	GND_SMC_AVSS	46 47 50
		THERMAL	SMC_CPU_1V5_ISENSE	46 50
		THERMAL	SMC_CPU_1V5_ISENSE_R	50
		THERMAL	SMC_CPU_1V5_VSENSE	46 50
		THERMAL	SMC_CPU_VSENSE	46 50
	VID_PHY	VR_CTL	VR_CPU_IOUT	12 65
	THERM_DIFF	THERMAL	VR_ISNS_CPU_P	50
	THERM_DIFF	THERMAL	VR_ISNS_CPU_N	50
		THERMAL	SNS_PS_CPU_ISNS	50
		THERMAL	SMC_CPU_ISENSE	46 50

SYNC_MASTER=TEMP SYNC_DATE=12/09/2009

PAGE TITLE

SMC Constraints

Apple Inc.

DRAWING NUMBER 051-8337 SIZE D

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BRANCH

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SWITCHNODE	SWITCHNODE	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	BGA_P1MM	BGA_P2MM
SWITCHNODE	GND	BGA_P1MM	BGA_P2MM
SWITCHNODE	*	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	*	6:1_SPACING
SWITCHNODE	GND	*	6:1_SPACING
SWITCHNODE	*	*	SWITCHNODE

POWER NET PROPERTIES

PHYSICAL	SPACING	VOLTAGE	NET
POWER	SWITCHNODE	1.5V	VR_CPU_PHASE1
POWER	SWITCHNODE	1.5V	VR_CPU_PHASE2
POWER	SWITCHNODE	1.5V	VR_CPU_PHASE3
POWER	SWITCHNODE	3.3V	P3V3S5_REG_PHASE
POWER	SWITCHNODE	5V	P5VS3_REG_PHASE
POWER	SWITCHNODE	1.1V	VTT_REG_PHASE
POWER	SWITCHNODE	3.4V	P3V42G3H_SW
POWER	SWITCHNODE	1.05V	PCHCORE_REG_PHASE
POWER	SWITCHNODE	1.5V	DDR_REG_PHASE
POWER	SWITCHNODE	1.8V	P1V8_REG_PHASE
POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA_A
POWER	POWER	1.5V	PP0V75_S3_MEM_VREFCA_B
POWER	POWER	1.5V	PP0V75_S3_MEM_VREFDQ_A
POWER	POWER	1.5V	PP0V75_S3_MEM_VREFDQ_B
POWER	POWER	12V	PP12V_AUD_SPKRAMP_PLANE
POWER	POWER	12V	PP12V_S0
POWER	POWER	12V	PP12V_S0_CPU_FLTRD
POWER	POWER	12V	PP12V_S0_FAN0_L
POWER	POWER	12V	PP12V_S0_FAN1_L
POWER	POWER	12V	PP12V_S0_FAN2_L
POWER	POWER	12V	PP12V_G3H
POWER	POWER	12V	PP12V_S5
POWER	POWER	12V	FW_PORT0_VP
POWER	POWER	12V	FW_PORT0_VP_F
POWER	POWER	12V	PPVP_FW_PHY_CPS
POWER	POWER	1.1V	PPVCORE_S0_CPU
POWER	POWER	1.1V	PPVCORE_S0_CPU_REG1
POWER	POWER	1.1V	PPVCORE_S0_CPU_REG2
POWER	POWER	1.1V	PPVCORE_S0_CPU_REG3
POWER	POWER	1.05V	PP1V05_S0
POWER	POWER	1.05V	PP1V05_S0_CK505_F
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB_F
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_EXP
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_FDI
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_SATA
POWER	POWER	1.05V	PP1V05_S0_PCH_VCCA_CLK
POWER	POWER	1.05V	PP1V05_SM_PCH_LAN
POWER	POWER	1.1V	PPVTT_S0
POWER	POWER	0.75V	PPVTT_S0_DDR
POWER	POWER	0.75V	PP0V75_S0
POWER	POWER	1.5V	PP1V5_S0
POWER	POWER	1.5V	PP1V5_S0_CK505_F
POWER	POWER	1.5V	PP1V5_S0_CK505_R
POWER	POWER	1.5V	PP1V5_S3
POWER	POWER	1.5V	PP1V5_CPU_MEM
POWER	POWER	1.5V	PP1V8R1V5_S0_PCH_VCCVRM
POWER	POWER	1.5V	PP1V5_FW_VDDA
POWER	POWER	1.5V	PP1V8_S0
POWER	POWER	1.96V	PP1V96_FW_PLLVDD
POWER	POWER	1.96V	PP1V95_FW_FWPHY

V_USB2_PORT1

POWER NET PROPERTIES

PHYSICAL	SPACING	VOLTAGE	NET
POWER	POWER	3.3V	PP3V3_S0
POWER	POWER	3.3V	PP3V3_S0_CK505_F
POWER	POWER	3.3V	PP3V3_S0_DPFUSE
POWER	POWER	3.3V	PP3V3_S0_DPPWR
POWER	POWER	3.3V	PP3V3_S0_HS_F
POWER	POWER	3.3V	PP3V3_S0_PCH_VCCA_DAC
POWER	POWER	3.3V	PP3V3_S0_TSENS_R
POWER	POWER	3.3V	PP3V3_S3
POWER	POWER	3.3V	PP3V3_S3_BT_FLT
POWER	POWER	3.3V	PP3V3_S3_SDCARD_FLT
POWER	POWER	3.3V	PP3V3_S5
POWER	POWER	3.3V	PPVTT_S3_DDR_BUF
POWER	POWER	3.3V	PPV_S0_MXM_PWRSRC
POWER	POWER	3.3V	PPVOUT_S0_PCH_DCPSST
POWER	POWER	3.3V	PPVOUT_S5_PCH_DCPSUS
POWER	POWER	3.3V	PPVOUT_S5_PCH_DCPSUSBYP
POWER	POWER	3.3V	PPVOUT_G3_PCH_DCPRTC
POWER	POWER	3.3V	PPVOUT_S0_PCH_VCCRTC_NCTF
POWER	POWER	3.3V	PPVBATT_G3_RTC
POWER	POWER	3.3V	PPVBATT_G3_RTC_R
POWER	POWER	3.3V	PP3V3_AUDIO_SPDIF_JACK
POWER	POWER	3.3V	PP3V3_FW_AVDD
POWER	POWER	3.3V	PP3V3_FW_ESD
POWER	POWER	3.3V	PP3V3_FW_PLLVDD
POWER	POWER	3.3V	PP3V3_FW_VDDA
POWER	POWER	3.3V	PP3V3_G3_RTC
POWER	POWER	3.3V	PP_ENET_CTRL12
POWER	POWER	3.4V	PP3V3_G3H_SMC_AVCC
POWER	POWER	3.3V	PP3V3_G3H_AVREF_SMC
POWER	POWER	3.42V	PP3V42_G3H
POWER	POWER	4.5V	4V5_REG_IN
POWER	POWER	4.5V	PP4V5_AUDIO_ANALOG
POWER	POWER	5V	PP5V_S0
POWER	POWER	5V	PP5V_S0_CPU_VCORE_VCC
POWER	POWER	5V	PP5V_S0_PCH_V5REF
POWER	POWER	5V	PP5V_S3
POWER	POWER	5V	PP5V_S3_DDR_REG_V5FILT
POWER	POWER	5V	PP5V_S3_CAMERA_FLT
POWER	POWER	5V	PP5V_S3_IR_FLT
POWER	POWER	5V	PP5V_S5
POWER	POWER	5V	PP5V_S5_PCH_V5REFSUS
POWER	POWER	5V	PP5V_USB2_PORT0
POWER	POWER	5V	PP5V_USB2_PORT0_F
POWER	POWER	5V	PP5V_USB2_PORT1
POWER	POWER	5V	PP5V_USB2_PORT1_F
POWER	POWER	5V	PP5V_USB2_PORT2
POWER	POWER	5V	PP5V_USB2_PORT2_F
POWER	POWER	5V	PP5V_USB2_PORT3
POWER	POWER	5V	PP5V_USB2_PORT3_F
POWER	POWER		DDR_REG_PGND
POWER	POWER		DDR_REG_CSGND

SENSING NET PROPERTIES

PHYSICAL	SPACING	NET
SNS_DIFF	THERMAL	VR_CPU_ISNS1_P
SNS_DIFF	THERMAL	VR_CPU_ISNS1_N
SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_P
SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_N
SNS_DIFF	THERMAL	VR_CPU_ISNS2_P
SNS_DIFF	THERMAL	VR_CPU_ISNS2_N
SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_P
SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_N
SNS_DIFF	THERMAL	VR_CPU_ISNS3_P
SNS_DIFF	THERMAL	VR_CPU_ISNS3_N
SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_P
SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_N
SNS_DIFF		VR_CPU_VSEN
SNS_DIFF		VR_CPU_RGND
SNS_DIFF		VR_CPU_VSNS_R_N
SNS_DIFF		VR_CPU_VSNS_R_P
SNS_DIFF		VR_CPU_VSNS_XW_P
SNS_DIFF		VR_CPU_VSNS_XW_N
SNS_DIFF		CPU_VCC_PKG_SENSE_P
SNS_DIFF		CPU_VCC_PKG_SENSE_N
SNS_DIFF		VTT_REG_ISNS_P
SNS_DIFF		VTT_REG_ISNS_N
SNS_DIFF		CPU_VTTSENSE_P
SNS_DIFF		CPU_VTTSENSE_N
SNS_DIFF		CPU_VTTSENSE_R_P
SNS_DIFF		CPU_VTTSENSE_R_N
SNS_DIFF		VTT_REG_VSEN
SNS_DIFF		VTT_REG_RGND
SNS_DIFF		VTT_REG_RTO1
SNS_DIFF		VTT_REG_RTR1

VR CTRL NET PROPERTIES

PHYSICAL	SPACING	NET
VR_CTL_PHY	VR_CTL	VR_CPU_PH1_SNUB
VR_CTL_PHY	VR_CTL	VR_CPU_PH2_SNUB
VR_CTL_PHY	VR_CTL	VR_CPU_PH3_SNUB
VR_CTL_PHY	VR_CTL	VR_CPU_PWM1
VR_CTL_PHY	VR_CTL	VR_CPU_PWM2
VR_CTL_PHY	VR_CTL	VR_CPU_PWM2_R
VR_CTL_PHY	VR_CTL	VR_CPU_PWM3
VR_CTL_PHY	VR_CTL	VR_CPU_PWM3_R
VR_CTL_PHY	VR_CTL	VR_CPU_PWM4_R
VR_CTL_PHY	VR_CTL	VR_CPU_REF
VR_CTL_PHY	VR_CTL	VR_CPU_SS
VR_CTL_PHY	VR_CTL	VR_CPU_TCOMP
VR_CTL_PHY	VR_CTL	VR_CPU_TM
VR_CTL_PHY	VR_CTL	VR_CPU_BOOT1_RC
VR_CTL_PHY	VR_CTL	VR_CPU_BOOT2_RC
VR_CTL_PHY	VR_CTL	VR_CPU_BOOT3_RC
VR_CTL_PHY	VR_CTL	VR_CPU_COMP
VR_CTL_PHY	VR_CTL	VR_CPU_COMP_R
VR_CTL_PHY	VR_CTL	VR_CPU_COMP_RC
VR_CTL_PHY	VR_CTL	VR_CPU_DAC
VR_CTL_PHY	VR_CTL	VR_CPU_DRV1_BOOT
VR_CTL_PHY	VR_CTL	VR_CPU_DRV1_GDSEL
VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV1_LGATE
VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV1_UGATE
VR_CTL_PHY	VR_CTL	VR_CPU_DRV2_BOOT
VR_CTL_PHY	VR_CTL	VR_CPU_DRV2_GDSEL
VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV2_LGATE
VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV2_UGATE
VR_CTL_PHY	VR_CTL	VR_CPU_DRV3_BOOT
VR_CTL_PHY	VR_CTL	VR_CPU_DRV3_GDSEL
VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV3_LGATE
VR_CTL_PHY	SWITCHNODE	VR_CPU_DRV3_UGATE
VR_CTL_PHY	VR_CTL	VR_CPU_FAN
VR_CTL_PHY	VR_CTL	VR_CPU_FB
VR_CTL_PHY	VR_CTL	VR_CPU_FB_R
VR_CTL_PHY	VR_CTL	VR_CPU_FS
VR_CTL_PHY	VR_CTL	VR_CPU_IMON
VR_CTL_PHY	VR_CTL	VR_CPU_IOUT_PD
VR_CTL_PHY	SWITCHNODE	PCHCORE_REG_UGATE
VR_CTL_PHY	SWITCHNODE	PCHCORE_REG_LGATE
VR_CTL_PHY	VR_CTL	PCHCORE_REG_VFB
VR_CTL_PHY	VR_CTL	PCHCORE_REG_TON
VR_CTL_PHY	VR_CTL	PCHCORE_REG_TRIP
VR_CTL_PHY	VR_CTL	PCHCORE_REG_BOOT
VR_CTL_PHY	VR_CTL	PCHCORE_REG_BOOT_R
VR_CTL_PHY	VR_CTL	VTT_REG_BOOT
VR_CTL_PHY	VR_CTL	VTT_REG_COMP
VR_CTL_PHY	VR_CTL	VTT_REG_FB
VR_CTL_PHY	VR_CTL	VTT_REG_FS
VR_CTL_PHY	VR_CTL	VTT_REG_COMP
VR_CTL_PHY	VR_CTL	VTT_REG_REF
VR_CTL_PHY	SWITCHNODE	VTT_REG_LGATE
VR_CTL_PHY	VR_CTL	VTT_REG_OCSET
VR_CTL_PHY	VR_CTL	VTT_OFS
VR_CTL_PHY	VR_CTL	VTT_REG_REF
VR_CTL_PHY	VR_CTL	VTT_REG_UGATE
VR_CTL_PHY	VR_CTL	VTT_REG_PH1_SNUB
VR_CTL_PHY	VR_CTL	P3V42G3H_BOOST
VR_CTL_PHY	VR_CTL	P3V42G3H_FB

VR CTRL NET PROPERTIES

PHYSICAL	SPACING	NET
VR_CTL_PHY	VR_CTL	DDR_REG_CS
VR_CTL_PHY	VR_CTL	DDR_REG_FB
VR_CTL_PHY	SWITCHNODE	DDR_REG_LGATE
VR_CTL_PHY	SWITCHNODE	DDR_REG_UGATE
VR_CTL_PHY	VR_CTL	DDR_REG_BOOT
VR_CTL_PHY	VR_CTL	DDR_REG_BOOT_R
VR_CTL_PHY	VR_CTL	DDR_REG_VDDQSNS
VR_CTL_PHY	VR_CTL	DDR_REG_VTTSNS
VR_CTL_PHY	VR_CTL	P1V8_REG_POR
VR_CTL_PHY	VR_CTL	P3V3S5_REG_BOOT
VR_CTL_PHY	VR_CTL	P3V3S5_REG_BOOT_R
VR_CTL_PHY	VR_CTL	P3V3S5_REG_FB
VR_CTL_PHY	VR_CTL	P3V3S5_REG_ISEN
VR_CTL_PHY	SWITCHNODE	P3V3S5_REG_LGATE
VR_CTL_PHY	VR_CTL	P3V3S5_REG_OCSET
VR_CTL_PHY	SWITCHNODE	P3V3S5_REG_UGATE
VR_CTL_PHY	VR_CTL	P3V3S5_REG_SNUB
VR_CTL_PHY	VR_CTL	P5VS3_REG_BOOT
VR_CTL_PHY	VR_CTL	P5VS3_REG_FB
VR_CTL_PHY	VR_CTL	P5VS3_REG_ISEN
VR_CTL_PHY	SWITCHNODE	P5VS3_REG_LGATE
VR_CTL_PHY	VR_CTL	P5VS3_REG_OCSET
VR_CTL_PHY	SWITCHNODE	P5VS3_REG_UGATE

VR VID NET PROPERTIES

PULL-UP STUB < 1-INCH
VID LENGTH SKEW < 1-INCH
VID LENGTH RANGE< 1 TO 15-INCH

PHYSICAL	SPACING	NET
VID_PHY	VR_CTL	CPU_VID<0>
VID_PHY	VR_CTL	CPU_VID<1>
VID_PHY	VR_CTL	CPU_VID<2>
VID_PHY	VR_CTL	CPU_VID<3>
VID_PHY	VR_CTL	CPU_VID<4>
VID_PHY	VR_CTL	CPU_VID<5>
VID_PHY	VR_CTL	CPU_VID<6>
VID_PHY	VR_CTL	CPU_VID<7>
VID_PHY	VR_CTL	CPU_PSI_L

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VID_PHY	*	39_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL	*	0.2MM	?

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
POWER CONSTRAINTS

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SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

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PM NET PROPERTIES
(PM, RESET, EN, PGOOD)

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	2:1_SPACING
PM_VTT	PM_VTT	*	2:1_SPACING
PM_VTT	*	*	3:1_SPACING
PM_VTT	GND	*	DEFAULT
PM	GND	*	DEFAULT

PHYSICAL	NET_TYPE SPACING	NET	PAGES
I2	PM	PLT_RESET_L	19 27
I1	PM_VTT	PLT_RESET_LS1V1_L	10
I4	PM	PM_ACDC_PS_ON	5
I3	PM	PM_BATLOW_L	14 18 46
I5	PM	PM_CLK32K_SUSCLK	8 46 85
I7	PM	PM_CLK32K_SUSCLK_R	8 18 85
I6	PM	PM_CLKRUN_L	14 18 46 48
I10	PM	PM_EXT_TS_L<0>	
I9	PM	PM_EXT_TS_L<1>	
I12	PM	PM_LAN_PWRGD	14 18
I117	PM_VTT	FSB_CPURSTOUT_L	10 24
I25	PM_VTT	PM_MEM_PWRGD	10 18
I24	PM	PM_ME_PWRGD	18 64
I30	PM	PM_MXM_PGOOD	64 75
I29	PM	PM_PCH_PWRGD	18 64
I31	PM	PM_PGOOD_DDRREG_S3	63 71
I33	PM	PM_PGOOD_PVCORE_CPU	25 64 65
I32	PM	PM_PWRBTN_L	18 24 46
I35	PM	PM_RSMRST_L	46 63
I34	PM	PM_RSMRST_PCH_L	18 63
I36	PM	PM_SLP_M_L	18 63
I37	PM	PM_SLP_S3_L	5 18 26 36 46 47 63 64 81
I120	PM	PM_SLP_S4_1_L	18 63
I39	PM	PM_SLP_S4_L	18 47
I41	PM	PM_SLP_S5_L	18 47
I43	PM	PM_SUS_PWR_ACK	18
I42	PM_VTT	PM_SYNC	10 18
I122	PM	SDCARD_PLT_RST_L	27 44
I45	PM	PM_SYSRST_L	18 27 46
I44	PM	PM_SYS_PWRGD	18 64
I46	PM_VTT	PM_THRMTRIP_L	10 20 47
I48	PM	RSMRST_PWRGD	46 64
I47	PM	RTC_RESET_L	17 91
I50	PM_VTT	CPU_PWRGD	10 20 24
I49	PM	CPU_RESET_L	10 27
I52	PM	PGOOD_1V8_S0_G1	64
I54	PM	PGOOD_1V8_S0_G2	64
I56	PM	PGOOD_CPU_GFX_DDR	64
I57	PM	PGOOD_P1V5_S0	10 73
I59	PM	PGOOD_P1V8_S0	64
I60	PM	PGOOD_P3V3_S0	64 73
I63	PM	PGOOD_P3V3_S3	34 73
I62	PM	PGOOD_P5V_S0	63 73
I64	PM	PGOOD_PCH_AND_P1V8	64
I66	PM	PGOOD_PCH_S0	64
I65	PM	PGOOD_SYSPWROK	64
I67	PM	PGOOD_SYSPWROK_R	64
I68	PM	RTC_RESET_L	17 91
I69	PM	P12V_S3_EN	
I70	PM	P1V5_S0_EN	63 73
I74	PM	P3V3S0_EN	63 73
I73	PM	P3V3S3_EN	63 73
I76	PM	P5VS0_EN	63 73
I75	PM	P5VS3_EN	63 70
I77	PM	PCHCORE_REG_EN	63 69
I79	PM	PCHCORE_REG_PGOOD	63 64 69
I78	PM	PEG_RESET_L	8 27
I80	PM	SDCARD_RESET	20 44 92

PHYSICAL	NET_TYPE SPACING	NET	PAGES
I82	PM	4V5_REG_EN	56
I81	PM	ALL_SYS_PWRGD_R	5 64
I84	PM	ALL_SYS_PWRGD_SMC	46 64
I83	PM	CK505_27MHZ_EN	25
I85	PM	CPUVTT_REG_EN	63 68
I86	PM_VTT	CPUVTT_REG_PGOOD	10 63 64 68
I87	PM	CPU_MEM_RESET_L	10 26
I88	PM	DDRVTT_EN	63 71
I89	PM	DEBUG_RESET_L	27 48
I90	PM	FWPHY_RESET_L	39
I91	PM	FWXIO_SNOOP_EN	39
I92	PM	FW_RESET_L	27 39
I95	PM	ENET_RESET_L	27 37
I97	PM	MEM_RESET_L	26 30 31
I96	PM	MINI_RESET_L	27 33
I98	PM	SMC_DELAYED_PWRGD	47 64
I100	PM	SMC_LRESET_L	27 46
I99	PM	SMC_RESET_L	46 47 48
I101	PM_VTT	XDP_CPUPWRGD	10 24
I103	PM_VTT	XDP_DBRESET_L	10 24 27
I104	PM_VTT	XDP_PWRGD	24

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE
PM RESETS ENABLES PGOOD CONST

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FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT

J4700 USB CAMERA

89 5 IN PP5V_S3 FUNC_TEST=TRUE MIN_ALLOWED_TPS=1
85 44 IN USB_CAMERA_L_P FUNC_TEST=TRUE
85 44 IN USB_CAMERA_L_N FUNC_TEST=TRUE
85 44 IN USB_BT_L_P FUNC_TEST=TRUE
85 44 IN USB_BT_L_N FUNC_TEST=TRUE
49 44 IN =SMB_ALS_SCL FUNC_TEST=TRUE
49 44 IN =SMB_ALS_SDA FUNC_TEST=TRUE

1 PP5V_S3_REG Testpoint near J4700
1 PP3V3_S3 TESTPOINT NEAR J4700
6 GROUND TESTPOINTS NEAR J4700

J4750 USB CARD READER

85 44 IN USB_SDCARD_L_P FUNC_TEST=TRUE
85 44 IN USB_SDCARD_L_N FUNC_TEST=TRUE
91 44 20 IN SDCARD_RESET FUNC_TEST=TRUE

1 PP3V3_S3 Testpoint near J4750
2 Ground Testpoints near J4750

J4780 IR BOARD

85 44 IN USB_IR_L_P FUNC_TEST=TRUE
85 44 IN USB_IR_L_N FUNC_TEST=TRUE
89 44 IN PP5V_S3_IR_FLT FUNC_TEST=TRUE

1 GROUND TESTPOINT NEAR J4780

J4520 SATA ODD (HIGH SPEED)

46 42 IN SMC_ODD_DETECT FUNC_TEST=TRUE

1 PP5V_S0 Testpoint near J4520
1 GROUND TESTPOINTS NEAR J4520

J5551 ODD TEMP SENSOR

88 52 IN SNS_ODD_P FUNC_TEST=TRUE
88 52 IN SNS_ODD_N FUNC_TEST=TRUE

J5600 ODD FAN

53 IN FAN_0_PWR_L FUNC_TEST=TRUE
53 IN FAN_TACH0_L FUNC_TEST=TRUE
89 53 IN PP12V_S0_FAN0_L FUNC_TEST=TRUE
53 IN FAN_0_GND FUNC_TEST=TRUE

J5700 CPU FAN

54 IN FAN_2_PWR_L FUNC_TEST=TRUE
54 IN FAN_TACH2_L FUNC_TEST=TRUE
89 54 IN PP12V_S0_FAN2_L FUNC_TEST=TRUE
54 IN FAN_2_GND FUNC_TEST=TRUE
88 54 52 IN SNS_AMB_P FUNC_TEST=TRUE
88 54 52 IN SNS_AMB_N FUNC_TEST=TRUE

1 GROUND TESTPOINT NEAR J5700

J5601 HD FAN

53 IN FAN_1_PWR_L FUNC_TEST=TRUE
53 IN FAN_TACH1_L FUNC_TEST=TRUE
89 53 IN PP12V_S0_FAN1_L FUNC_TEST=TRUE
53 IN FAN_1_GND FUNC_TEST=TRUE

J5400 HDD TEMP SENSOR

88 51 IN HDD_OOB_TEMP_FILT FUNC_TEST=TRUE

1 GROUND TESTPOINTS NEAR J5400

J5560 SKIN TEMP SENSOR

88 52 IN SNS_SKIN_P FUNC_TEST=TRUE
88 52 IN SNS_SKIN_N FUNC_TEST=TRUE

J6601 AUDIO MICROPHONE

60 IN AUD_MIC_IN1_N_CONN FUNC_TEST=TRUE
60 IN GND_AUDIO_MIC1_CONN FUNC_TEST=TRUE
60 IN AUD_MIC_IN1_P_CONN FUNC_TEST=TRUE

1 Ground Testpoint near J6601

J6602 AUDIO RIGHT SPEAKER

85 60 58 IN AUD_SPKR_OUTLO2R_POUT FUNC_TEST=TRUE
85 60 58 IN AUD_SPKR_OUTLO2R_NOUT FUNC_TEST=TRUE
85 60 58 IN AUD_SPKR_OUTLO1R_POUT FUNC_TEST=TRUE
85 60 58 IN AUD_SPKR_OUTLO1R_NOUT FUNC_TEST=TRUE

J6603 AUDIO LEFT SPEAKER

85 60 59 IN AUD_SPKR_OUTLO2L_POUT FUNC_TEST=TRUE
85 60 59 IN AUD_SPKR_OUTLO2L_NOUT FUNC_TEST=TRUE
85 60 59 IN AUD_SPKR_OUTLO1L_POUT FUNC_TEST=TRUE
85 60 59 IN AUD_SPKR_OUTLO1L_NOUT FUNC_TEST=TRUE

J6600 AUDIO AUXILIARY CONNECTOR

89 60 IN PP3V3_AUDIO_SPDIF_JACK 2 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=2
60 IN AUD_LI_DET_JACK FUNC_TEST=TRUE
60 IN AUD_LI_R_JACK FUNC_TEST=TRUE
60 IN AUD_LI_GND_JACK FUNC_TEST=TRUE
60 IN AUD_LI_L_JACK FUNC_TEST=TRUE
60 IN HS_MIC_LO_JACK FUNC_TEST=TRUE
60 IN HS_MIC_HI_JACK FUNC_TEST=TRUE
60 IN AUD_HP_L_JACK FUNC_TEST=TRUE
60 IN AUD_HP_GND_JACK FUNC_TEST=TRUE
60 IN AUD_HP_R_JACK FUNC_TEST=TRUE
60 IN AUD_HP_TYPEDET_JACK FUNC_TEST=TRUE
60 IN AUD_IP_PERPH_JACK FUNC_TEST=TRUE
60 IN AUD_HP_TIPDET_JACK FUNC_TEST=TRUE
60 IN AUD_SPDIFIN_JACK FUNC_TEST=TRUE
60 IN AUD_SPDIF_OUT_JACK FUNC_TEST=TRUE

4 GROUND TESTPOINTS NEAR J6600

GND 17 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=17
89 5 IN PP3V3_S3 2 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=2
70 5 IN PP5V_S3_REG 1 TP'S FUNC_TEST=TRUE MIN_ALLOWED_TPS=1
89 5 IN PP5V_S0 FUNC_TEST=TRUE MIN_ALLOWED_TPS=1

SYNC_MASTER=K74_MASTER SYNC_DATE=N/A
PAGE TITLE
K74/K75 ICT/FCT
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